

Naming Elementary Embeddings

Atom Components, Replacement,
and Reflection

Exact criteria and sharp cofinality thresholds

Prepared for Vladimir Reshetnikov

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Research status. Complete mathematical arguments are supplied for the stated results. The kernel-ideal construction is established prior work, explicitly credited below. The component-profile criterion and its finite-signature refinements are proposed contributions of this manuscript; independent review and literature-priority verification remain outstanding. No Lean verification is claimed. Elliot Glazer and Bokai Yao are cited researchers, not authors or endorsers of this manuscript.

[write] Filed in Provelt's research-report collection as `ordinals-and-order-types/naming-elementary-embeddings` (batch 89, manuscript 05; label prefix `nee:`). Provenance, formal status, the relation to the repository and the notation table are in Sections 1.4 and 1.5.

[write] Since 4 October 2026 the report has two Parts: Part I (Sections 1–11 and Appendices A–C; batch 89, labels nee:) is the original report, with its numbering unchanged; Part II (from Section 12; batch 93, labels nee:ca:) treats commuting actions and uniformly named groups, answers Question 10.4 of Part I, and answers Question 10.1 for commuting permutations and the fixed-action part of Question 10.10. The abstract below is Part I’s; Part II has its own. One bibliography serves both Parts. [Added 4 October 2026, batch 96: since then a third Part, Part III (from Section 24, with Appendices F–G; batch 96, labels nee:aa: and nee:ci:), merges two manuscripts on relations among named injections. It answers Question 23.3 of Part II and, with pure real parameters only, Questions 23.1 and 23.2 there and Question 10.1 of Part I. It has its own abstracts; the one bibliography serves all three Parts.]

Abstract

Let A be an external set of atoms, let κ be an infinite cardinal with $|A| \geq \kappa$, and let $M_\kappa(A)$ consist of the objects whose transitive closures involve fewer than κ atoms. This is a standard kernel-ideal model of replacement-based set theory with urelements and choice. Every injection $f : A \rightarrow A$ has a canonical, amenable elementary lift $j_f : M_\kappa(A) \rightarrow M_\kappa(A)$ fixing every pure set. Nevertheless, naming these lifts can invalidate Replacement.

For a finite tuple f of atom injections, consider the connected components of its labeled functional graph. Write B_t for the union of all components of isomorphism type t , and put

$$D_\kappa(f) = \bigcup \{B_t : |B_t| < \kappa\}.$$

We prove that for $\kappa > \omega$, Replacement in the expanded language holds exactly when $|D_\kappa(f)| < \kappa$. For $\kappa = \omega$, it holds exactly when this union is finite and all components are finite. The proof combines uniform internal coding of component types with a fresh-component automorphism argument.

There are countably many one-injection component types, but continuum many types for two injections, already for two permutations. Consequently every one-symbol expansion preserves Replacement exactly when $\text{cf}(\kappa) > \omega$, whereas every expansion by $r \geq 2$ fixed, finitely many symbols preserves it exactly when $\text{cf}(\kappa) > 2^{\aleph_0}$. At the cutoff $\aleph_2$, the latter universal preservation property is equivalent, externally, to the continuum hypothesis. This is a characterization, not a resolution, of that hypothesis.

We also classify the initial elementary self-embeddings, determine precisely which canonical lifts are parameter-definable, and prove that full Collection and full transitive reflection in an expanded model hold exactly when κ is a successor cardinal and fewer than κ component types are realized. At limit cutoffs, the exact domain-size threshold for Collection is $\text{cf}(\kappa)$, with failure already in the unexpanded bounded formula fragment. The final sections identify concrete interfaces with Provel’s first-order set theory and bounded-consistency developments and propose twelve further research directions.

Keywords. Urelements; elementary self-embeddings; amenable classes; expanded-language Replacement; Collection; reflection; component types; continuum hypothesis; formalization.

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Part I

Atom components, Replacement, and reflection

1 Motivation, scope, and the contribution boundary

1.1 Why this intersection is substantive

Glazer and Yao's *Reflection Principles in ZFU* studies how Collection and several forms of reflection diverge when urelements are admitted, especially without choice [1]. Glazer's work on global choice over Zermelo set theory emphasizes another source of strength: adjoining a class function and allowing its symbol in an axiom scheme is not merely a change of notation [2]. These two themes meet in a concrete question:

When an elementary self-embedding is named, which set-existence schemes continue to hold in the expanded language?

ProveIt supplies a complementary perspective. Its first-order ZF development represents Separation and Replacement as constructions on formulas, while its bounded-consistency project explicitly distinguishes external families of formulas from single internally quantified assertions [4, 5]. A new symbol changes the formulas over which a scheme ranges. This manuscript studies that change semantically in models where the obstruction can be computed exactly.

The repository also contains a substantial, explicitly unformalized report on surreal self-embeddings [6]. Here the embeddings are of a set-theoretic universe with atoms, not of the surreal ordered field. They fix the entire pure universe. Thus they provide a useful control example: nontrivial elementary self-embeddings need not move ordinals when atoms are present, but they do not produce nontrivial embeddings of the usual pure-set presentations of surreal numbers.

1.2 The mathematical question

An elementary map preserves every formula of a specified language. An amenable map has a set-sized restriction to each internal set. Neither statement says that *every function definable using a name for the map has an internal range set*. That third assertion is expanded-language Replacement.

The difference is not cosmetic. We will construct an automorphism j for which every restriction $j \upharpoonright x$ belongs to the model, yet Replacement fails once j is named. More sharply, at the cutoff $\aleph_1$ we construct two automorphisms such that each one-symbol expansion satisfies Replacement, Collection, and reflection, while their joint expansion fails Replacement.

The main classification is local to a precisely specified family of models. It is not a classification of all models of set theory with urelements, all amenable classes, or all elementary embeddings of the universe.

1.3 What is known and what is proposed here

The construction using an ideal of atom kernels is not new. Yao's Definition 24 and Theorem 26 give it in substantially greater generality; his Theorem 27 contains relevant finite, countable, and

larger kernel-bound examples [3]. We reproduce the elementary symmetry argument needed for the later results, rather than treating that construction as an original discovery.

Part of the manuscript	Status and role
Kernel-ideal models	Established construction, credited to Yao; reviewed for self-containment.
Canonical atom lifts	Explicit structural development using standard permutation symmetry; no claim that the underlying lifting idea is new.
Component-profile criterion	Principal proposed contribution: an exact criterion for full expanded-language Replacement.
One-name versus two-name thresholds	Derived proposed contribution, including an explicit coding construction and a CH characterization.
Collection and reflection spectrum	A detailed calibration for the expanded family; the unexpanded cutoff phenomena build on the established kernel-ideal method.
Formalization	A concrete design and dependency audit, not a completed Lean development.

No assertion below is advertised as the solution of an open problem explicitly posed by Glazer. Instead, the paper formulates and completely answers the model-family classification questions above. The available literature search establishes the prior-art boundary just described; it does not establish publication priority for every refinement.

1.4 Provenance and place in the collection

[write] Added 3 October 2026, batch 89. This report is a single manuscript, printed in full: manuscript 05 of batch 89 of Provelt’s incoming-reports intake, delivered as the archive `Naming_Elementary_Embeddings_Research.zip` (330,868 bytes; inner directory `Naming_Elementary_Embeddings/`, main file `article.tex`, 1041 lines, a 27-page US Letter PDF) in the arrival commit `8ea27d6c0`, and placed in commit `23adb85f9` in the research-report collection as `ordinals-and-order-types/naming-elementary-embeddings`. It contains no surreal mathematics and continues the formal projects `SetTheory/ZF` and `SetTheory/BoundedConsistency`, so it was filed in the collection, in the category that already holds Provelt’s other set-theoretic reports, rather than in either project. The author lines read “Prepared for Vladimir Reshetnikov” and “Research manuscript prepared with ChatGPT”; the report is AI-assisted, unrefereed and not formalized. Its four sister manuscripts of batch 89 became additions to surreal-collection reports: Part XV of `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders` (manuscript 01), Part X of `.../polish-models-of-omnific-arithmetic` (02), and Parts VI and VII of `.../birthday-cutoffs-and-hereditary-sets` (03 and 04). Only manuscript 03 shares a theorem with this report (see below). Being a single source, the report needed no merge choices. The write prefixed all 64 delivered labels with `nee:` and updated every reference (`38 \cref keys`, `13 \eqref`); added the two labels `nee:sec:provenance` and `nee:sec:notation`; added these dated **[write]** notes, the title-page line and the `pageanchor` switch around the title page (a delivered build gave a duplicate `page.1` destination). No statement, proof, number or non-claim of the manuscript was changed; no section, theorem or equation number moved. [Added 4 October 2026, batch 93: this paragraph describes what is now Part I. Batch-93 manuscript 01, *Commuting Actions and Replacement*, written against this report, became Part II (Section 12.4 gives its provenance); wrapping the text above as Part I moved no number, and the single source of Part I still needed no merge choices.]

Pin and repository claims. The pin `2b7b388ba81a3355b19a7c2d2fe797e92f572355` (Section 9 and the repository entries of the bibliography) is the Provelt commit “Catalogue batches 81 to 86” of 3 October 2026, an ancestor of the placement. The three repository files the manuscript read are byte-identical at the pin and

at the write: SetTheory/ZF/Lean/ZF/Zf.lean (blob b33db57cf), SetTheory/BoundedConsistency/README.md (ebb5d1bfa) and Algebra/SurrealNumbers/docs/surreal/surreal-self-embeddings/README.md (554357db9). What Sections 1, 8 and 9 say about them is accurate: Zf.lean declares Sep_form, Func_form, Image_form, Repl_form, ZFax, ZFprov and ZFax_s, with schema constructors over the formula type Form, semantic bridges and an internal finite-recursion theorem; the bounded-consistency README states a numeralwise endpoint, chooses partial satisfaction over Levy reflection, and records that its quantifier-group rank, in a syntax without primitive bounded quantifiers, is strictly finer than the Levy hierarchy; the self-embedding report is marked unrefereed and not formalized. No repository statement is contradicted, and nothing needed retraction.

Formal status. No statement of this report is formalized in Lean or Rocq, and its place beside the formal projects confers no formal status. What those projects do formalize is pure membership set theory: SetTheory/ZF/Lean/ZF/Zf.lean (a Lean port of the Rocq file SetTheory/ZF/Coq/Zf.v, which declares Sep_form, Repl_form, ZFax and ZFax_s likewise) proves the bridges bridge_Sep and bridge_Repl and, inside every model of Extensionality, Separation, Pairing, Union, Infinity and Replacement, the closure theorem ClosuresF0_of_ZF; SetTheory/BoundedConsistency proves, by partial satisfaction, the numeralwise endpoint BoundedZFConsistency.Endpoint.zfc_proves_conZFC ($ZFC \vdash \text{Con}_n(ZFC)$ for each metatheoretic n). Their formula type Form (Logic/FirstOrder/Lean/FirstOrder/Fol.lean) is purely relational, with the atomic formulas $\in$ and $=$ only. Neither the base language $\mathcal{L}_U = \{\in, \text{At}\}$ nor the expanded language $\mathcal{L}_r$ of this report exists there, so the “first task” of Section 9 and all six layers that follow it are outstanding. No urelement theory is formalized anywhere in Provel.

Review in the Hilbert’s-tenth research tree. Before placement, another session routed all five batch-89 archives (commit 6aadaa4f2): Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_topology_surreal_arrivals_20261003.md, with the receipt review_topology_surreal_arrivals_20261003.json beside it. For this manuscript it read the delivered README and source audit, the recorded finite-check scope, and the delivered lines 118–133, 183–211, 807–865, 911–927 and 988–994 of article.tex (the abstract, Section 2 through Remark 2.1, Section 9, the last two questions of Section 10 with the conclusion, and Appendix C). It records that “finitely many names” is a language parameter, not a bound on Diophantine witnesses, that the finite checks are regression evidence, not verification of the infinite theorems, and that the proposed formalization is not an implemented compiler; it finds nothing that changes the Hilbert’s-tenth programme’s 84-operation universal polynomial. It claims no theorem audit, literature check or replay of the finite checks, and it made no correctness finding; nothing here needed a change of scope.

The same Collection theorem in a surreal report. Batch-89 manuscript 03, *Surreal Cuts, Replacement, and Atom-Support Spectra*, written against the same pin and placed in the same commit, is Part VI of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (labels hset:zr:; written concurrently with this report). It studies the same kernel models, which it writes M_κ^{ker} (manuscript 03’s M_κ , renamed there), and proves, in its delivered numbering (Part VI numbering in brackets, added 4 October 2026, after that write was committed): kernel confinement of a unique output (its Lemma 9.1 [Lemma 32.1, hset:zr:lem:confinement], our lemma 2.3); the base model theorem (its Theorem 9.2 [Theorem 32.2, hset:zr:thm:supportbase], our theorem 2.4; this report credits the construction to Yao [3], manuscript 03 to Yao and to Hamkins–Yao with Lévy); and the Collection spectrum of these models (its Lemma 10.1, Theorems 10.2–10.3 and Corollary 10.4 [Lemma 33.1, Theorems 33.2–33.3 and Corollary 33.4, hset:zr:cor:classification]): Collection holds on domains of size below $\text{cf } \kappa$ for every infinite κ , fails on a domain of size $\text{cf } \kappa$ when κ is a limit cardinal (ω included), and holds in full when κ is a successor. That is the unexpanded case of lemma 7.1, lemma 7.2, corollary 7.3 and theorem 7.5 (take $f = (\text{id}_A)$, so that $|I(f)| = 1$). The two proofs are independent and both stand; neither manuscript cites the other, and neither claims priority for the unexpanded spectrum. They differ as the note after corollary 7.6 records. Part VI records the overlap in its Remark 24.6 (hset:zr:rem:nee) and prints manuscript 03’s own proofs. Part XV of surreal-well-orders (manuscript 01) finds the same cofinality threshold $\text{cf } \lambda$ for class Replacement in the full-class rank models $(V_\lambda, \mathcal{P}(V_\lambda))$; those are different models, and no proof is shared.

Neighbouring reports. The surreal report birthday-cutoffs-and-hereditary-sets also proves, in its subsection

“Transfer of Kunen’s inconsistency” (Theorem `hset : thm : kunen`), that the birthday-expanded surreal class has no nontrivial class elementary self-embedding, under the Kunen hypotheses that allow the embedding as a class parameter in Separation and Replacement. This report shows how strong that hypothesis is in another setting: in the urelement models $M_\kappa(A)$ nontrivial amenable elementary self-embeddings exist (theorem 3.2), and naming them may or may not preserve Replacement (theorem 5.1). There is no conflict: every j_f fixes the whole pure universe, and Kunen’s theorem concerns embeddings of a pure universe with choice. The self-embedding report cited as [6] makes the analogous point in its Remark `sse : pf : rem : kunen`. In this category, ordinals-and-order-types/measurable-box-games also answers a Glazer paper and also cites [2], but shares no theorem or notation, and ordinals-and-order-types/lexicographic-well-orderings-of-reals concerns pure ZFC. Apart from Parts VI and VII of birthday-cutoffs-and-hereditary-sets (manuscripts 03 and 04; Part VII’s Fraenkel–Mostowski permutation models are choiceless models with atoms, unlike the models here), no report in ProveIt treats urelements or cites [1] or [3]. [Added 4 October 2026, batch 90: Part VIII of the same surreal report (batch-90 manuscript 05) also works in finite-kernel models with atoms and cites [1] (see the note at the end of section 5), and Part XII of polish-models-of-omnific-arithmetic (batch-90 manuscript 03) cites [1] as context for its class theory without Foundation; neither cites [3].] [Added 6 October 2026, batch 114: ordinals-and-order-types/bounded-width-power-set-compression also cites [1], as motivation only (Glazer’s work with Yao separates principles that coincide under Choice but diverge in ZF with urelements); it works in ZF without atoms, names this report only for that shared citation, does not cite [3], and shares no theorem with this report. The sentence above, “no report in ProveIt . . . cites [1]”, is therefore no longer complete.]

The cited papers, checked at the write. On 3 October 2026 the arXiv record of [1] listed a single version, v1 of 25 February 2026 (math.LO), with the cited title and authors; the record of [3] lists three versions, the last (v3, the version cited) of 18 June 2023; the record of [2] lists three versions, the last of 7 January 2024. The page references to Yao’s Definition 24 and Theorems 26–27 were not rechecked.

1.5 Notation used in this report

[write] **Added 3 October 2026, batch 89.** No symbol was renamed. Several symbols of this report occur with other meanings in neighbouring reports; this table fixes the readings here, with the tempting false one where confusion is likely.

Symbol	Meaning here (and what it is not)
$M_\kappa(A)$	the kernel-cutoff model (2.1); Part VI of birthday-cutoffs-and-hereditary-sets writes M_κ^{ker} for the same class (manuscript 03's M_κ). <i>Not</i> the full-class rank model $M_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$ of Part XV of surreal-well-orders, nor the Presburger models M_α of polish-models-of-omnific-arithmetic
$\mathcal{N}_{\kappa,f}$	the expansion of $M_\kappa(A)$ by the named lifts; not the natural numbers
$\ker(x)$	the atom kernel $A \cap \text{TC}(\{x\})$; not the kernel of a homomorphism
$\text{mov}(f)$	the atoms moved by f (theorem 3.6); not a support in the sense of permutation models
$\text{ZFCU}_R, \text{ZFU}_R$	urelement set theory with Replacement and without Collection; the subscript R names Replacement, not the reals
$B_t, D_\kappa(f), I(f)$	the type block of t , the union of the small blocks, the set of realized types. B_t is not the birthday-cutoff structure B_κ of birthday-cutoffs-and-hereditary-sets; the B_n, B_i of the examples are atom blocks
H	in (5.4) and lemmas 5.2 and 7.4, a protected union of components of size below κ ; not the hereditary-size universe H_κ , and not manuscript 03's hierarchy axiom H
$V, V_\alpha(C)$	the pure universe (Example 3.8); the cumulative hierarchy over the atom set C
$\text{Collection}_{\leq \mu}(J)$	Collection on domains of size <i>at most</i> μ ; manuscript 03 writes $\text{Collection}_{< \tau}$ for size <i>below</i> τ
$\text{Rob}_r(\kappa, A), \text{RP}(J)$	universal Replacement preservation for r names (Section 6); full transitive reflection in the expanded language
$\mathfrak{c}$	$2^{\aleph_0}$
limit cardinal	includes ω (lemma 7.2); “successor” in theorem 7.5 means an uncountable successor

2 Ambient foundations and the kernel-cutoff model

2.1 Two universes must not be confused

Work in an external, well-founded set universe satisfying choice, with a *set* A of atoms. The full cumulative universe of sets over A is denoted by $U(A)$. This setup can be implemented in ordinary ZFC using tagged atoms and tagged sets; membership in the coded universe is then the interpreted membership relation. No inaccessible cardinal or external proper class of atoms is required. Whenever witnesses are chosen below, external Collection first bounds them inside a set; reservoir choices and component isomorphisms are also choices from sets. No global class-choice principle is assumed.

An atom has no members and is distinguished from the empty set by a unary predicate At . Put $\text{Set}(x) \equiv \neg \text{At}(x)$. Extensionality is restricted to sets. Define

$$\ker(x) = \{a \in A : a \in \text{TC}(\{x\})\}.$$

In particular, $\ker(a) = \{a\}$ for an atom a . An object is *pure* when its kernel is empty. The pure part is a copy of the full external pure-set universe.

Fix throughout an infinite cardinal κ and a set A satisfying $|A| \geq \kappa$. Our model is the definable external class

$$M_\kappa(A) = \{x \in U(A) : |\ker(x)| < \kappa\}. \quad (2.1)$$

All cardinal inequalities in the definition are evaluated externally. Every individual atom belongs to $M_\kappa(A)$, but A itself is not a set of the model. This distinction is the engine of the examples.

For any external $C \subseteq A$, let $V_\alpha(C)$ be the usual cumulative hierarchy over C :

$$V_0(C) = C, \quad V_{\alpha+1}(C) = \mathcal{P}(V_\alpha(C)) \cup C, \quad V_\lambda(C) = \bigcup_{\alpha < \lambda} V_\alpha(C).$$

Every object whose kernel is contained in C belongs to some $V_\alpha(C)$. Each $V_\alpha(C)$ is an external set, transitive in the urelement sense, and has kernel contained in C .

Remark 2.1 (Schemas, not a truth predicate). For a fixed first-order formula φ , the assertion $M_\kappa(A) \models \varphi$ means its external relativization to the definable class (2.1). We use one such relativization at a time. We do not postulate a first-order predicate uniformly defining truth for all formulas of a proper-class structure. Claims of elementarity and reflection are corresponding metatheoretic schemes. Quantification over externally given class embeddings is likewise metatheoretic, not an internal set of all class maps.

2.2 The base axioms and internal cardinalities

Let ZFCU_R denote set theory with atoms, Extensionality for sets, Foundation, Pairing, Union, Powerset, Infinity, Separation, Replacement, and the assertion that every set can be well-ordered. Collection is *not* included as an axiom.

Lemma 2.2 (Kernel arithmetic). *For sets and objects in $U(A)$, the following hold:*

1. If $y \in x$, then $\ker(y) \subseteq \ker(x)$.
2. $\ker(\{x, y\}) = \ker(x) \cup \ker(y)$.
3. If x is a set, then $\ker(\bigcup x) \subseteq \ker(x)$ and $\ker(\mathcal{P}(x)) \subseteq \ker(x)$.
4. If Y is a set of objects, then $\ker(Y) = \bigcup_{y \in Y} \ker(y)$.

Consequently $M_\kappa(A)$ is transitive, contains every pure set, and is closed under subsets of its sets and under the ordinary finite set operations.

Proof. Membership chains witnessing occurrence of an atom concatenate through an element of a set. This proves the first, second, and fourth assertions. A membership chain through a subset of x introduces no atom not already occurring through x , giving the powerset assertion; the union assertion is similar. A finite union of cardinals less than an infinite cardinal κ is still less than κ , whether or not κ is regular. $\square$

Any permutation π of A extends recursively to $U(A)$ by

$$\hat{\pi}(a) = \pi(a), \quad \hat{\pi}(x) = \{\hat{\pi}(y) : y \in x\} \quad (\text{Set}(x)).$$

The extension is an automorphism of $M_\kappa(A)$, because it preserves kernel cardinalities. It fixes an object whenever it fixes all atoms in its kernel.

Lemma 2.3 (No new atoms in a unique base-language output). *Suppose $x, p \in M_\kappa(A)$ and*

$$M_\kappa(A) \models \exists! y \varphi(x, y, p),$$

where φ belongs to $\mathcal{L}_U = \{\in, \text{At}\}$. For its unique witness y ,

$$\ker(y) \subseteq \ker(x) \cup \ker(p).$$

Proof. Assume b belongs to the left side but not the right. Since the union of the three kernels has cardinality less than κ , choose an atom c outside that union. The transposition of b and c fixes x, p , preserves the model and its language, and changes the kernel of y . Its lift therefore supplies a different witness to the same formula, contradicting uniqueness. $\square$

Theorem 2.4 (Known kernel-ideal theorem, specialized). $M_\kappa(A) \models \text{ZFCU}_R$. *Its class of atoms is internally proper. It has the same pure ordinals and pure cardinal arithmetic as the ambient pure universe. For each internal set x , its cardinality computed internally equals its external cardinality.*

Proof. The structural axioms and Separation follow from [lemma 2.2](#). Given a functional formula on a set w , external Replacement produces its range Y . By [lemma 2.3](#),

$$\ker(Y) \subseteq \ker(w) \cup \ker(p),$$

so Y is internal. This proves every instance of base-language Replacement.

An external well-order of x is a subset of $x \times x$ with kernel contained in $\ker(x)$, hence is internal. Its ordinal enumeration and any external bijection between x and a pure ordinal have the same kernel bound. Thus internal and external cardinalities of x agree, and choice holds internally. All pure objects and their relations are present, so pure cardinal calculations agree as well. Finally, a set containing all atoms would have kernel A , contradicting the cutoff. $\square$

The theorem is a specialization of Yao's ideal construction [[3](#), Definition 24 and Theorem 26]. Notice that the proof bounds the atoms in a uniquely specified output. It does *not* choose witnesses for an arbitrary relation. The latter problem is Collection and behaves differently.

3 Canonical elementary lifts and definability

3.1 Every atom injection gives an elementary map

For any external injection $f : A \rightarrow A$, define its canonical lift by well-founded recursion:

$$j_f(a) = f(a) \quad (a \in A), \quad j_f(x) = \{j_f(y) : y \in x\} \quad (\text{Set}(x)). \quad (3.1)$$

This is an external definable class function with parameter f ; the graph of f need not belong to $M_\kappa(A)$.

Lemma 3.1 (Extending a small partial bijection). *If $P \subseteq A$, $|P| < \kappa$, and $h : P \rightarrow A$ is injective, then h extends to a permutation of A .*

Proof. Both $A \setminus P$ and $A \setminus h[P]$ have cardinality $|A|$, since fewer than $\kappa \leq |A|$ points were removed. Choose a bijection between the complements and combine it with h . $\square$

Theorem 3.2 (Elementary and amenable lifts). *The map j_f restricts to an elementary embedding of $(M_\kappa(A), \in, \text{At})$ into itself. It preserves rank, fixes every pure set, and satisfies*

$$\ker(j_f(x)) = f[\ker(x)]. \quad (3.2)$$

For every internal set x , the graph of $j_f \upharpoonright x$ is internal. Its range as a class is

$$j_f[M_\kappa(A)] = \{x \in U(A) : \ker(x) \subseteq f[A], |\ker(x)| < \kappa\}.$$

Proof. External well-founded induction gives injectivity, preservation of membership and atomhood, rank preservation, and (3.2). The kernel identity shows that the map preserves the cutoff. The same induction shows that it fixes pure objects.

To prove elementarity, fix one formula and a finite tuple $\bar{x}$ of parameters. Let P be the union of their kernels. By lemma 3.1, $f|P$ extends to a permutation π of A . The automorphism $\hat{\pi}$ agrees with j_f on $\bar{x}$. Thus the formula has the same truth value at $\bar{x}$ and at $j_f(\bar{x})$. This is the required elementary-embedding scheme.

External Replacement constructs the restriction graph on x , whose kernel is contained in $\ker(x) \cup f[\ker(x)]$, a set of size less than κ . This proves amenability. Finally, the inverse bijection $f[A] \rightarrow A$ lifts on objects supported on $f[A]$, giving the stated range. $\square$

Definition 3.3. An elementary self-embedding is *initial* here when its range is transitive: if $z \in j(x)$, then z belongs to the range of j .

Theorem 3.4 (Classification of initial embeddings). *Every initial elementary self-embedding of $(M_\kappa(A), \in, \text{At})$ is j_f for a unique injection $f : A \rightarrow A$. Conversely every j_f is initial. Composition corresponds to composition of atom injections.*

Proof. An elementary embedding restricts on atoms to an injection f . If $z \in j(x)$, initiality gives $z = j(y)$ for some y , and preservation and reflection of membership give $y \in x$. Hence, for every set x ,

$$j(x) = \{j(y) : y \in x\}.$$

External well-founded induction identifies j with (3.1). The range description in theorem 3.2 proves the converse. Uniqueness and the composition law follow from the recursion. $\square$

The initiality hypothesis is essential to the scope of this theorem. We do not classify elementary self-embeddings with nontransitive range.

Proposition 3.5 (Elementary subuniverses over smaller atom reservoirs). *For an external $C \subseteq A$, put*

$$M_\kappa(C) = \{x \in U(A) : \ker(x) \subseteq C, |\ker(x)| < \kappa\}.$$

Then $M_\kappa(C) \preceq M_\kappa(A)$ exactly when $|C| \geq \kappa$.

Proof. Suppose $|C| \geq \kappa$. For parameters supported on $P \subseteq C$ and an existential witness supported on K , move $K \setminus P$ into $C \setminus P$ by a permutation fixing P . The required partial bijection extends because all prescribed supports have size less than κ . The moved witness belongs to $M_\kappa(C)$ and satisfies the same formula. The Tarski–Vaught argument, applied one formula at a time, proves elementarity.

If $|C| < \kappa$, the set C itself is in the smaller model. The formula asserting an atom outside C is true in $M_\kappa(A)$ and false in $M_\kappa(C)$. $\square$

3.2 Exactly which lifts are definable?

Theorem 3.6 (Small movement is equivalent to parameter-definability). *Let $\text{mov}(f) = \{a \in A : f(a) \neq a\}$. The canonical lift j_f is definable in the base-language structure $M_\kappa(A)$ from an internal parameter if and only if*

$$|\text{mov}(f)| < \kappa. \tag{3.3}$$

If p is a parameter defining j_f , then $\text{mov}(f) \subseteq \ker(p)$.

Proof. Suppose p defines j_f and put $P = \ker(p)$. Every atom permutation fixing P pointwise preserves the definition, so f commutes with every such permutation. For $a \notin P$, a permutation fixing $P \cup \{a\}$ can move any atom outside $P \cup \{a\}$. Therefore $f(a) \in P \cup \{a\}$. If $f(a) \in P$, interchange a with a different $b \notin P$. Commutation gives $f(b) = f(a)$, contrary to injectivity. Thus $f(a) = a$ outside P .

Conversely put $S = \text{mov}(f)$ and assume $|S| < \kappa$. Injectivity implies $f[S] \subseteq S$: an image outside S would collide with that outside atom's fixed image. The graph $g = f \upharpoonright S$ is an internal set. It defines f on atoms, with identity elsewhere.

There is a uniform first-order definition of the recursive lift from g . For clarity, on $D = \text{TC}(\{x\})$ ask for a set function h that uses the prescribed atom map and satisfies

$$h(z) = \{h(u) : u \in z\} \quad \text{for each set } z \in D.$$

External well-founded recursion supplies the unique such function. Its graph and every value have kernels contained in $\ker(x) \cup S$, hence are internal. The displayed recursion can be expressed by ordinary first-order quantification over the graph h . It therefore defines $j_f(x)$ internally from g , uniformly in x . $\square$

Corollary 3.7. *The only parameter-free canonical self-embedding is the identity. At $\kappa = \omega$, every parameter-definable initial elementary self-embedding is an automorphism. At every $\kappa > \omega$, there is a parameter-definable proper initial elementary self-embedding. If j_f is defined by p and $j_f(p) = p$, then j_f is an automorphism.*

Proof. A definition without parameters is invariant under all atom permutations, so the proof above has $P = \emptyset$. Finite movement makes an injection a finite permutation, giving the ω case. For $\kappa > \omega$, shift an internal countably infinite set of atoms forward by one and fix its complement; the moved set is countable and the map is not onto.

For the last assertion, (3.2) gives $f[P] = P$ when $j_f(p) = p$. Since f fixes $A \setminus P$ and maps P onto itself, it is surjective. $\square$

Example 3.8 (An elementary map with exactly the pure fixed points). Identify A externally with $B \times \omega$ and let $f(b, n) = (b, n + 1)$. A nonempty subset of A cannot satisfy $f[S] = S$: in a nonempty fiber choose its least second coordinate. If $j_f(x) = x$, (3.2) forces the kernel to be empty. Thus

$$\text{Fix}(j_f) = V,$$

where V is the pure universe. Also $\bigcap_{n < \omega} j_f^n[M_\kappa(A)] = V$. Each range is elementary by [proposition 3.5](#), whereas their intersection is not elementary because it has no atoms. Every ordinal is fixed; there is no critical ordinal and no large-cardinal conclusion.

4 Expanded languages and countable component types

4.1 What naming a map asserts—and what it does not

Fix a positive integer r and external injections $f = (f_1, \dots, f_r)$ on A . Write

$$\mathcal{L}_r = \{\in, \text{At}, J_1, \dots, J_r\}, \quad \mathcal{N}_{\kappa, f} = (M_\kappa(A), \in, \text{At}, j_{f_1}, \dots, j_{f_r}).$$

Here J_i is a unary function symbol. We write $\text{Replacement}(J)$ and $\text{Collection}(J)$ for the schemes whose formulas may contain any of these symbols. The notation does not mean only the special instance asserting the existence of $J_i''x$.

Every instance of expanded Separation holds: externally separate a subset of the given internal set, then use the kernel bound. All the other base axioms except potentially expanded Replacement remain true for the same reason or because they have no new schema instances. Every named map is amenable by [theorem 3.2](#).

If an atom permutation π commutes with every f_i , its lift is an automorphism of $\mathcal{N}_{\kappa, f}$. This follows first on atoms and then by recursion on sets. Such simultaneous symmetries, rather than arbitrary atom permutations, control expanded Replacement.

4.2 Functional graphs and their pure codes

View $(A; f_1, \dots, f_r)$ as a labeled directed graph, with an edge labeled i from a to $f_i(a)$. A *component* means a connected component after directions are forgotten. There are at most $2r$ neighbors at a vertex, since each f_i is injective. Thus every component is finite or countably infinite.

Let $\mathcal{C}_r$ be the pure set of codes for connected structures with r total injections and domain either a nonzero natural number or ω . Isomorphism respects every function label and direction. Let $\mathcal{T}_r$ be the quotient of $\mathcal{C}_r$ by isomorphism. An element of $\mathcal{T}_r$ is taken literally as a set of codes in $\mathcal{C}_r$; this avoids a choice of canonical representatives. All these objects are pure sets and therefore belong to $M_\kappa(A)$.

For $t \in \mathcal{T}_r$, define externally

$$B_t = \{a \in A : \text{the component of } a \text{ has type } t\}, \quad I(f) = \{t \in \mathcal{T}_r : B_t \neq \emptyset\}. \quad (4.1)$$

The B_t are pairwise disjoint. They need not be internal sets; $I(f)$, being pure, always is.

Lemma 4.1 (Number of possible component types). *For one injection, $\mathcal{T}_1$ is countably infinite. Its types are finite directed cycles, a one-sided ray, and a two-sided chain. For every finite $r \geq 2$,*

$$|\mathcal{T}_r| = 2^{\aleph_0} = \mathfrak{c},$$

and the lower bound is already realized using permutations.

Proof. For one injection, each vertex has one outgoing edge and at most one incoming edge. A periodic component is precisely a finite cycle. A nonperiodic component either has a first vertex, giving a ray, or has a predecessor at every point, giving a two-sided chain. These are all the possibilities.

There are at most continuum many finite or countable codes in any fixed finite language. For the lower bound, for each $S \subseteq \omega$ define two permutations of $\mathbb{Z}$. The first is $F(n) = n + 1$. The second, G_S , swaps 0 with 1, swaps

$$3(m+1) \text{ with } 3(m+1)+1 \quad (m \in S), \quad (4.2)$$

and fixes all other integers. The transpositions are disjoint. An isomorphism respecting F must be a translation of $\mathbb{Z}$. The least integer moved by G_S is 0 for every S , so that translation must fix 0 and hence be the identity. It then respects G_S, G_T exactly when $S = T$. This gives continuum many different types. For $r > 2$, add identity permutations. $\square$

4.3 Definability without assuming the scheme being tested

Lemma 4.2 (Finite paths exist internally). *Reachability by a specified finite word in the operations f_i and their partial inverses is uniformly first-order definable in $\mathcal{L}_r$. Every successful finite path has an internal graph. These facts require no expanded Replacement.*

Proof. A forward letter i requires $J_i(a) = b$; a backward letter requires $J_i(b) = a$. Injectivity makes each backward step unique when it exists. A finite sequence of vertices witnessing the word is coded as an ordinary set function on a finite ordinal. It uses only finitely many atoms and therefore belongs to $M_\kappa(A)$, even when $\kappa = \omega$. Quantification over such finite graphs gives one first-order formula with the word as a pure parameter. The natural numbers here are the well-founded external natural numbers, not a nonstandard model's pseudo-finite sequence lengths. $\square$

Lemma 4.3 (Uniform recognition of component types). *If $\kappa > \omega$, there is a formula $\text{Type}_r(t, a)$ in $\mathcal{L}_r$ such that, for every $t \in \mathcal{T}_r$ and atom a ,*

$$\mathcal{N}_{\kappa, f} \models \text{Type}_r(t, a) \iff a \in B_t.$$

The same conclusion for finite component types holds for $\kappa = \omega$.

Proof. Ask for a code $c \in t$ and a set function h from its pure domain into the atoms, injective and with a in its range. Require h to preserve every labeled function. Require also that its range is closed under incoming edges: for each i ,

$$\forall b (\text{At}(b) \wedge J_i(b) \in \text{ran}(h) \longrightarrow b \in \text{ran}(h)).$$

Forward closure already follows from preservation of the code functions. The code is connected, so its image is connected; closure in both directions and inclusion of a make that image exactly the component of a .

Conversely, enumerate the actual component by a finite ordinal or by ω . Its code is pure. Its enumeration graph has only countably many atoms and is therefore internal when $\kappa > \omega$; for a finite component it is internal at every cutoff. The required witnesses exist. All conditions are first-order assertions about the set code and the set graph h . No appeal has been made to expanded Replacement or to a satisfaction predicate for arbitrary formulas. $\square$

Definition 4.4 (The small-type part). The small-type part of the atom graph is

$$D_\kappa(f) = \bigcup \{B_t : t \in \mathcal{T}_r, |B_t| < \kappa\}. \quad (4.3)$$

The adjective “small” applies to the *whole block of all components of one type*, not to an individual component. Every component is countable, but its type block may be arbitrarily large.

This distinction is decisive. Infinitely many copies of one finite cycle can form a large homogeneous block, while one copy of each of infinitely many different finite cycles forms a union of small, individually distinguishable blocks.

5 The exact Replacement criterion

Theorem 5.1 (Component-profile criterion). *Let $r \geq 1$ be finite and let f be an r -tuple of injections on A .*

1. *If $\kappa > \omega$, then*

$$\mathcal{N}_{\kappa, f} \models \text{Replacement}(J) \iff |D_\kappa(f)| < \kappa. \quad (5.1)$$

2. *If $\kappa = \omega$, then*

$$\mathcal{N}_{\omega, f} \models \text{Replacement}(J) \iff \begin{cases} \text{every component of } (A; f) \text{ is finite,} \\ D_\omega(f) \text{ is finite.} \end{cases} \quad (5.2)$$

5.1 Necessity: collecting uniquely definable type blocks

Proof of necessity for $\kappa > \omega$. Use the internal pure domain $\mathcal{T}_r$. In the expanded language define $F(t)$ as follows: if there is a set whose members are exactly the atoms satisfying $\text{Type}_r(t, a)$, let $F(t)$ be that set; if there is no such set, let $F(t) = \emptyset$. Extensionality for sets makes this a total functional definition. By lemma 4.3, its external interpretation is

$$F(t) = \begin{cases} B_t, & |B_t| < \kappa, \\ \emptyset, & |B_t| \geq \kappa. \end{cases}$$

Indeed, a set of atoms belongs to $M_\kappa(A)$ precisely when its cardinality is less than κ .

Expanded Replacement yields an internal range set for F on $\mathcal{T}_r$. Taking its union produces exactly $D_\kappa(f)$. The kernel of a set of atoms is the set itself, so this union must have cardinality less than κ . $\square$

Proof of necessity for $\kappa = \omega$. First let a be an atom. The pure set of finite words in the $2r$ forward and backward letters is countable and belongs to the model. By lemma 4.2, a word either has a unique endpoint starting at a , or has no endpoint because a backward step is unavailable. Define its value to be that endpoint when it exists and a otherwise. This is a total functional definition in the expanded language. Its range is exactly the component of a .

Expanded Replacement would therefore make every component an internal set of atoms, hence finite. Once this is known, repeat the preceding type-block argument over the pure set $\mathcal{T}_r^{\text{fin}}$ of finite component types. The finite-type recognition formula is correct at the finite-kernel cutoff, and its union of small blocks is $D_\omega(f)$. That union must be finite. $\square$

The first argument uses formulas substantially more complicated than $y = J_i(x)$. Amenability alone says nothing about collecting all the type blocks selected by that formula.

5.2 Sufficiency: the fresh-component argument

For $P \subseteq A$, let $\text{cl}_f(P)$ be the union of all components meeting P . It is closed under all the f_i and all their available inverse steps. If $\kappa > \omega$ and $|P| < \kappa$, then

$$|\text{cl}_f(P)| \leq \max(\aleph_0, |P|) < \kappa. \quad (5.3)$$

For $\kappa = \omega$ the corresponding closure is finite whenever P is finite and all components are finite.

Lemma 5.2 (Fresh component swap). *Let $H \subseteq A$ be a union of components with $|H| < \kappa$. Let $y \in M_\kappa(A)$, and suppose $b \in \ker(y) \setminus H$ belongs to a type block B_t of cardinality at least κ . If $\kappa > \omega$, there is an atom permutation π commuting with every f_i , fixing H pointwise, and satisfying $\hat{\pi}(y) \neq y$. The same holds for $\kappa = \omega$ when the components are finite.*

Proof. The component C of b is disjoint from H . For uncountable κ , B_t contains at least κ distinct components: fewer than κ countable sets cannot have union of cardinality at least κ , since their sizes are uniformly bounded by $\aleph_0$. The set $H \cup \ker(y)$ meets fewer than κ components. Choose a component C' of type t disjoint from that set.

Choose a labeled-graph isomorphism $u : C \rightarrow C'$. Define π to be u on C , u^{-1} on C' , and the identity elsewhere. It commutes with each f_i because components are invariant in both directions and u respects the functions. It fixes H and moves b to an atom outside $\ker(y)$, so it changes that kernel and hence changes y .

At $\kappa = \omega$, a fixed finite type has a fixed finite component size. An infinite block of that type has infinitely many components, while $H \cup \ker(y)$ meets only finitely many. The same construction applies. $\square$

Proof of sufficiency in theorem 5.1. Assume the right side of the appropriate criterion. Fix a formula $\varphi(x, y, p)$ of $\mathcal{L}_r$ that defines a unique output for every x in an internal set w . Set

$$P = \ker(w) \cup \ker(p), \quad H = D_\kappa(f) \cup \text{cl}_f(P). \quad (5.4)$$

By the hypothesis and (5.3), H has size less than κ . At the finite cutoff this follows instead from finiteness of both terms. It is a union of components.

We claim that every uniquely defined output y has kernel contained in H . Otherwise choose $b \in \ker(y) \setminus H$. Its type block cannot be small, because every small block is included in $D_\kappa(f)$. Apply lemma 5.2. The resulting lifted automorphism preserves the *expanded* structure, fixes x and p , and changes y . It contradicts uniqueness.

External Replacement now collects all these unique outputs into a set Y . Its kernel is contained in H , independently of the cardinality of the input set w . Thus $Y \in M_\kappa(A)$, as required. The argument works for every fixed expanded-language formula and every internal parameter. $\square$

Remark 5.3 (Why the external construction is legitimate). The ambient parameters A, κ, f are sets, the model is definable, and each j_{f_i} has an external recursive definition. A fixed expanded formula therefore has an ordinary external first-order translation. External Separation and Replacement can be applied to it. The proof then checks whether the resulting set meets the internal kernel cutoff. It does not assume any expanded Replacement inside $\mathcal{N}_{\kappa, f}$.

5.3 Boundary examples for one symbol

Example 5.4 (A named automorphism can destroy Replacement). Let $\kappa = \aleph_\omega$ and choose a subset of A partitioned as $\coprod_{n < \omega} B_n$, with $|B_n| = \aleph_n$. On B_n , let f consist of cycles of length $n + 2$. Fix all remaining atoms. Every B_n is an internal set and is precisely the block of atoms of that cycle length. Their union has cardinality $\aleph_\omega$. Thus $D_\kappa(f)$ is not small and $\text{Replacement}(J)$ fails, although j_f is an amenable automorphism fixing every pure set.

A direct failed instance is the unique definition sending n to the set of all atoms of exact period $n + 2$. An internal range set for this definition would contain, in its kernel, the whole union of the B_n .

Example 5.5 (The two finite-cutoff obstructions are independent). At $\kappa = \omega$, a shift on a two-sided infinite atom chain has no small nonempty type block, but Replacement fails by collecting the iterates of one atom. Conversely, take one finite cycle of each length $n + 2$ on a countable subset of atoms and fix any remainder. Every component is finite, but the union of the finite cycle-type blocks is infinite, so Replacement fails by collecting those blocks. Both clauses in (5.2) are necessary.

Example 5.6 (Infinitely many atoms need not cause a failure). An arbitrary number of copies of one finite cycle type, with an infinite total block, is harmless at the finite cutoff provided the union of all the other finite blocks is finite. The main theorem applies because a unique output cannot select a new component from the homogeneous infinite block. The issue is distinguishability, not simply the number of atoms moved.

[write] **The finite-cutoff criterion for group actions, 4 October 2026, batch 90.** Batch-90 manuscript 05, *Named Symmetries and Replacement*, pinned to commit 8dc2592e9 (an ancestor of this report’s placement, so it does not cite this report), is printed as Part VIII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (its source 12, labels `hset:ns:`). It works at the finite cutoff $M_\omega(A)$ with a uniformly named action of an arbitrary pure set-sized group Γ on the atoms. Its Theorem 58.1 (`hset:ns:thm:full`): expanded Replacement holds iff every Γ -orbit is finite and the union of the finite orbit types that occur only finitely often is finite. For finitely many named permutations this is part (2) of [theorem 5.1](#): naming the atom permutations and naming their canonical lifts give definitionally equivalent languages (its Propositions 62.1 and 62.4), the orbits of the generated free-group action are the components here, and its exceptional set is $D_\omega(f)$ once every component is finite. The proofs are independent; this report also covers non-surjective injections and every uncountable cutoff, which that part does not. New there: arbitrary pure group actions; the graded levels R_k (Replacement for outputs with at most k atoms), which hold for $k \geq 1$ iff every orbit is finite and only finitely many atoms have orbit of size at most k under the centralizer of the named action (its Theorem 59.1); one cycle of each length, as in [example 5.5](#), satisfies every R_k while failing Replacement (its Theorem 60.1 and Example 60.2); and, for every $d \geq 1$, two involutions, each harmless alone and with all joint orbits finite, whose joint expansion satisfies exactly the levels R_k with $k < d$ (its Theorem 61.3), a finite-cutoff, graded counterpart of [corollary 7.9](#). It also proves that pure-valued Collection, hence the whole pure sign-sequence surreal universe, survives every such naming (its Theorem 63.1). Its Remark 54.4 (`hset:ns:rem:nee`) records the overlap from its side. AI-assisted, unrefereed, not formalized; neither manuscript is claimed first.

6 Sharp thresholds for one name and for two names

Define $\text{Rob}_r(\kappa, A)$ to mean:

For every external r -tuple of injections on A , its canonical expanded model satisfies full Replacement in the r -symbol language.

This is an external property of a definable family of models. Its meaning is the same if “injections” is replaced by “permutations” in the equivalence below. The negative examples use permutations, and the positive arguments cover injections.

Theorem 6.1 (Finite-signature robustness thresholds). *For every infinite κ , every $|A| \geq \kappa$, and every fixed finite $r \geq 1$,*

$$\text{Rob}_r(\kappa, A) \iff \begin{cases} \text{cf}(\kappa) > \omega, & r = 1, \\ \text{cf}(\kappa) > \mathfrak{c}, & r \geq 2, \end{cases} \quad \mathfrak{c} = 2^{\aleph_0}. \quad (6.1)$$

6.1 The positive bounds

Proof of the forward preservation bounds. If $\text{cf}(\kappa) > \omega$, a countable union of sets of cardinality less than κ still has cardinality less than κ . There are only countably many one-injection types. Thus $D_\kappa(f)$ is small for every f , and [theorem 5.1](#) proves Rob_1 .

More generally, for $r \geq 2$ there are at most $\mathfrak{c}$ component types. If $\text{cf}(\kappa) > \mathfrak{c}$, the union of all the small blocks has cardinality less than κ . Again apply [theorem 5.1](#). These hypotheses automatically put us at an uncountable cutoff. $\square$

6.2 Sharpness for one symbol

Failure when $\text{cf}(\kappa) = \omega$. At $\kappa = \omega$, use a two-sided infinite shift on a countable subset of A , fixing the rest. Its infinite component violates (5.2).

For uncountable κ of countable cofinality, choose infinite cardinals $\lambda_n < \kappa$ cofinal in κ . On disjoint atom blocks B_n of sizes λ_n , use cycles of distinct lengths $n + 2$, and fix the remaining atoms. The union of these small type blocks has size κ . The argument of [example 5.4](#) applies without any special property of the aleph sequence. This proves the converse for $r = 1$. $\square$

6.3 Sharpness for two symbols: a real-coded construction

Lemma 6.2 (Cardinal decomposition). *If $\kappa > \omega$ and $\delta = \text{cf}(\kappa)$, there are infinite cardinals $\lambda_i < \kappa$ for $i < \delta$ with*

$$\sum_{i < \delta} \lambda_i = \kappa.$$

Proof. For regular κ , take $\delta = \kappa$ and every $\lambda_i = \aleph_0$. For singular κ , take an increasing cofinal sequence of infinite cardinals below κ . Its cardinal sum is κ . $\square$

Assume now $\kappa > \omega$ and $\delta = \text{cf}(\kappa) \leq \mathfrak{c}$. Choose pairwise distinct pure real codes $S_i \subseteq \omega$, $i < \delta$, and cardinals as in [lemma 6.2](#). On a subset A_0 of A of cardinality κ , use the atom indexing

$$A_0 = \{a_{i,\xi,n} : i < \delta, \xi < \lambda_i, n \in \mathbb{Z}\}. \quad (6.2)$$

The subscripts are external labels, not internal atom coordinates. Define permutations f, g by

$$f(a_{i,\xi,n}) = a_{i,\xi,n+1}, \quad g(a_{i,\xi,n}) = a_{i,\xi,G_{S_i}(n)},$$

where G_S is the involution in (4.2). On $A \setminus A_0$, let both maps be the identity.

Each set $\{a_{i,\xi,n} : n \in \mathbb{Z}\}$ is one component. Different i give nonisomorphic components by [lemma 4.1](#). For each i , the entire type block

$$B_i = \{a_{i,\xi,n} : \xi < \lambda_i, n \in \mathbb{Z}\}$$

has cardinality $\lambda_i < \kappa$. Their union is A_0 , of size κ . Hence $D_\kappa(f, g)$ is not small, and expanded Replacement fails.

For completeness, the failure can be expressed directly without the general quotient of component codes. On any constructed component, an atom a is the *root* exactly when $g(a) \neq a$ and every point strictly before a in the f -chain is fixed by g . This is uniformly first-order expressible using

finite positive and negative iterates, as in [lemma 4.2](#). The root is its coordinate-0 atom. Its real label is recovered from

$$m \in S \iff g(f^{3(m+1)}(a)) \neq f^{3(m+1)}(a). \quad (6.3)$$

Thus, with the pure parameter $q = \langle S_i : i < \delta \rangle$, there is a single formula defining B_i as the set of atoms whose f -chain has a root with label S_i . Every B_i exists internally and is unique. Replacement for $i \mapsto B_i$ on the pure set δ would collect all of A_0 into a kernel of size κ , which is impossible.

At $\kappa = \omega$ a shift already defeats one symbol, and adding identity symbols cannot repair its failed instance. For $r > 2$, pad the two-symbol construction with identities. This completes the converse of [theorem 6.1](#).

Corollary 6.3 (A continuum-hypothesis characterization). *For any external atom set A of cardinality at least $\aleph_2$,*

$$\text{CH} \iff \text{Rob}_2(\aleph_2, A).$$

Equivalently, CH holds exactly when every pair of atom permutations has a joint canonical expansion of $M_{\aleph_2}(A)$ satisfying Replacement.

Proof. By [theorem 6.1](#), the preservation property is equivalent to $\aleph_2 > \mathfrak{c}$, since $\aleph_2$ is regular. By choice and Cantor's theorem, this is equivalent to $\mathfrak{c} = \aleph_1$. $\square$

This statement neither proves nor refutes CH. It translates CH into a precise preservation property for a family of language expansions. In particular, changing the ambient continuum can change a universal expansion property even though every one-symbol expansion at the same successor cutoff is harmless.

Corollary 6.4 (Countably many named maps). *The universal preservation threshold for a countable sequence of atom injections, with one symbol for each lift and ordinary finitary first-order formulas, is also $\text{cf}(\kappa) > \mathfrak{c}$.*

Proof. Each formula uses only finitely many symbols. Under the stated inequality its Replacement instance is valid by the finite-symbol theorem. Conversely the two-symbol counterexamples embed in a countable expansion by adding identities. $\square$

7 Collection, reflection, and their exact spectra

7.1 Collection on small domains

For an external cardinal μ , let $\text{Collection}_{\leq \mu}(J)$ denote all Collection instances on internal sets of cardinality at most μ , with arbitrary internal parameters. These cardinalities agree with their external values by [theorem 2.4](#).

Lemma 7.1 (Automatic small-domain Collection). *For every finite tuple f , if $\mu < \text{cf}(\kappa)$ then*

$$\mathcal{N}_{\kappa, f} \models \text{Collection}_{\leq \mu}(J).$$

This conclusion does not assume expanded Replacement.

Proof. Fix a set w of size at most μ and a total relation on it. In the ambient universe, first bound ranks of witnesses for this fixed formula and then choose a witness y_x for each $x \in w$. Equivalently, use ambient Collection followed by ambient choice. The union of at most $\mu < \text{cf}(\kappa)$ kernels, each of size less than κ , has size less than κ . Hence the external witness set $\{y_x : x \in w\}$ is internal and supplies Collection. $\square$

Lemma 7.2 (The limit-cutoff obstruction is already bounded). *Suppose κ is an infinite limit cardinal, including ω . There is a Collection instance with a Δ_0 matrix in the base language, on a pure domain of size $\text{cf}(\kappa)$, that fails in $M_\kappa(A)$ and in every expansion under consideration.*

Proof. Choose a pure set w of ordinals below κ , of size $\delta = \text{cf}(\kappa)$, whose cardinalities are unbounded below κ . For $\kappa = \omega$, take $w = \omega$. For uncountable limit κ , use a cofinal family of cardinals below κ .

Let $\text{InjectAtoms}(\alpha, s)$ assert that s is the graph of an injection with domain α and with atom values. It has a Δ_0 definition in $\{\in, \text{At}\}$; an explicit bounded formulation is given in [section A](#). For each $\alpha \in w$, an external injection into A has graph with $|\alpha| < \kappa$ atoms, hence exists internally.

If a single internal set v contained suitable graphs for all $\alpha \in w$, its kernel would have cardinality at least $|\alpha|$ for all those α . This contradicts $|\ker(v)| < \kappa$ and the unboundedness of their cardinalities. $\square$

Corollary 7.3 (Exact local spectrum at limit cutoffs). *For limit κ , every finite tuple f , and every cardinal μ ,*

$$\mathcal{N}_{\kappa, f} \models \text{Collection}_{\leq \mu}(J) \iff \mu < \text{cf}(\kappa).$$

Proof. Combine the preceding two lemmas. Failure on one domain of cardinality $\text{cf}(\kappa)$ implies failure for every larger domain bound. $\square$

7.2 A reservoir lemma at successor cutoffs

Lemma 7.4 (Compression into a fixed component reservoir). *Let $\kappa = \nu^+$ with ν infinite, and suppose $|I(f)| \leq \nu$. For every component-closed $P \subseteq A$ of cardinality at most ν , there is a component-closed $H \supseteq P$, $|H| \leq \nu$, such that every object $y \in M_\kappa(A)$ can be carried to an object supported on H by an expanded-structure automorphism fixing P pointwise.*

Proof. First add all small type blocks to P . There are at most ν such blocks and each has size at most ν , so their union is still of size at most ν . Call the result P^* .

Every remaining nonempty block has size at least ν^+ and therefore has at least ν^+ components. For each such type choose ν components disjoint from P^* . Let H consist of P^* and all these selected reservoirs. There are at most ν types and all components are countable, so $|H| \leq \nu$.

The kernel of y meets at most ν components of each type. Leave components in P^* fixed. For each large type, map the components meeting that kernel outside P^* injectively into the chosen ν reservoir components. This partial matching extends to a permutation of the components outside P^* : its domain and image have size at most ν , whereas the whole component family has cardinality at least ν^+ . Choose compatible isomorphisms on matched components and arbitrary isomorphisms on the remainder. These component permutations together define an atom permutation commuting with all the f_i and fixing P^* . Its lift has the required properties. $\square$

Theorem 7.5 (Full Collection criterion). *For any infinite κ and finite nonempty tuple f ,*

$$\mathcal{N}_{\kappa,f} \models \text{Collection}(J) \iff \kappa \text{ is an uncountable successor cardinal and } |I(f)| < \kappa. \quad (7.1)$$

Proof. At limit cutoffs, including ω , [lemma 7.2](#) rules out Collection. Suppose $\kappa = \nu^+$. If $|I(f)| \geq \kappa$, choose a pure subset $I_0 \subseteq I(f)$ of cardinality κ . By [lemma 4.3](#), for every $t \in I_0$ there is an atom satisfying $\text{Type}_r(t, a)$. A witness set for all these types must contain at least κ atoms, since type blocks are disjoint. It cannot be internal. Thus Collection fails.

Conversely assume $|I(f)| < \nu^+$. Fix a total expanded relation $\varphi(x, y, p)$ on an internal set w . Close $\ker(w) \cup \ker(p)$ under components and apply [lemma 7.4](#). For each $x \in w$, any witness can be moved by an automorphism fixing x, p to a witness with kernel contained in the same H . Externally choose such witnesses and bound their ranks, or use ambient Collection and choice. Their range set has kernel contained in H , hence is internal. This proves full expanded Collection. $\square$

Corollary 7.6 (All local Collection cases). *For a successor κ , if $|I(f)| < \kappa$ then Collection holds on every internal domain. If $|I(f)| \geq \kappa$, then*

$$\mathcal{N}_{\kappa,f} \models \text{Collection}_{\leq \mu}(J) \iff \mu < \kappa.$$

Together with [corollary 7.3](#), this gives the exact domain-size spectrum in every case.

Proof. The positive statement is [theorem 7.5](#). In the second case the preceding proof fails Collection on a domain of size κ , while [lemma 7.1](#) proves it below $\text{cf}(\kappa) = \kappa$. $\square$

[write] A second proof of the unexpanded spectrum, 3 October 2026, batch 89. For the unexpanded model (one name, $f = \text{id}_A$, a single realized type) [lemma 7.1](#), [lemma 7.2](#), [corollary 7.3](#) and [theorem 7.5](#) give the Collection spectrum of $M_\kappa(A)$ that batch-89 manuscript 03 proves independently in Part VI of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (its delivered Lemma 10.1, Theorems 10.2–10.3 and Corollary 10.4; Lemma 33.1, Theorems 33.2–33.3 and Corollary 33.4 there; see Section 1.4). Both proofs stand. They differ in three places. (i) *The failing instance at a limit cutoff.* Manuscript 03 collects sets of atoms of prescribed cardinalities μ_i cofinal in κ , a relation that quantifies over bijections; [lemma 7.2](#) collects injection graphs $\alpha \rightarrow A$ with the Δ_0 matrix of Appendix A, so the failure is already one of Δ_0 Collection. That bound, and [proposition 8.1](#) built on it, are this report's own. (ii) *The successor case.* Manuscript 03 moves each witness, by an atom permutation fixing the parameters' kernels, into one fixed reservoir of ν atoms and bounds the stages by Replacement inside the model; [lemma 7.4](#) must use permutations that commute with the named maps, so it moves whole components into a reservoir of components, one per realized type, and bounds ranks externally. For $f = \text{id}_A$ the two arguments coincide in substance. (iii) *What is named.* Manuscript 03's Theorem 11.2 (Theorem 34.2 there, `hset:zr:thm:expandedthreshold`) names an external enumeration $e : \kappa \rightarrow A_0$ of κ atoms by a predicate E ; then Replacement and Collection both hold below $\text{cf } \kappa$ and both fail on a domain of size $\text{cf } \kappa$, so at $\kappa = \aleph_1$ naming e destroys Replacement, whereas every one-name expansion by a canonical lift keeps full Collection and reflection there (the paragraph after [theorem 7.7](#)). There is no conflict: e is not an elementary lift, and every atom permutation preserving it fixes A_0 pointwise, so the fresh-component symmetries of Section 5 are unavailable. Manuscript 03's observation that its small-domain bound needs no invariance under atom permutations is the argument of [lemma 7.1](#), which likewise holds in every expansion. Its Corollary 10.5 (Corollary 33.5 there; each infinite regular τ is the first failure size of some $M_{\aleph_\tau}$) is [corollary 7.3](#) at $\kappa = \aleph_\tau$, a case this report does not state.

7.3 Full transitive reflection

In this section, full reflection in $\mathcal{L}_r$ has the following explicit meaning. For every finite set Φ of formulas and every internal set w , there is an internal transitive set t with $w \subseteq t$, closed under the named functions, such that each formula of Φ is absolute between the full expanded model and its restriction to t , for all its parameter tuples from t . Call this scheme $\text{RP}(J)$.

Theorem 7.7 (Exact reflection criterion). *In the present model family,*

$$\text{RP}(J) \iff \text{Collection}(J) \iff [\kappa \text{ is an uncountable successor and } |I(f)| < \kappa].$$

Proof. Full reflection implies Collection directly. Given a total relation on w , reflect a finite family containing its matrix and its totality assertion into a transitive set containing w, p . The restricted totality assertion supplies witnesses in that set, and absoluteness of the matrix makes them genuine witnesses in the full structure. Thus [theorem 7.5](#) proves necessity.

For sufficiency, write $\kappa = \nu^+$ and assume $|I(f)| \leq \nu$. Enlarge the finite family Φ by subformulas, after expressing universal quantifiers through negation and existential quantifiers. We construct externally an increasing sequence

$$X_n = V_{\alpha_n}(H_n), \quad n < \omega,$$

where H_n is a component-closed atom set of size at most ν , the ordinals α_n increase, $w \subseteq X_0$, and every true existential subformula with parameters in X_n has a witness in X_{n+1} .

Begin by closing the atoms of w under components and choosing a sufficiently large α_0 . At stage n , apply [lemma 7.4](#) to H_n . This gives a component-closed $H_{n+1} \supseteq H_n$, of size at most ν , into which witnesses can be compressed while all parameters in X_n are fixed. There are only set-many relevant parameter tuples and finitely many subformulas. External Collection yields a common rank bound for witnesses after compression. Choose $\alpha_{n+1} > \alpha_n$ large enough that all these witnesses lie in $V_{\alpha_{n+1}}(H_{n+1})$.

Let $t = \bigcup_{n < \omega} X_n$. Its kernel is contained in $\bigcup_n H_n$, of size at most ν , so t is internal. It is transitive and closed under every J_i , since the H_n contain whole components and canonical lifts preserve rank. A finite parameter tuple in t lies in some X_n . The witness property and structural induction on the formulas now prove the required absoluteness. Atomic formulas agree because the restricted structure has the same membership, atom predicate, and function values; the Boolean steps are immediate; the existential step is exactly the witness property. $\square$

Remark 7.8 (Why one fixed atom reservoir is not a reflecting universe). A single $V_\alpha(H)$ can contain H as an element and consequently regard the atoms it sees as exhausted by one set. The full model has no set of all atoms. The reflection construction therefore grows the atom reservoirs along with the rank bounds. It does not incorrectly identify a fixed-reservoir hierarchy with a reflecting substructure of the entire atom universe.

For one named injection, at most countably many types are realized. Thus every one-symbol expansion at an uncountable successor cutoff satisfies full Collection and reflection. The unexpanded model is the special case $f = \text{id}_A$, and its full Collection/reflection criterion reduces to the familiar successor-versus-limit kernel cutoff.

7.4 Two individually safe names need not be jointly safe

Corollary 7.9 (Failure of compositional safety). *For $|A| \geq \aleph_1$, there are atom permutations f, g such that each of*

$$(M_{\aleph_1}(A), j_f), \quad (M_{\aleph_1}(A), j_g)$$

satisfies expanded Replacement, Collection, and full transitive reflection, but $(M_{\aleph_1}(A), j_f, j_g)$ fails expanded Replacement.

Proof. Use the construction (6.2) with $\delta = \aleph_1$ and every $\lambda_i = \aleph_0$. It is available because $\aleph_1 \leq \mathfrak{c}$. The joint expansion fails Replacement as proved in section 6. Each individual expansion is a one-symbol expansion at a successor cutoff, so theorems 7.5 and 7.7 apply; Replacement follows as well. $\square$

Example 7.10 (Joint Replacement can survive while joint Collection fails). Again put $\kappa = \aleph_1$ and use $\aleph_1$ distinct real labels. This time take $\aleph_1$ copies of the countable component for each label, on a subset of A of size $\aleph_1$. Each realized nontrivial type block has size $\aleph_1$. Any additional fixed-atom block is either large or is one small block. Hence $D_\kappa(f, g)$ has size less than κ , and joint Replacement holds. But there are $\aleph_1$ realized types, so Collection and full reflection fail. Both individual one-symbol expansions still have Collection and reflection.

Property	Controlling quantity	Exact criterion
Replacement, $\kappa > \omega$	Union of small type blocks	$ D_\kappa(f) < \kappa$
Replacement, $\kappa = \omega$	Component sizes and small blocks	All components finite and $D_\omega(f)$ finite
Full Collection	Cutoff and realized type count	κ successor and $ I(f) < \kappa$
Full reflection	Same as full Collection	Same criterion, with named-function-closed transitive reflectors
Universal one-name safety	Countably many possible types	$\text{cf}(\kappa) > \omega$
Universal finite multi-name safety	Continuum many possible types	$\text{cf}(\kappa) > \mathfrak{c}$ for at least two names

8 Partial reflection and a quantifier-normalization trap

Full reflection above requires agreement on a finite family for all parameters in the reflecting structure. A weaker question asks only that a single true assertion with its given parameters hold in some transitive set containing those parameters. In weak set theories it matters greatly which syntactic assertions are allowed.

Proposition 8.1 (Two superficially similar reflection schemes). *Over replacement-based set theory with atoms:*

1. *Every true assertion of the form $\exists y \delta(y, p)$, where δ is Δ_0 in the base language, is true in some transitive set containing p .*

2. The scheme of partial transitive reflection for assertions

$$\forall x \in w \exists y \delta(x, y, p), \quad (8.1)$$

with Δ_0 matrices δ , is equivalent to Δ_0 Collection.

Proof. For the first assertion, choose a witness y and take a transitive closure containing p, y . Bounded formulas are absolute between transitive sets and the universe: their quantifiers inspect only elements of already present parameters. Atomhood is unchanged. Hence the witness still works in the transitive set.

For the forward direction of the second assertion, reflect a true instance (8.1) into a transitive set t containing w, p . Every $x \in w$ belongs to t , and the reflected assertion supplies a witness $y \in t$. Bounded absoluteness makes that witness correct in the full universe. Thus t is a collection set.

Conversely, Δ_0 Collection supplies a set b of witnesses. A transitive closure containing w, p, b contains suitable witnesses for every $x \in w$. Bounded absoluteness makes (8.1) true in that closure. $\square$

At a limit cutoff, the instance from [lemma 7.2](#) therefore has no such transitive reflector, even though each of its individual existential assertions does. This gives a precise warning about terminology. In the presence of Collection, bounded universal quantification in front of an existential assertion can often be converted into one existential quantifier over a set of witnesses. Without Collection, that conversion is precisely what is at issue.

It is therefore unsafe to identify a strict existential–bounded normal form with every bounded–quantifier closure customarily placed at the same level of a hierarchy, unless the required normalization has been proved in the base theory. ProveIt’s bounded-consistency documentation explicitly separates its actual quantifier-group measure from the Levy hierarchy [5]; the present models give concrete semantic tests for such distinctions.

Remark 8.2 (No conflict with the Glazer–Yao separation results). The results of [section 7](#) concern a particular well-founded family with ambient choice, an explicit cutoff, and finitely many canonical function symbols. They do not reproduce the choiceless separation diagram in [1], and they do not assert that all reflection notions coincide over arbitrary urelement theories. Even within this paper, [proposition 8.1](#) distinguishes two restricted partial-reflection schemes that must not be silently conflated.

9 A concrete ProveIt formalization route

9.1 Inspected interfaces and the repository snapshot

The repository material used here is pinned to commit

2b7b388ba81a3355b19a7c2d2fe797e92f572355.

In particular, the inspected beginning of

SetTheory/ZF/Lean/ZF/Zf.lean

contains the formula constructors `Sep_form`, `Func_form`, `Image_form`, and `Repl_form`, and the axiom sets `ZFax` and `ZFax_s`. Its schema constructors act on the current formula type `Form`. Its introductory documentation describes the bridge from object-language axioms to semantic axioms and the internal finite-recursion development [4].

The bounded-consistency README states a numeralwise endpoint, one derivation for each externally fixed complexity bound, rather than an object-language universal consistency claim. It also records the choice of partial satisfaction rather than a reflection-based implementation [5]. We have inspected those interfaces and descriptions, not rebuilt or independently audited the repository’s claimed formal results.

The self-embedding report’s README explicitly marks its contents as unrefereed and not formalized [6]. That status matters: adjacency to verified Lean code does not convert a mathematical report into a kernel-checked theorem. The same boundary applies to this manuscript.

[write] Formal status at the write, 3 October 2026, batch 89. The interfaces described in this subsection are unchanged at the write (Section 1.4). None of the six layers below exists in Provel: the shared formula type `Form` of `Logic/FirstOrder/Lean/FirstOrder/Fol.lean`, used by `SetTheory/ZF` and `SetTheory/BoundedConsistency`, has only the atomic formulas $\in$ and $=$, with no atom predicate and no function symbols, and no urelement theory, kernel model or canonical lift is formalized. The module names and interfaces of this section are proposals.

9.2 First task: make the language parameter explicit

A port should not merely append an “embedding axiom” to the existing pure-membership theory. It needs a genuine expanded syntax. Two reasonable routes are available.

The direct route introduces a language with a unary atom predicate and r unary function symbols. Terms, renaming, substitution, satisfaction, and all formula-valued schema arguments must then be extended. This is mathematically transparent but requires term infrastructure if the selected existing syntax is purely relational.

A relational route instead uses binary graph predicates $E_i(x, y)$, together with totality and functionality axioms. It can be closer to the repository’s current relational syntax. Formulas containing function terms can be translated to graph formulas, but the translation may introduce quantifiers. Therefore an equivalence of theories is not automatically a bound-preserving equivalence of proof-complexity measures.

For either route, there must be separate semantic assertions

$$\text{ModelsReplacement}(\mathcal{L}_U, M) \quad \text{and} \quad \text{ModelsReplacement}(\mathcal{L}_r, M, J).$$

The second quantifies over a larger formula type. The counterexamples of [section 6](#) should become regression tests against any interface that mistakenly identifies these assertions.

9.3 A proof dependency plan

A practical implementation can be divided into six layers.

Layer 1: well-founded objects with atoms. Implement a tagged cumulative model, or a well-founded extensional relation with distinguished atoms, and define its kernel. Prove kernel arith-

metic, the hereditary cutoff property, and the existence of all pure codes used later. Keep the ambient cardinal assumptions visible.

Layer 2: recursive lifts and semantic automorphisms. Define the lift of an atom injection by well-founded recursion. Prove its kernel equation and functoriality. Prove the small-partial-bijection extension lemma, then base-language elementarity by local agreement with automorphisms. Do not derive this by assuming the lift is surjective.

Layer 3: component combinatorics. Define finite labeled paths, undirected reachability, and component closure. Prove the countable component bound from finite degree, and prove the component-swap lemma with an explicit isomorphism witness. This layer is independent of the set-theoretic Replacement scheme.

Layer 4: internal recognition of types. Define a set of pure graph codes and its isomorphism quotient. Prove the satisfaction specification for the type-recognition formula. The cardinal hypothesis $\omega < \kappa$ must be attached exactly where a countable enumeration graph is imported into the model. At the finite cutoff, use finite codes only after proving that expanded Replacement forces components to be finite.

Layer 5: schemes and spectrum theorems. Represent the type-block selector as one expanded formula and prove that Replacement for it imports the union of small blocks. Prove sufficiency using the kernel bound (5.4). Then formalize the marker construction, the cardinal decompositions, and the universal thresholds. Collection needs its own relation and reservoir arguments; it should not be inferred from Replacement.

Layer 6: reflection and proof-theoretic interfaces. Implement reflection only for an externally fixed finite set of formulas, using the growing-reservoir construction. Relate semantic instances to the chosen object theory through explicitly proved soundness or completeness bridges. No uniform truth predicate or internal assertion of full elementarity is needed.

9.4 Expected hard points and trust boundaries

The finite graph swap is elementary; the demanding work is likely to be the internal syntax and the exact specifications of set-coded component isomorphisms. There are four particularly important failure modes to test.

First, a port must not use expanded Replacement to construct the finite paths or countable enumeration graphs later used to *test* Replacement. Their membership in the cutoff model is proved externally from their kernels. Second, a cardinal-kernel bound is a bound on the atoms in a whole set of outputs, not on each output independently. Third, class elementarity must remain a scheme unless the metatheory explicitly supplies an appropriate satisfaction class. Fourth, the reflecting sets must be closed under every named function, not merely under membership.

The supplementary Python program checks finite component swaps and the real-marker coding on finite test families. It does not prove any infinite-cardinal theorem, verify a first-order derivation,

or replace the proposed Lean development. Its purpose is to catch combinatorial implementation mistakes before a formal port.

10 Further research questions

The following are proposed projects arising from the present results. They are not all claimed to be previously published open problems. Their formulation records where the proved classification ends.

Question 10.1 (An intrinsic definable-closure theorem). For a fixed finite tuple f , determine the exact atom kernel of the definable closure of a parameter set in the expanded model. The proof of [theorem 5.1](#) bounds it by component closure plus $D_\kappa(f)$, but this bound can be larger than necessary. Which atoms or entire finite blocks are actually definable, rather than merely protected from a fresh-component swap?

[write] **Answered for commuting permutations, 4 October 2026, batch 93.** For a finite tuple of *commuting permutations* the bound is exact: [theorem 17.1](#) of Part II proves, at every infinite cutoff including $\kappa = \omega$, that the atom kernel of the objects uniquely definable from $\bar{a}$ is exactly $G \cdot \bigcup_i \ker(a_i) \cup D_\kappa(p)$, the orbit saturation (here the component closure) of the parameter kernels together with the whole small-type part, and that pure parameters add nothing. [Example 17.4](#) shows that this formula fails as stated for a non-surjective injection at $\kappa = \omega$ (the root of a ray is definable). The question stays open for injections, for non-commuting tuples, and for the atom-valued part of definable closure; Questions [23.1](#) and [23.2](#) of Part II continue it.

[write] **Answered with pure real parameters, 4 October 2026, batch 96.** The note above says that the question stays open “for injections, for non-commuting tuples, and for the atom-valued part of definable closure”. Part III (source 03 there) answers all three *with pure real parameters*, that is, for objects definable from the parameters together with finitely many pure reals. For every finite tuple of injections, commuting or not, and every infinite κ , [theorem 27.1](#) gives the exact atom kernel of those objects: $\text{Sat}(B) \cup D_\kappa^{\text{pt}}$, the component closure of the parameter kernels together with the atoms whose orbit under the centralizer $\mathcal{Z}$ of the named maps has fewer than κ elements. For $\kappa > \omega$, $D_\kappa^{\text{pt}} = D_\kappa(f)$ ([corollary 27.2](#)), so the bound named in the question, component closure plus $D_\kappa(f)$, is exact in this sense; at $\kappa = \omega$ the second term must be replaced by D_ω^{pt} , as the ray of [example 17.4](#) shows. The atom-valued part is [theorem 27.3](#) (atom parameters) and [corollary 28.1](#) (group actions, through $N_G(H)/H$). Ordinary definability, without pure parameters, is open: at $\kappa = \omega$, [proposition 27.4](#) reduces it to the definability of rooted codes, which is [Question 37.4](#). The question is re-scoped to that ordinary form, not closed.

Question 10.2 (Noninitial elementary embeddings). Classify elementary self-embeddings of $M_\kappa(A)$ without requiring transitive range. Can one obtain a useful dichotomy separating canonical atom lifts from embeddings that alter the pure core or introduce noncanonical members into images of sets? Any such theorem must state its external class-existence assumptions explicitly.

Question 10.3 (Fixed-parameter rigidity beyond canonical lifts). For canonical lifts, a proper parameter-definable map cannot fix a parameter defining it. Investigate analogous statements for noninitial maps, for embeddings between different cutoffs, and for richer predicates on atoms. The key issue is whether the stabilizer argument still forces the atom action to be small.

Question 10.4 (Commuting named automorphisms). The continuum-sized type family used in the two-name lower bound is noncommutative. Determine the sharp threshold for a fixed number of *commuting* atom permutations. Their components are transitive actions of a finitely generated abelian group, whose stabilizer structure is much more restricted. This suggests that commutation

can lower the type-count threshold, but the uniform internal recognition and all boundary cases should be proved separately.

[write] **Answered, 4 October 2026, batch 93.** [Theorem 15.3](#) of Part II answers this question in full: for every fixed $r \geq 1$, every infinite κ and every atom set with $|A| \geq \kappa$, every r -tuple of commuting atom permutations keeps expanded Replacement if and only if $\text{cf}(\kappa) > \omega$, the threshold of a single name. The boundary cases are covered: at $\kappa = \omega$ a shift fails, and at uncountable cutoffs of countable cofinality the distinct-cycle construction of the proof of [theorem 6.1](#) for $r = 1$ ([example 5.4](#) at $\aleph_\omega$), padded by identities, fails. The uniform internal recognition the question asks for is [lemma 15.2](#) (stabilizer classes are parameter-free definable at every cutoff); the type count is [lemma 15.1](#). Part II goes further: commuting permutations together with t commuting injections have threshold $\text{cf}(\kappa) > \omega$ for $t \leq 1$ and $\text{cf}(\kappa) > \mathfrak{c}$ for $t \geq 2$ ([theorem 16.3](#)); for a finitely generated group named by generators the threshold is $\text{cf}(\kappa) > \tau(G)$, the number of conjugacy classes of subgroups, at uncountable κ , and finiteness of G at $\kappa = \omega$ ([theorem 18.2](#)). For finitely many commuting involutions, and more generally a finite group, at $\kappa = \omega$ the answer was already in the repository as Corollary 58.2 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:cor:finitigroup`; see the note after [corollary 15.4](#)).

Question 10.5 (Other bounded-degree signatures). Extend the component-profile criterion to relational structures on atoms of uniformly bounded or countable degree. Identify hypotheses ensuring that component types have pure set codes recognizable internally without the Replacement scheme under examination. The fresh-component sufficiency argument is flexible; necessity depends on the definability of the complete type blocks.

Question 10.6 (Uncountably many names and infinitary syntax). With ordinary finitary formulas, even arbitrarily many names are tested one finite sublanguage at a time. What changes for a specified infinitary logic that admits countable conjunctions or longer formulas? Component sizes, available type codes, satisfaction specifications, and cofinality bounds should all be recalibrated together.

Question 10.7 (General atom ideals). Replace $[A]^{<\kappa}$ by a downward-closed, finite-union-closed atom ideal. Find an exact analogue of [theorem 5.1](#) in terms of ideal-small unions of isomorphism blocks and ideal-small component closures. Cardinal symmetry was used both to extend partial matchings and to find fresh components; those two requirements may separate for a general ideal.

Question 10.8 (Removing ambient choice). Which parts of the criteria survive without ambient choice? Enumeration of countable components, extension of partial bijections, and the conversion from a bound on witnesses to an internal set each require separate analysis. This is the most direct route toward the choiceless reflection phenomena studied by Glazer and Yao, rather than merely staying in a choice-based special case.

Question 10.9 (Minimal complexity of failed instances). Determine the least syntactic complexity of a formula witnessing failure of expanded Replacement in the real-coded two-name construction. The limit-cutoff Collection obstruction is already Δ_0 , but the component-block selector uses unbounded quantification. Can the two-name obstruction be lowered to a sharp prenex level without importing Collection through normalization?

Question 10.10 (Forcing and preservation properties). The CH characterization suggests a systematic forcing analysis of $\text{Rob}_r(\kappa, A)$. Track which cutoffs, atom reservoirs, and external class parameters are held fixed when the pure continuum changes. Distinguish preservation of one already given expanded structure from universal preservation over all tuples in a forcing extension.

[write] **Fixed-action part answered, 4 October 2026, batch 93.** [Theorem 21.1](#) of Part II answers the fixed-action part: for a ground tuple of injections on a ground atom set, set forcing that keeps κ a cardinal does not change whether the rebuilt expansion satisfies expanded Replacement (component types and the small-type part $D_\kappa(f)$ are absolute). It does not assert that the old class model or the truth value of each formula is unchanged, nor that Collection is absolute under collapses below κ . Open: what happens when κ is collapsed and the model is rebuilt at the new cutoff, and the universal properties $\text{Rob}_r(\kappa, A)$, whose counterexamples may be new actions of the extension ([Question 23.11](#) of Part II).

Question 10.11 (Bounded consistency with atoms and named maps). Can ProveIt's externally indexed partial-satisfaction construction be adapted to a replacement-based urelement theory without accidentally adding Collection? After that, determine exactly which enlarged-language schema instances are needed to formalize the component-profile proof. A theorem about an external family of bounded-consistency derivations must remain distinct from its universal closure in the object language.

Question 10.12 (Machine-checked independence certificates). Develop a reusable Lean interface that packages a definable class model, its valid schemes, and one explicit failed instance of another scheme. Use the one-name singular-cutoff example and the two-name successor-cutoff example as test cases. A good interface should make it impossible to transfer Replacement across a language expansion without providing a proof of the corresponding new instances.

[write] **Status of the questions, 3 October 2026, batch 89.** None of the twelve questions is answered in ProveIt. The two that name the repository remain open there: SetTheory/BoundedConsistency proves its numeralwise endpoint for pure ZFC only, with no atoms and no named maps, and no Lean interface of the kind the last question asks for exists. On the question about removing ambient choice: Part VII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (batch-89 manuscript 04) works in choiceless permutation models with atoms, but about surreal coding and universality; it says nothing about named lifts, Replacement or Collection in the models here. [Added 4 October 2026, batch 90:] Part VIII of the same surreal report (batch-90 manuscript 05) asks the finite-cutoff counterparts of four of these questions for named group actions: its Question 65.1 is the question on general atom ideals, its Question 65.3 the one on other bounded-degree signatures, its Question 65.6 the one on minimal complexity of failed instances, and its Question 65.12 the one on machine-checked certificates; for the question on commuting named automorphisms, its Question 65.4 notes that finitely many commuting involutions generate a finite group and so never destroy Replacement at the finite cutoff (its Corollary 58.2). None of these is answered there. Conversely, [theorem 5.1\(1\)](#) and [theorem 6.1](#) answer in part its Question 65.2 (infinite kernel bounds), and [theorems 7.5](#) and [7.7](#) its Question 65.11 (reflection and Collection in expanded models), for finitely many named injections; dated notes after those two questions there record this. See also the note at the end of [section 5](#). [Added 4 October 2026, batch 93: the first sentence of this note is no longer true. Part II of this report (batch-93 manuscript 01) answers [Question 10.4](#) in full, [Question 10.1](#) for commuting permutations and the fixed-action part of [Question 10.10](#); dated notes after those three questions give the details. Questions 10.1 and 10.10 stay open beyond those parts, and the other nine questions are untouched. Part II's own questions (Section 23) continue Questions 10.1, 10.6, 10.7, 10.8, 10.9, 10.10 and 10.12, as the note after them records.] [Added 4 October 2026, batch 96: Part III answers [Question 10.1](#) with pure real parameters, for every finite tuple of injections (the second dated note after that question); in its ordinary form, without pure parameters, it stays open, and nothing else in this note changes. Part III's questions (Section 37) continue Questions 10.1, 10.5, 10.7, 10.8, 10.9, 10.10 and 10.12.]

11 Conclusion and proof-status summary

The central invariant is not the number of atoms moved by an elementary map, nor the existence of all its set-sized restrictions. It is the union of the small, internally recognizable component-type blocks of the atom structure obtained after the maps are named.

For uncountable cutoffs, that union is small exactly when full expanded Replacement holds. At the finite cutoff, finiteness of all components is an additional necessary and sufficient condition. Counting the available component types gives the sharp distinction between one name and two: the thresholds are ω and $2^{\aleph_0}$ in the cofinality comparison. Counting the *realized* types, together with the successor/limit nature of the cutoff, gives the exact Collection and full-reflection criterion.

These statements are proved for an explicitly constructed family over an external set of atoms. The foundational construction is known; the proposed new content is the expansion analysis and its sharp classifications. The article does not settle CH, does not contradict results about elementary embeddings of pure universes, and does not claim that an external semantic argument has already been turned into a Lean proof.

The most concrete next mathematical tests are the commuting-automorphism case and the minimal complexity of the two-name failure. The most concrete implementation test is whether a formal library can distinguish base-language Replacement, amenability of named maps, and Replacement over the expanded formula type throughout the entire development.

[write] Added 4 October 2026, batch 93. The commuting-automorphism case is settled in Part II (theorem 15.3); the minimal complexity of the two-name failure remains open (Question 23.10).

A An explicitly bounded injection-to-atoms formula

Here is a formulation supporting the Δ_0 claim in lemma 7.2. All bounded quantifiers below are ordinary abbreviations for quantification restricted by membership. No function symbols for pairing or union are required.

Define the bounded predicates

$$\begin{aligned} \text{Sing}(u, a) &\equiv \text{Set}(u) \wedge a \in u \wedge \forall z \in u (z = a), \\ \text{Double}(v, a, b) &\equiv \text{Set}(v) \wedge a \in v \wedge b \in v \wedge \forall z \in v (z = a \vee z = b), \\ \text{OP}(p, a, b) &\equiv \text{Set}(p) \wedge \exists u \in p \exists v \in p (\text{Sing}(u, a) \wedge \text{Double}(v, a, b) \\ &\quad \wedge \forall z \in p (z = u \vee z = v)). \end{aligned}$$

Thus $\text{OP}(p, a, b)$ says that $p = \langle a, b \rangle$ in Kuratowski coding.

The formula $\text{InjectAtoms}(\alpha, s)$ is the conjunction of $\text{Set}(s)$ and the following three bounded conditions.

Every graph entry has the right shape.

$$\forall p \in s \exists i \in \alpha \exists q \in p \exists a \in q (\text{At}(a) \wedge \text{OP}(p, i, a)).$$

Every domain element is represented.

$$\forall i \in \alpha \exists p \in s \exists q \in p \exists a \in q (\text{At}(a) \wedge \text{OP}(p, i, a)).$$

Functionality and injectivity. For all $p, p' \in s, q \in p, q' \in p', a \in q, a' \in q'$, and $i, i' \in \alpha$, require

$$(\text{OP}(p, i, a) \wedge \text{OP}(p', i', a')) \longrightarrow (i = i' \leftrightarrow a = a').$$

Every displayed quantifier is bounded. The first clause excludes spurious graph entries; the second gives totality on α ; the last simultaneously imposes single-valuedness and injectivity. For $\alpha = 0$, the clauses force $s = \emptyset$. Because the domain elements used in the application are pure ordinals, atom values cannot be confused with those indices.

Transitive absoluteness of this formula is immediate from structural induction on bounded formulas. In particular, a transitive set containing an alleged injection graph sees its actual entries and their atom values, rather than some truncated graph with hidden range atoms.

B A dependency audit of the principal proofs

Result	Positive mechanism	Point that must not be assumed
Base Replacement	Fresh atom transposition fixes inputs and moves a proposed new output atom.	Arbitrary witnesses need not be uniquely definable.
Elementary lift	On each finite parameter tuple, the atom injection agrees with an atom permutation.	The global injection need not extend to a global permutation agreeing everywhere.
Parameter-definability	Stabilizers of parameter kernels force identity off the kernel; internal graph codes give the converse.	External definability from the full atom map is not internal parameter-definability.
Type recognition	A countable component enumeration graph lies below an uncountable kernel cutoff.	Expanded Replacement is not used to construct that graph.
Necessity in the profile theorem	Replacement applied to one uniformly defined selector collects all small type blocks.	The blocks need not already form an internal family.
Sufficiency in the profile theorem	A fresh isomorphic component yields an automorphism commuting with every named map.	Separate commutation with only one of several names is insufficient.
Two-name lower bound	Marker roots and their real labels distinguish continuum many countable component types.	No continuum hypothesis is used to obtain $\aleph_1$ distinct reals.
Successor Collection	A common reservoir of at most ν components per type bounds all witness kernels.	A rank bound alone does not bound the atoms in the witnesses.
Full reflection	Grow both rank bounds and component reservoirs for a fixed finite subformula family.	One fixed atom reservoir need not reflect the absence of a set of all atoms.

C Reproducibility and supplementary finite checks

The source package is self-contained: `article.tex` contains its bibliography and compiles with pdfLaTeX. The build script runs three passes to settle the table of contents and cross-references. No external mathematical software or bibliography processor is needed for the manuscript.

The supplementary `verify_finite.py` uses only Python's standard library. It enumerates all permutations on at most seven points for the one-map swap checks, all ordered pairs of permutations on at most five points for the two-map checks, and all marker codes using six bits. It constructs component isomorphisms explicitly, verifies that the induced swaps commute with all named maps, and checks the finite-support marker involutions and their translation-rigidity property.

These are exact finite regression tests of the combinatorial constructions, not statistical experiments and not a formal verification of the theorems. They cannot establish a claim about an infinite cutoff, Replacement, a proper-class elementary embedding, or CH. The accompanying JSON file records the executed checks; the source audit records the repository pin and the literature used.

[write] Shipped layout and replay, 3 October 2026, batch 89. In ProvelIt the program is `code/verify_finite.py`, the build script `code/build.sh`, and the recorded run and the delivered build audit are `data/verification_results.json` and `data/BUILD_AUDIT.json`; the source audit is `SOURCE_AUDIT.md` beside this article. The delivered `build.sh` builds the `article.tex` of its own directory, so in this layout it must be run on a copy with the article beside it. At the write the program was rerun on a scratch copy (Python 3.14.4): it passed, and its report equals the recorded one except for the field `python_version` (3.14.4 against the delivered 3.13.5). The delivered `BUILD_AUDIT.json` describes the delivered 27-page PDF and lists the hashes of the delivered files, among them the delivered PDF and README, which are not shipped; the article and PDF in ProvelIt are the written versions. The report's README gives the replay commands.

Part II

Commuting actions and Replacement

Exact cofinality thresholds, definable kernels, and uniform naming

A solution to ProveIt Question 10.4 with further classification theorems

Research manuscript prepared with ChatGPT, 4 October 2026

Status and scope. This article supplies complete arguments for its stated results in an explicitly specified family of models. The atom-kernel construction and the component-profile criterion are prior work, credited and reproved where needed. The commuting, mixed-injection, definable-kernel, and uniform-naming refinements are the contributions developed here. The work answers a question recorded as open in the inspected ProveIt report; publication priority beyond that report is not established. These are mathematical proofs, not machine-checked proofs. Elliot Glazer and Bokai Yao are cited researchers, not authors or endorsers of this manuscript.

Abstract. Let A be an external set of atoms, let κ be an infinite cardinal with $|A| \geq \kappa$, and retain exactly those sets whose transitive closures involve fewer than κ atoms. Naming an atom injection by its canonical elementary lift can change which instances of Replacement hold. We determine exactly what commutation contributes to this phenomenon. Every fixed finite nonempty family of commuting atom permutations preserves expanded Replacement for every action precisely when $\text{cf}(\kappa) > \omega$. More generally, for commuting signatures with s designated permutations and t additional injections, the threshold is ω when $t \leq 1$ and is $2^{\aleph_0}$ when $t \geq 2$. The latter bound is witnessed by explicit lattice staircases; the two coordinate maps can individually preserve Replacement and have only pure fixed points while their joint expansion fails Replacement.

For a finitely generated group G , the uncountable-cutoff threshold is the number of conjugacy classes of subgroups of G , with a separate and exact finite-cutoff criterion. For commuting permutations we also determine the complete atom kernel of definable closure, including the finite-cutoff case. Finally, an action of $\bigoplus_{\omega} C_2$ on two-point orbits shows that naming every group element separately can preserve Replacement at every cutoff while a single amenable binary evaluation map for the same action destroys it. The article explains the relation to Glazer–Yao reflection theory, gives explicit failed schema instances, distinguishes established results from the new refinements, and proposes a focused research program.

[write] Part II of this report since 4 October 2026: batch-93 manuscript 01 of ProveIt’s incoming-reports intake, printed in full (label prefix `nee:ca:`). It continues Part I and answers its Question 10.4 ([Question 10.4](#)), its Question 10.1 for commuting permutations ([Question 10.1](#)) and the fixed-action part of its Question 10.10 ([Question 10.10](#)). Section n of the manuscript is Section $n + 11$ here, so its item $n.k$ is item $(n + 11).k$ (its Theorem 4.3 is [theorem 15.3](#)); its Appendices A and B are Appendices D and E; its equations, numbered consecutively there, are numbered within sections here. “ProveIt Question 10.4” and every other number of Part I that the manuscript quotes are unchanged. Provenance, the overlap with Part VIII of birthday-cutoffs-and-hereditary-sets (prior in the repository), the imported group-theoretic fact and the notation are in Sections [12.4](#) and [12.5](#).

12 The research intersection and the results

The subject is the interaction between symmetry and axiom schemes in set theory with atoms. An atom is an object which may belong to sets but has no members of its own. Once atoms are

permitted, a universe may satisfy Replacement and the Axiom of Choice while lacking a set that contains all its atoms. This makes it possible to examine exactly how extra named structure changes definability.

Glazer and Yao's *Reflection Principles in ZFU* develops the interaction of Collection, reflection, and choice in urelement set theory, including constructions controlled by small atom kernels [1]. The directly relevant ProvelIt report, *Naming Elementary Embeddings*, studies canonical lifts of atom maps in a particularly explicit kernel model [7]. Its Question 10.4 asks for the sharp Replacement threshold when finitely many named atom permutations commute. The same report's Question 10.1 asks how sharp its bound on definable kernels can be.

We answer Question 10.4 for every positive finite number of names. We also give an exact answer to the kernel question in the commuting-permutation setting and extend the analysis to commuting injections and uniformly named group actions. The setting is deliberately concrete: the conclusions below concern the specified models, rather than all models of urelement set theory or all elementary embeddings.

12.1 The basic model and the meaning of preservation

The atom kernel $\ker(x)$ is the set of atoms occurring anywhere in the transitive closure of $\{x\}$. In an ambient universe with Choice, define

$$M_\kappa(A) = \{x \in U(A) : |\ker(x)| < \kappa\}. \quad (12.1)$$

Here $U(A)$ is the cumulative universe over the external set A of atoms. The model contains all pure sets, so it contains arbitrarily large pure sets even when only finitely many atoms may occur in any one of its objects.

An injection $f : A \rightarrow A$ extends recursively to a map j_f on sets. The base model regards j_f as an external elementary map; adjoining a symbol J for it creates new first-order formulas. *Expanded Replacement* means that Replacement is valid for every formula of this enlarged language, with arbitrary internal set parameters. It is stronger than the assertion that the graph of j_f restricted to each internal set is internal.

Throughout the article $\mathfrak{c} = 2^{\aleph_0}$ is computed in the ambient pure universe. A statement of *universal preservation* quantifies over all actions of the indicated kind on the fixed atom set A . Failure of this universal statement means that a counterexample action exists; it does not mean that every action fails.

12.2 Main classification in one table

The following table summarizes the universal statements. Every row assumes κ infinite and $|A| \geq \kappa$.

Named structure	Exact condition for universal Replacement
A fixed finite nonempty family of commuting permutations	$\text{cf}(\kappa) > \omega$
Commuting permutations together with at most one additional injection, with at least one name	$\text{cf}(\kappa) > \omega$
At least two additional commuting injections	$\text{cf}(\kappa) > \mathfrak{c}$
An action of a fixed finite group, named by generators	Every infinite κ
An action of a finitely generated G , with $\kappa > \omega$	$\text{cf}(\kappa) > \tau(G)$
An action of a finitely generated G , with $\kappa = \omega$	G is finite

Here $\tau(G) = |\text{Sub}(G)/G|$, where G acts on its subgroups by conjugation. In the mixed rows, an “additional injection” is not required to be surjective; it is permitted to be a permutation. The lower bounds can be made genuinely nonsurjective where stated below.

The first row answers the repository question. The second and third identify a finer boundary: one noninvertible direction is still controlled by countably many types, whereas two such directions admit a binary sequence of boundary data. The group formulation explains why a number of generators alone cannot determine the answer.

12.3 What is inherited and what is proved here

Result or ingredient	Provenance and role
The model $M_\kappa(A)$ and base Replacement	A specialization of the kernel-ideal constructions in urelement set theory; see Yao [3] and Glazer–Yao [1]. We give the needed symmetry proof.
The component-profile criterion	Theorem 5.1 of the inspected ProvelT report [7]. It is stated and reproved in section 14 , and is not claimed as a new result.
Transitive group actions as coset actions	Standard group theory [9]. The new use is the exact Replacement classification by the resulting type count.
Commuting and mixed signatures	Theorems proved here, including the boundary cases requested by ProvelT Question 10.4.
Definable atom kernels	An exact refinement proved here for commuting permutations, addressing Question 10.1 in that setting.
Separate names versus uniform evaluation	An explicit separation proved here. It uses ordinary finitary first-order syntax on both sides.
Collection and reflection	The underlying criteria are already in [7]; we explain their commuting-action consequences without claiming a new general reflection theorem.
Fixed-action forcing stability	A profile-based refinement proved here, answering the fixed-action part of Question 10.10 when the cutoff remains a cardinal.

The repository source was inspected at commit a9ab9a69872c2fb7464097af32fd7b76c75cb030.

The source’s dated status note says that its twelve proposed questions remain unanswered there. This is a precise, limited novelty claim: the present work resolves the commuting question recorded in that source. No claim is made to have settled an open problem expressly posed in Glazer and Yao’s paper, or to have established historical priority for every corollary.

12.4 Provenance and place in this report

[write] Added 4 October 2026, batch 93. *Source.* This Part is a single manuscript, printed in full: manuscript 01 of batch 93 of ProveIt’s incoming-reports intake, delivered as the archive `commuting_actions_replacement.zip` (690,120 bytes; inner directory `commuting_actions_replacement/`, main file `commuting_actions_replacement.tex`, 1,283 lines, 61 labels, 8 bibliography entries, a 31-page US Letter PDF) in the arrival commit `2faa3b37a`, and placed in commit `47a77daba` as Part II of this report (files prefixed `02-commuting-actions-`, labels `nee:ca:`). It was filed here rather than beside Part VIII of `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets` because it is written as a continuation of Part I and answers Part I’s questions. The author line reads “Research manuscript prepared with ChatGPT”, 4 October 2026; the Part is AI-assisted, unrefereed and not formalized. Being a single source, it needed no merge choices: every statement, proof, example, remark, question and non-claim of the manuscript is printed. The write prefixed all 61 labels with `nee:ca:` and updated every reference; gave the twelve questions of Section 23 labels; replaced the manuscript’s title page and abstract environment by the Part heading, the status box and the paragraph “Abstract”; dropped its separate table of contents; set its questions on the shared theorem counter (their numbers do not change, since Section 23 contains nothing else numbered); numbered its equations within sections (in the manuscript they run consecutively: its (1) is (12.1), (2)–(3) are (13.1)–(13.2), (4)–(5) are (14.1)–(14.2), (6) is (15.1), (7)–(11) are (16.1)–(16.5), (12) is (17.1), (13)–(14) are (18.1)–(18.2) and (15)–(17) are (19.1)–(19.3)); pointed its figure at the shipped file; merged its bibliography into the report’s (below); and added these dated **[write]** notes, the notation table of Section 12.5 and Question 23.13. Item $n.k$ of the manuscript is item $(n + 1).k$ here; its Appendices A and B are Appendices D and E. No statement, proof or non-claim was changed.

Pin and repository claims. The pin `a9ab9a69872c2fb7464097af32fd7b76c75cb030` (Section 12 above, Appendix E and the bibliography entries [7, 8]) is the ProveIt commit “Audit eight arithmetic compiler reports and the complete degree-12 matrix DAG” of 4 October 2026, an ancestor of the placement. The manuscript’s quoted blob `5426fc8da5f470efc448c3dff041751d0d7dc4a9` is exactly this report’s `article.tex` at the pin; it differs from Part I as printed only by the two dated notes of the batch-90 reciprocal commit `e50dde15b`, so every number of [7] that the manuscript quotes (Theorems 5.1, 6.1, 7.5, 7.7; Questions 10.1, 10.4, 10.10) is the number in Part I: [theorems 5.1, 6.1, 7.5 and 7.7](#) and [Questions 10.1, 10.4 and 10.10](#). The manuscript’s other repository statements are true at the pin and at this write: `SetTheory/ZF/Lean/ZF/Zf.lean` (blob `b33db57cf`, the same at Part I’s pin `2b7b388ba`, at this pin and at the write) declares `Sep_form`, `Func_form`, `Image_form`, `Rep1_form`, `ZFax` and `ZFprov`, and no urelement model is formalized; the dated note of Part I said that its twelve questions were unanswered, which was true until this Part. No repository statement is contradicted, and nothing needed retraction.

Formal status. No statement of this Part is formalized in Lean or Rocq; the formal-status note of Section 1.4 applies unchanged (the formula type of `SetTheory/ZF` has no atom predicate and no function symbols), and the milestones of Section 22 are proposals.

Prior in the repository: Part VIII of the surreal report. Batch-90 manuscript 05, *Named Symmetries and Replacement*, is Part VIII of `birthday-cutoffs-and-hereditary-sets` (its source 12, labels `hset:ns:`). It works only at the finite cutoff, with a uniformly named action of an arbitrary pure set-sized group. Its text was written in commit `1404038df` (4 October 2026, 10:57), twelve minutes after this manuscript’s pin (10:45; the batch-90 placement `12076b2e8`, an ancestor of the pin, staged only its files), so this manuscript does not cite it. Several results here meet it at $\kappa = \omega$; Part VIII is prior in the repository, the proofs are independent, and both stand. At $\kappa = \omega$: [corollary 15.4](#) is its Corollary 58.2 (`hset:ns:cor:finitigroup`); the finite-cutoff clause of [theorem 18.2](#) follows from its Theorem 58.1 (`hset:ns:thm:full`) with its Proposition 62.1 (`hset:ns:prop:finitewords`); [proposition 19.1](#) is its Proposition 62.2 (`hset:ns:prop:language`) with

Corollary 58.2; [proposition 19.2](#) is Theorem 58.1, which there holds for every pure set-sized group; [proposition 19.4](#) is Proposition 62.1, there for free-group words; and [theorem 19.3](#) makes the separation of its Theorem 62.3 (`hset:ns:thm:evaluator`) with the same group $\oplus_{\omega} C_2$ but a different witness (below). New here: everything at uncountable cutoffs ([theorem 18.2](#) for $\kappa > \omega$, [proposition 19.2](#) for $\kappa > \omega$, [corollary 19.5](#)), and the witness of [theorem 19.3](#), whose orbits have at most two atoms, which works at every κ with $\text{cf}(\kappa) \leq \mathfrak{c}$, whose evaluator is amenable, and which has a converse for $\text{cf}(\kappa) > \mathfrak{c}$; Part VIII’s witness is the regular action, with infinite orbits, at $\kappa = \omega$ only. Dated notes after each of these results record the overlap, and a note after [Question 23.12](#) relates the questions. The rest of the Part (commuting and mixed signatures, staircases, definable kernels, the forcing theorem) has no counterpart there.

Relation to Part I. The manuscript reproves and credits Part I’s foundations: [lemma 13.2](#) is [lemma 2.3](#), [proposition 13.3](#) is [theorem 2.4](#) (Yao’s construction, specialized; also Theorem 32.2 of Part VI of birthday-cutoffs-and-hereditary-sets, `hset:zr:thm:supportbase`), [proposition 13.4](#) is part of [theorem 3.2](#), [lemma 14.1](#) is [lemma 4.3](#) with [lemma 4.2](#), and [theorem 14.2](#) is [theorem 5.1](#). [Lemma 14.4](#) abstracts the sharpness constructions of [theorem 6.1](#) and [lemma 6.2](#); [theorem 16.4](#) is a commuting, proper-injection variant of [corollary 7.9](#), whose two permutations do not commute; [theorem 20.1](#) is a consequence of [theorems 7.5](#) and [7.7](#), as the manuscript says, and its failure at limit cutoffs is [lemma 7.2](#).

An imported fact. [Remark 18.4](#) rests on an existence theorem of Ivanov and Ol’shanskii [12], read in Smith’s paper [11]; it is not proved here. The note after that remark records what was checked, and [Question 23.13](#) records what is missing. Nothing else in this Part is stated without proof, and no claim of the manuscript was found to be wrong.

The bibliography. The manuscript’s entries for [1] and [3] are Part I’s and are printed once (the manuscript adds the full-text address <https://arxiv.org/html/2602.21970v1> and dates Yao’s dissertation “2023” with arXiv v3 “June 2023”). Its entry RepoZF, pinned at a9ab9a698, is cited as [8], so that Part I’s entry [4], pinned at 2b7b388ba, stays as it was; the file is the same blob at both pins. The entries [7, 9, 10, 11, 12] are new.

Non-claims of the delivery README (not shipped), beyond those in the text: the manuscript is not peer reviewed, and the “independent review” within its preparation (Appendix E) is not external review; it claims the resolution of the repository’s question only, not worldwide priority, and not the settlement of an open problem posed by Glazer and Yao, whose work is a credited source, not an endorsement or coauthorship; the scripts supplement the proofs and establish no infinite classification, cardinal arithmetic, forcing theorem, class-model theorem or axiom scheme; and the upstream repository was inspected, not modified.

Checks at the write. The delivered finite checks were rerun on a scratch copy (Python 3.14.4, standard library, about 2 s): they pass, with 347,278 explicit checks, and the output is byte-identical to the recorded one. The delivered figure generator, run on a scratch copy with Matplotlib 3.10.8, regenerates a byte-identical `staircases.pdf` (TrueType fonts, no Type 3) and a pixel-identical `staircases.png` (RGBA, 1970×1659 ; differently compressed). Section 22 and Appendix E name the delivered files; in ProvelT they are `code/02-commuting-actions-verify_examples.py`, `code/02-commuting-actions-make_figure.py`, `data/02-commuting-actions-verification_output.txt` and `figures/02-commuting-actions-staircases.pdf` (with its PNG preview). The generator writes `figures/staircases.*` beside itself, so it must be run on a copy; the report’s README gives the commands. Appendix E says that a LinkedIn page identifies Glazer; it gives no address, and none is recorded here.

12.5 Notation of Part II

[write] Added 4 October 2026, batch 93. No symbol was renamed; the only changes are the label prefix and one citation key (above). The objects of Part I keep their meaning: $M_{\kappa}(A)$ is $M_{\kappa}(A)$ of (2.1), and $\ker, j_f, \mathcal{N}_{\kappa,f}, \mathcal{T}_r, B_t, D_{\kappa}(f), V_{\alpha}(C), \mathfrak{c}$, Replacement, Collection, RP are as there. Several letters carry more than one meaning in this Part:

Symbol	Meaning here (and what it is not)
$\mathbb{N}$	the natural numbers. <i>Not</i> Part I's macro $\mathcal{N}_{\kappa,f}$, the expanded model
U_S	the staircase (16.4) coded by $S \subseteq \mathbb{N}$; <i>not</i> the cumulative universe $U(A)$
F, G	in Section 16, the two coordinate maps of a staircase; elsewhere G is a group: $\mathbb{Z}^r$ in Section 17, a finitely generated group in Section 18, a countable group or $\bigoplus_{\omega} C_2$ in Section 19. In lemma 4.1 of Part I, G_S is an involution of $\mathbb{Z}$
H	in Section 16, the group $\mathbb{Z}^S$ acting through the permutations; in Section 17, $H(\bar{a})$ is the protected union $G \cdot P \cup D_{\kappa}(p)$, the counterpart of (5.4); in Sections 18–19, a subgroup; in Section 20, a reservoir of atoms, as in lemma 7.4. Never the hereditary-size universe H_{κ}
K	a subgroup (a stabilizer) in Sections 15–17 and proposition 19.1; in the proof of theorem 16.4, a set of atoms
p, p_i, p	in Sections 16–17, named commuting permutations; elsewhere p is a parameter tuple, as in Part I
B_K	the atoms with stabilizer exactly K : the type block B_i of $t = \mathbb{Z}^r / K$
$E(g, x)$	the uniform evaluator (19.1) of a group action. It is Part VIII's evaluator $T(i, a, b)$ (hset:ns:thm:evaluator) and its named action graph Act (source 12's E) of birthday-cutoffs-and-hereditary-sets; <i>not</i> the enumeration predicate E of Part VI there, which names an injection of κ into the atoms (see the note after corollary 7.6)
$\tau(G), \tau_{\text{fin}}(G)$	numbers of conjugacy classes of all subgroups, of finite-index subgroups; <i>not</i> Part VIII's involution τ nor Part VI's $\tau = \text{cf } \kappa$ there
$\mathcal{K}(\text{dcl}(\bar{a}))$	scheme notation for the kernels of uniquely definable objects (before theorem 17.1); no truth predicate
universal preservation	preservation for every action of the stated kind on A : for r arbitrary injections this is Part I's $\text{Rob}_r(\kappa, A)$

13 Foundations, kernels, and canonical lifts

13.1 External and internal sets

Work in a well-founded ambient set theory with the Axiom of Choice and an external set A of urelements. Equivalently, ordinary ZFC can implement the entire construction with tagged atoms, tagged sets, and an interpreted membership relation. The use of $U(A)$ as a cumulative class does not require an inaccessible cardinal. The cardinal bounds used below are external cardinal bounds.

Write $\text{At}(x)$ for “ x is an atom” and $\text{Set}(x)$ for its negation. Extensionality applies to sets. For every object x put

$$\ker(x) = \{a \in A : a \in \text{TC}(\{x\})\}.$$

Thus $\ker(a) = \{a\}$ for an atom, whereas a pure set has empty kernel. A set of atoms $B \subseteq A$ belongs to $M_{\kappa}(A)$ exactly when $|B| < \kappa$.

The distinction between “ B is an external set” and “ B is an internal set of $M_{\kappa}(A)$ ” is essential. Every atom is internal. The totality A is an external set but is not an internal set, because $|A| \geq \kappa$. Conversely, every pure set is internal, with no restriction on its cardinality or rank.

For an atom set C , use the hierarchy

$$V_0(C) = C, \quad V_{\alpha+1}(C) = \mathcal{P}(V_{\alpha}(C)) \cup C, \quad V_{\lambda}(C) = \bigcup_{\alpha < \lambda} V_{\alpha}(C).$$

Every object supported on C occurs in some $V_\alpha(C)$. Each displayed stage is an external set. When $|C| < \kappa$, every object in every stage is internal.

Remark 13.1 (The semantic convention). All assertions about a formula holding in $M_\kappa(A)$ mean the external relativization of that *fixed* formula to the definable class (12.1). Elementarity, Replacement, reflection, and definable-closure assertions are interpreted as schemes. We do not posit an internal truth predicate for a proper-class structure or an internal set of all class embeddings. Whenever a proof collects witnesses, it uses external Collection for the fixed translated formula and then checks the kernel bound. This convention also governs the definable-kernel theorem in section 17.

13.2 Why the base model satisfies Replacement

The following elementary identities organize the argument:

$$\begin{aligned} y \in x &\implies \ker(y) \subseteq \ker(x), \\ \ker(\{x, y\}) &= \ker(x) \cup \ker(y), \\ \ker(\mathcal{P}(x)) &\subseteq \ker(x), \\ \ker(Y) &= \bigcup_{y \in Y} \ker(y) \quad \text{for an external set } Y. \end{aligned} \tag{13.1}$$

A finite union of sets of cardinality below an infinite κ still has cardinality below κ , even if κ is singular. It follows that $M_\kappa(A)$ is transitive, closed under subsets of its internal sets, and closed under Pairing, Union, and Powerset.

Every permutation π of A lifts recursively to an automorphism $\hat{\pi}$ of $U(A)$ and of $M_\kappa(A)$. If π fixes $\ker(x)$ pointwise, then $\hat{\pi}(x) = x$.

Lemma 13.2 (Base-language confinement). *If a formula of $\{\in, \text{At}\}$ uniquely defines y from internal parameters x, p , then*

$$\ker(y) \subseteq \ker(x) \cup \ker(p).$$

Proof. Suppose an atom b occurs in y but not in x or p . Since the union of their three kernels has size below κ , there is $c \in A$ outside that union. Interchanging b and c fixes x, p and changes the kernel of y . Its lift is a base-language automorphism and therefore produces a second output for the same uniquely defining formula. This is impossible. $\square$

Proposition 13.3 (The base theory). *The model $M_\kappa(A)$ satisfies the usual urelement axioms with Separation, Replacement, and the assertion that every set can be well-ordered. Its pure part is the full ambient pure universe. Its internal cardinality for each internal set agrees with the external cardinality.*

Proof. The structural axioms follow from (13.1). For a functional formula on an internal domain w , external Replacement supplies its range Y . Applying lemma 13.2 separately to each input gives

$$\ker(Y) \subseteq \ker(w) \cup \ker(p),$$

so Y is internal. Separation holds because every externally separated subset of an internal set has no new atoms.

An external well-order of w has kernel contained in $\ker(w)$, and hence belongs to the model. The graph of any external bijection between w and its pure cardinal has the same kernel bound. All pure objects are present, so there are no additional restrictions on pure ordinals, cardinal arithmetic, or the pure coding sets used below. $\square$

Collection is not included in the conclusion. In urelement set theory with an internally proper class of atoms, it is a distinct issue; [section 20](#) states the relevant distinction explicitly.

13.3 Elementary and amenable lifts

For an injection $f : A \rightarrow A$, define

$$j_f(a) = f(a), \quad j_f(x) = \{j_f(z) : z \in x\} \quad \text{if } \text{Set}(x).$$

Then

$$\ker(j_f(x)) = f[\ker(x)]. \quad (13.2)$$

The recursive map preserves ranks and maps $M_\kappa(A)$ to itself. It fixes every pure object. Composition of atom maps corresponds to composition of their lifts; in particular commuting atom maps have commuting lifts.

Proposition 13.4 (Canonical lifts). *Every j_f is a base-language elementary self-embedding of $M_\kappa(A)$. For each internal set w , the restriction graph $j_f \restriction w$ is internal. If f is a permutation, j_f is an automorphism.*

Proof. Let $\bar{x}$ be a finite tuple of parameters and put $P = \bigcup_i \ker(x_i)$. The partial bijection $f \restriction P : P \rightarrow f[P]$ extends to a permutation π of A : both complements have cardinality $|A|$, because $|P| < \kappa \leq |A|$. On every object in the tuple, $\hat{\pi}$ agrees with j_f . An automorphism preserves the truth of any fixed base-language formula, which proves elementarity on $\bar{x}$.

The graph of $j_f \restriction w$ has kernel contained in $\ker(w) \cup f[\ker(w)]$, a set of size below κ . This is the asserted amenability. An inverse atom permutation lifts to an inverse automorphism when f is bijective. $\square$

Definition 13.5 (The expanded model). For a finite tuple $f = (f_1, \dots, f_r)$ of atom injections, let

$$\mathcal{N}_{\kappa, f} = (M_\kappa(A), \in, \text{At}, j_{f_1}, \dots, j_{f_r}).$$

The symbols $J_1, \dots, J_r$ are genuine function symbols. Expanded Replacement permits them inside every formula of the scheme. A lifted atom permutation is an automorphism of this expanded structure whenever its atom action commutes with each f_i .

Expanded Separation always holds: an externally defined subset of an internal set remains internal. The issue for Replacement is that a set of individually small outputs may collectively involve too many atoms.

14 The component-profile criterion

This section supplies the central prior tool in a self-contained form. Its content is Theorem 5.1 of [7]. The later classifications depend on the number and definability of its component types.

14.1 Codes and entire type blocks

Make the atoms into a labelled directed graph, with an i -edge from a to $f_i(a)$. A component is a connected component of the underlying undirected graph. There are at most $2r$ neighbors at each vertex, since an injection gives at most one predecessor of each label. Every component is therefore finite or countably infinite.

Let $\mathcal{T}_r$ be the pure set of isomorphism types of connected structures with r total injections on a nonempty finite ordinal or on ω . A type may be represented by the set of all its codes, so no definable choice of representative is needed. Define

$$B_t = \{a \in A : \text{the component of } a \text{ has type } t\}, \quad D_\kappa(f) = \bigcup_{\substack{t \in \mathcal{T}_r \\ |B_t| < \kappa}} B_t. \quad (14.1)$$

The word “block” always refers to all copies of one component type. A countable component can belong to a block of arbitrarily large cardinality.

Lemma 14.1 (Recognition without expanded Replacement). *For $\kappa > \omega$, membership $a \in B_t$ is uniformly first-order definable in the expanded language, with the pure type t as a parameter. At $\kappa = \omega$ the same holds for finite component types. A specified finite path is definable at every cutoff.*

Proof. A path is represented by a function on a finite ordinal. A forward step of label i requires $J_i(a) = b$; a backward step requires $J_i(b) = a$. The graph of a successful finite path uses finitely many atoms, so it is internal even at $\kappa = \omega$.

For a full type, ask for a pure code $c \in t$ and an injective set function h from its domain into the atoms. Require a in its range, preservation of every labelled operation, and closure of its range under incoming edges. The image is connected, closed under both directions of every edge, and contains a , so it is exactly the component of a . Conversely an enumeration of the actual component has a graph with at most countably many atoms. It is internal when $\kappa > \omega$, and it is internal at $\kappa = \omega$ when the component is finite. These are external kernel checks on particular graphs, not applications of expanded Replacement. $\square$

Theorem 14.2 (Component-profile criterion; prior work). *For a nonempty finite tuple of atom injections,*

$$\mathcal{N}_{\kappa, f} \models \text{Replacement} \iff |D_\kappa(f)| < \kappa \quad (\kappa > \omega).$$

At $\kappa = \omega$, expanded Replacement holds exactly when every component is finite and $D_\omega(f)$ is finite.

Proof. Necessity at an uncountable cutoff. On the pure domain $\mathcal{T}_r$, define $F(t)$ to be the set of all atoms of type t if such an internal set exists, and $\emptyset$ otherwise. By lemma 14.1 this is a single first-order definition. Set Extensionality makes its output unique. Its external value is B_t exactly when $|B_t| < \kappa$, and is empty otherwise. Replacement would collect the range of F into an internal set. The union of that range is $D_\kappa(f)$, so this union must have size below κ .

The additional finite-cutoff obstruction. Fix an atom a . On the pure set of finite words in the forward and backward labels, define the value to be the endpoint of the path starting at a , using a as a default when a backward step does not exist. The endpoint is uniquely specified by a finite graph. Its range is the entire component of a . Expanded Replacement at $\kappa = \omega$ would make this component finite. Once every component is finite, the preceding selector argument over the pure finite types proves that D_ω is finite.

Sufficiency. For a set P of atoms let $\text{cl}(P)$ be the union of components meeting P . If $\kappa > \omega$ and $|P| < \kappa$, then

$$|\text{cl}(P)| \leq \max(\aleph_0, |P|) < \kappa. \quad (14.2)$$

At the finite cutoff the closure of a finite P is finite when every component is finite. Suppose a formula uniquely specifies y_x for each x in an internal domain w , with parameter tuple p . Put

$$H = \text{cl}(\ker(w) \cup \ker(p)) \cup D_\kappa(f).$$

The relevant right-hand conditions make H small. It is a union of components.

If $b \in \ker(y_x) \setminus H$, its type block has size at least κ . For an uncountable κ this block has at least κ components: each has at most $\aleph_0$ vertices, and a family of fewer than κ uniformly countable sets has union smaller than κ . Fewer than κ components meet $H \cup \ker(y_x)$, so there is a fresh component of the same type disjoint from that set. At $\kappa = \omega$, the same conclusion holds because this particular type has a fixed finite size and its block has infinitely many copies.

Interchange the component of b with the fresh component by a labelled isomorphism and its inverse, fixing every other component. This is an atom permutation commuting with every f_i . It fixes x, p , and it moves b outside the old kernel of y_x . Its lift changes the unique output, a contradiction. Therefore every output kernel is contained in H . External Replacement collects the outputs into a set with kernel contained in H , which is internal. This proves every expanded instance. $\square$

Remark 14.3 (Where singular cardinal arithmetic is used). The proof does not assert that an arbitrary union of fewer than κ small sets is small. That assertion can fail when κ is singular. In (14.2) there is a *uniform countable bound* on component sizes. Later unions of whole type blocks require the stronger comparison with $\text{cf}(\kappa)$.

Lemma 14.4 (From a supply of types to a sharp bound). *Suppose a class of finite-signature atom structures has at most θ countable connected types, is closed under disjoint unions, and admits a fixed harmless type for unused atoms. At an uncountable cutoff, $\text{cf}(\kappa) > \theta$ ensures universal Replacement. If θ is infinite and there are θ distinct types available independently, then every $\kappa > \omega$ with $\text{cf}(\kappa) \leq \theta$ admits a counterexample.*

Proof. For the positive direction, a union of fewer than $\text{cf}(\kappa)$ sets, each of size below κ , has size below κ . Apply this to the small type blocks.

For the converse put $\delta = \text{cf}(\kappa)$. Choose δ distinct nonreserved types t_i and infinite cardinals $\lambda_i < \kappa$ with $\sum_{i < \delta} \lambda_i = \kappa$. If κ is regular use $\delta = \kappa$ and $\lambda_i = \aleph_0$; if it is singular use a cofinal sequence of infinite cardinals below it. Give type t_i exactly λ_i copies. Its block has size λ_i because every component is nonempty and countable. Realize their disjoint union on κ atoms and use the reserved type on any remainder. The small blocks together contain κ atoms. The failed instance is the type-block selector in the proof of [theorem 14.2](#). $\square$

15 Commuting permutations: the exact answer

Fix $r \geq 1$. A tuple of r commuting permutations gives an action of the free abelian group $\mathbb{Z}^r$, where the standard basis vectors act by the named permutations. The action may have a kernel; faithfulness is not required.

15.1 Orbit classification and finite subgroup descriptions

Lemma 15.1. *Connected structures of r commuting permutations are exactly the transitive $\mathbb{Z}^r$ -sets $\mathbb{Z}^r / K$ for subgroups $K \leq \mathbb{Z}^r$. Two such labelled structures are isomorphic exactly when their subgroups are equal. There are countably infinitely many types.*

Proof. For an atom a , the orbit map $u \mapsto u \cdot a$ identifies its orbit with $\mathbb{Z}^r / \text{Stab}(a)$. In a transitive action stabilizers change by conjugation; for an abelian group they do not change at all. Hence a labelled isomorphism preserves K , and equal K gives the evident isomorphism.

Every subgroup of $\mathbb{Z}^r$ is finitely generated. Here is a proof that also records the finitary information we need. Induct on r . Project a subgroup $K \leq \mathbb{Z}^{r-1} \times \mathbb{Z}$ to its last coordinate. Its image is $d\mathbb{Z}$. If $d = 0$, use induction in $\mathbb{Z}^{r-1}$. If $d > 0$, choose $v \in K$ with last coordinate d . Subtracting a suitable integer multiple of v from any member of K leaves an element of $K \cap (\mathbb{Z}^{r-1} \times \{0\})$, which is finitely generated by induction. Those generators together with v generate K . Thus the set of subgroups is the image of a countable set of finite integer matrices. It is infinite, as witnessed by $n\mathbb{Z} \times \mathbb{Z}^{r-1}$ for positive n . $\square$

The proof classifies labelled actions, not merely the abstract quotient groups. Changing the basis of $\mathbb{Z}^r$ would change the interpretation of the names, and is not an allowed isomorphism here.

Lemma 15.2 (Stabilizers can be recognized at every cutoff). *For each fixed subgroup $K \leq \mathbb{Z}^r$, the class of atoms with stabilizer exactly K is parameter-free definable in the expanded language. The definition is correct even at $\kappa = \omega$.*

Proof. Fix a finite integer matrix generating K . Its entries can be built into a formula by fixed numerals, and membership in its integer span is a first-order assertion about pure integers. A vector $u \in \mathbb{Z}^r$ determines a finite word in the named permutations and their inverses. Its value on a is expressed by a finite path graph, which is internal by [lemma 14.1](#). The required predicate says

$$\text{At}(a) \wedge \forall u \in \mathbb{Z}^r (u \cdot a = a \iff u \in K). \quad (15.1)$$

This abbreviates a first-order formula using the finite matrix and finite-path definition. It quantifies over words individually; it does not first assemble an orbit or a simultaneous table of all their values. All pure integer coding is available in $M_\kappa(A)$. $\square$

Theorem 15.3 (Commuting-permutation threshold). *For every fixed $r \geq 1$, every infinite cardinal κ , and every atom set A with $|A| \geq \kappa$, the following are equivalent:*

- (i) *Every r -tuple of commuting atom permutations has a canonical expansion of $M_\kappa(A)$ satisfying expanded Replacement.*
- (ii) $\text{cf}(\kappa) > \omega$.

Proof. If $\text{cf}(\kappa) > \omega$, then $\kappa > \omega$. By [lemma 15.1](#) the small-type part is a countable union of sets of size below κ , so it is small. Apply [theorem 14.2](#).

If $\kappa = \omega$, let the first permutation shift a two-sided countably infinite orbit and let all other names act identically. Replacement on finite iterates of an atom would collect an infinite atom set, so it fails.

If $\kappa > \omega$ and $\text{cf}(\kappa) = \omega$, choose infinite cardinals $\lambda_n < \kappa$ cofinal in κ . On disjoint blocks B_n of size λ_n , let the first permutation consist of cycles of exact length $n + 2$, and make every other

permutation the identity. Fix all atoms outside these blocks. The B_n are precisely the entire blocks of those cycle types. Each is internal, but $|\bigcup_n B_n| = \kappa$. The formula sending n to the complete set of atoms of exact period $n + 2$ is total and single-valued and has no internal range. This is the required failed instance. $\square$

This proves the threshold and both boundary cases requested in ProveIt Question 10.4. An arbitrary fixed finite number of commuting permutations therefore has the countable type threshold of one permutation.

[write] **Question 10.4 answered, 4 October 2026, batch 93.** “ProveIt Question 10.4” is [Question 10.4](#) of Part I; a dated note there records the answer. The positive half (countably many types, [lemma 15.1](#)) is new; the two counterexamples are those of [theorem 6.1](#) for one name, padded by identity names.

Corollary 15.4 (Finite group relations). *If the named permutations generate an action of a fixed finite group, expanded Replacement holds at every infinite cutoff. In particular this is true of finitely many commuting permutations with fixed finite orders.*

Proof. A finite group has finitely many subgroups and all of its orbits are finite. Hence there are finitely many types and D_κ is a finite union of small blocks. Both clauses of the finite-cutoff criterion are satisfied as well. $\square$

Example 15.5 (Why a fixed order bound matters). One arbitrary permutation may have cycles of every finite length and destroy Replacement at $\aleph_\omega$. An involution has only fixed-point and two-cycle types, so it never does. A finite tuple of commuting involutions acts through a finite elementary abelian 2-group and has the same unconditional preservation property.

[write] **Prior in the repository at the finite cutoff, 4 October 2026, batch 93.** At $\kappa = \omega$, [corollary 15.4](#) and the example above are Corollary 58.2 of Part VIII of birthday-cutoffs-and-hereditary-sets (hset:ns:cor:finitegroup: every action of a finite pure group preserves Replacement, in particular one permutation of finite order), and that Part’s Question 65.4 (hset:ns:q:generators) notes that commuting involutions generate a finite group. Part VIII was written after this manuscript’s pin (Section 12.4); the proofs are independent. For $\kappa > \omega$ the corollary is immediate from [theorem 14.2](#), as the proof shows.

Remark 15.6 (Sharpness persists with faithful actions). For uncountable-cutoff counterexamples one can require the $\mathbb{Z}^r$ -action to be faithful. Reserve a countable regular $\mathbb{Z}^r$ -orbit, and use the cycle construction on the remaining κ atoms. Its stabilizer type differs from the finite-cycle types, so the bad small blocks are unchanged. At the finite cutoff a regular orbit already witnesses failure. Faithfulness therefore does not improve [theorem 15.3](#).

16 Commuting injections and the role of invertibility

Fix nonnegative integers s, t with $s + t > 0$. Consider pairwise commuting atom maps

$$p_1, \dots, p_s, f_1, \dots, f_t,$$

where the p_i must be permutations and the f_j need only be injections. Thus the maps give an action of the commutative monoid $\mathbb{Z}^s \times \mathbb{N}^t$ in which every acting map is injective. The classification depends on t , rather than on $s + t$.

16.1 One injection over a finitely generated abelian action

Let $H = \mathbb{Z}^s$ act through the p_i , and let f be one commuting injection. The case $s = 0$ is included by taking H trivial.

Lemma 16.1 (Orbit quotient and its normal forms). *A connected structure $(C; p_1, \dots, p_s, f)$ is determined, up to labelled isomorphism, by the following data:*

- (i) a subgroup $K \leq H$;
- (ii) a quotient shape: a one-sided ray, a two-sided chain, or a cycle of some positive length n ;
- (iii) in the cycle case, an element $v \in H/K$.

In particular there are countably many possible connected types.

Proof. Commutation makes f equivariant for the H -action. It sends the orbit $H \cdot a$ onto the orbit $H \cdot f(a)$. The induced map on the set of H -orbits is injective: if $f(b) = h \cdot f(a)$, then $f(b) = f(h \cdot a)$, so $b = h \cdot a$.

Moreover $\text{Stab}_H(a) = \text{Stab}_H(f(a))$. One inclusion follows from equivariance. If $h \cdot f(a) = f(a)$, injectivity of f gives $h \cdot a = a$, proving the other inclusion. Stabilizers are constant along H -orbits because H is abelian. Thus every fiber in a connected quotient component is a copy of the same $Q = H/K$.

The quotient graph of f is connected and has one outgoing edge and at most one incoming edge at each vertex. It is a finite cycle, a ray, or a two-sided chain. Along each edge, f gives an equivariant bijection between two copies of Q .

For a ray choose a point a_0 in its initial fiber and put $a_i = f^i(a_0)$. The map

$$Q \times \mathbb{N} \longrightarrow C, \quad (q, i) \longmapsto q \cdot a_i$$

identifies the structure with the normal form in which H translates the first coordinate and $f(q, i) = (q, i + 1)$. For a two-sided chain, choose a point in one fiber and use the unique inverse fiber maps to define a_i for negative i as well. The same formula gives $Q \times \mathbb{Z}$ with the shift in its second coordinate. These choices only prove existence of an isomorphism; no canonical internal choice is being asserted.

For a quotient cycle of length n , choose a_0 and set $a_i = f^i(a_0)$ for $0 \leq i < n$. There is a unique $v \in Q$ with $f^n(a_0) = v \cdot a_0$. In the resulting coordinates,

$$f(q, i) = \begin{cases} (q, i + 1), & i < n - 1, \\ (q + v, 0), & i = n - 1. \end{cases} \quad (16.1)$$

Changing the chosen point in a fiber does not change v , since translations commute. Moving the starting fiber around the cycle also preserves the same n -step translation. Conversely the displayed data construct the structure and distinguish its labelled isomorphism type.

There are countably many K by the finite-generation argument in [lemma 15.1](#), countably many possible n , and countably many cosets v for each K . This proves countability of the type set. There are infinitely many types already among the finite cycles with all p_i trivial. $\square$

The element v is the residual translation after one turn around the cycle, often called its holonomy. No such datum survives on a ray or a chain because the successive choices of fiber coordinates remove each edge translation without encountering a cycle.

16.2 Two noninvertible directions encode a real

We next build continuum many connected types of two commuting injections. A slightly stronger construction than is needed makes both individual maps look as simple as possible.

For $S \subseteq \mathbb{N}$, define an integer sequence d_S on $\mathbb{Z}$ by

$$d_S(m) = \begin{cases} 1, & m < 0, \\ 3, & m = 0, \\ 1 + \mathbf{1}_S(m-1), & m > 0. \end{cases} \quad (16.2)$$

Let $h_S(0) = 0$ and extend uniquely to all integers by

$$h_S(m+1) = h_S(m) - d_S(m). \quad (16.3)$$

The function is strictly decreasing, tends to $+\infty$ toward the left, and tends to $-\infty$ toward the right. Set

$$U_S = \{(m, n) \in \mathbb{Z}^2 : n \geq h_S(m)\}, \quad F(m, n) = (m+1, n), \quad G(m, n) = (m, n+1). \quad (16.4)$$

Lemma 16.2 (Staircase rigidity). *For every $S \subseteq \mathbb{N}$, $F, G : U_S \rightarrow U_S$ are commuting proper injections and their joint graph is connected. The unary components of each map are all one-sided rays. As structures with the two fixed function labels,*

$$(U_S; F, G) \cong (U_T; F, G) \iff S = T.$$

Proof. Since h_S is decreasing, moving one step to the right preserves the defining inequality in (16.4); moving one step upwards does so as well. Both maps are restrictions of coordinate translations, so they are injective and commute. The point $(m, h_S(m))$ has no G -predecessor. At each column the strict drop leaves points with no F -predecessor. Thus both maps are nonsurjective.

Given two points, choose an integer pair coordinatewise above both. It belongs to U_S and is reached from each point by positive horizontal and vertical steps. This proves connectedness of the underlying undirected graph.

Each fixed column is a vertical ray. For a fixed height n , strict monotonicity and the two limits of h_S imply that

$$\{m \in \mathbb{Z} : n \geq h_S(m)\} = \{m \in \mathbb{Z} : m \geq m_n\}$$

for one integer m_n . Hence every horizontal component is a ray too.

Suppose $\phi : U_S \rightarrow U_T$ is a labelled isomorphism. Preservation of each edge gives

$$\phi(z + e_i) - (z + e_i) = \phi(z) - z \quad (i = 1, 2)$$

whenever the edge occurs. Connectedness makes the displacement constant. Thus $\phi(z) = z + (a, b)$ for some integers a, b , and surjectivity implies

$$h_T(m+a) = h_S(m) + b \quad (m \in \mathbb{Z}).$$

Taking consecutive differences gives $d_T(m+a) = d_S(m)$. The drop 3 occurs exactly at $m = 0$ for every code, forcing $a = 0$. The normalization $h_S(0) = h_T(0) = 0$ forces $b = 0$. Finally the drops for $m > 0$ recover every bit of S , so $S = T$. Equality of the codes gives the identity isomorphism in the other direction. $\square$

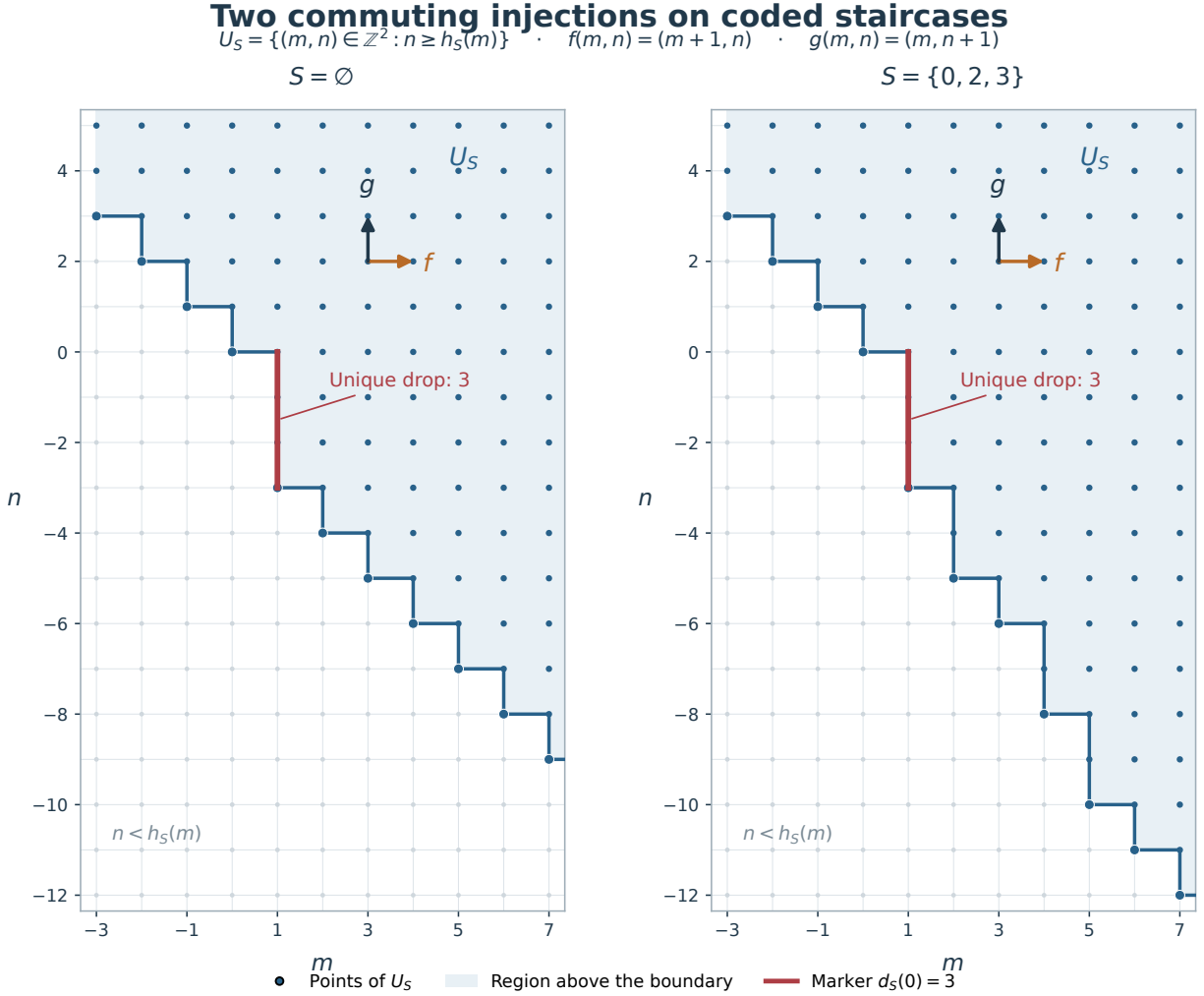


Figure 1: Finite windows in two of the infinite lattice structures U_S . Every lattice point on or above the indicated staircase belongs to U_S . The two coordinate injections commute. The unique height drop of 3 fixes an isomorphism's horizontal translation, while the successive drops of 1 or 2 encode the bits of S . Only the displayed windows are finite; the definitions and the proof concern all of $\mathbb{Z}^2$.

This is a classification of labelled graphs. Swapping the two operation names is not permitted. It is also a statement about countably infinite structures: any fixed finite window sees only finitely many bits, which is why the verification program is explicitly described as a finite check rather than a proof of the continuum claim.

Theorem 16.3 (Complete mixed-signature threshold). *Let $s, t \geq 0$ with $s + t > 0$. For every infinite κ and every A with $|A| \geq \kappa$, universal preservation of expanded Replacement for pairwise commuting s permutations and t additional injections is equivalent to*

$$\begin{cases} \text{cf}(\kappa) > \omega, & t \leq 1, \\ \text{cf}(\kappa) > \mathfrak{c}, & t \geq 2. \end{cases} \quad (16.5)$$

Proof. When $t = 0$, apply [theorem 15.3](#). When $t = 1$, [lemma 16.1](#) gives countably many types. The positive bound follows from [theorem 14.2](#). At uncountable countable-cofinality cutoffs, the distinct-cycle construction remains available because an injection name is allowed to act bijectively. At $\kappa = \omega$, a shift or a ray gives an infinite component. This proves sharpness for $t = 1$.

When $t \geq 2$, there are at most $\mathfrak{c}$ countable structures in the fixed finite language. This proves the positive bound. By [lemma 16.2](#) there are already $\mathfrak{c}$ distinct connected types for two commuting injections. The block construction of [lemma 14.4](#), with every other name the identity, gives failure at each uncountable cutoff with $\text{cf}(\kappa) \leq \mathfrak{c}$. At the finite cutoff one ray suffices, or one may use a single staircase component. This proves the remaining boundary case. $\square$

16.3 Individually preserving maps with a joint failure

Theorem 16.4 (Two commuting proper lifts with incompatible joint Replacement). *Suppose $\kappa > \omega$, $|A| \geq \kappa$, and $\text{cf}(\kappa) \leq \mathfrak{c}$. There are commuting proper injections $f, g : A \rightarrow A$ such that:*

- (i) *each of j_f, j_g is an amenable elementary self-embedding of the base model;*
- (ii) *each one-symbol expansion satisfies full expanded Replacement;*
- (iii) *the expansion naming both maps fails expanded Replacement;*
- (iv) *the fixed objects of either lift are exactly the pure objects.*

Proof. Reserve one staircase code first. Put $\delta = \text{cf}(\kappa)$ and choose distinct codes S_i for $i < \delta$ from the remaining $\mathfrak{c}$ codes, together with infinite $\lambda_i < \kappa$ whose cardinal sum is κ . Partition κ atoms into λ_i copies of U_{S_i} for each i . If $|A| = \kappa$, use all of A for these blocks. If $|A| > \kappa$, partition the remaining atoms into copies of the reserved staircase. This is possible because every staircase is countably infinite and the remainder can have cardinality $|A|$.

Let f, g be the coordinate injections on each copy. They commute and are both proper. Every unary component of either map is a ray. Thus, for each unary reduct, its sole nonempty type block is A , of size at least κ , and its small-type part is empty. The uncountable-cutoff profile criterion proves (ii). For the two-symbol structure, the block associated with S_i has size $\lambda_i < \kappa$, and their union has size κ . The same criterion proves (iii). Part (i) is [proposition 13.4](#).

For (iv), each unary atom map is a disjoint union of rays. There is no nonempty subset $K \subseteq A$ with $f[K] = K$: an atom in such a set would have predecessors of all finite lengths in its ray, contradicting its finite distance from the root. If $j_f(x) = x$, [\(13.2\)](#) gives $f[\ker(x)] = \ker(x)$, so $\ker(x) = \emptyset$. Conversely every pure object is fixed. The argument for g is identical. $\square$

The hypotheses concern a fixed action as well as a universal bound. At a cutoff such as $\aleph_\omega$, some single maps already fail Replacement, but the particular two maps in [theorem 16.4](#) do not. Their joint geometry creates the failure. At $\aleph_1$, the example works in every ambient universe with Choice, since $\aleph_1 \leq \mathfrak{c}$.

17 The exact atom kernel of definable closure

The profile proof gives a uniform bound for unique outputs. For commuting permutations this bound is exact in a strong sense, and the result remains valid when $\kappa = \omega$.

17.1 What is being measured

Fix commuting permutations $\mathbf{p} = (p_1, \dots, p_r)$, let $G = \mathbb{Z}^r$, and let $\bar{a}$ be a finite tuple of internal parameters. Put

$$P = \bigcup_i \ker(a_i), \quad G \cdot P = \{g \cdot b : g \in G, b \in P\}, \quad H(\bar{a}) = G \cdot P \cup D_\kappa(\mathbf{p}).$$

We use the notation

$$\mathcal{K}(\text{dcl}(\bar{a})) = \bigcup \{\ker(y) : y \text{ is uniquely definable from } \bar{a}\}$$

only in the scheme sense of [remark 13.1](#). An equality with $H(\bar{a})$ abbreviates two assertions: every fixed uniquely defining formula has its output kernel contained in $H(\bar{a})$, and every atom of $H(\bar{a})$ occurs in an explicitly exhibited definable object's kernel. No uniform satisfaction predicate is assumed.

An atom occurring in a definable set need not itself be definable. For example, a unique two-element atom block can be definable even though its two atoms are interchanged by an automorphism. The object under study is the kernel of definable objects, not just the atom-valued part of ordinary definable closure.

Theorem 17.1 (Exact definable-kernel formula). *For every infinite κ , every finite tuple of commuting atom permutations, and every finite tuple $\bar{a}$ of internal parameters,*

$$\mathcal{K}(\text{dcl}(\bar{a})) = G \cdot \bigcup_i \ker(a_i) \cup D_\kappa(\mathbf{p}). \quad (17.1)$$

Allowing additional pure parameters does not enlarge this atom kernel.

Proof. Every atom on the right occurs in a definable object. For a fixed group word g , the object $j_g(a_i)$ is definable from a_i . Negative letters are expressed by the unique inverse of the corresponding permutation lift. Its kernel is $g[\ker(a_i)]$. This exhibits every atom of $G \cdot P$ without requiring the entire saturation to be an internal set.

For $K \leq G$, the block

$$B_K = \{b \in A : \text{Stab}_G(b) = K\}$$

is parameter-free definable as a class by [lemma 15.2](#). Whenever $|B_K| < \kappa$, it is an internal set uniquely specified by

$$\text{Set}(y) \wedge \forall b (b \in y \longleftrightarrow \text{At}(b) \wedge \text{Stab}_G(b) = K).$$

The explicit set condition matters when the block is empty: it excludes atoms, which also have no members. Thus every atom of every small block occurs in a parameter-free definable object's kernel, proving the lower inclusion.

No uniquely defined object introduces an atom outside the right side. Suppose y is uniquely defined from $\bar{a}$ and $b \in \ker(y) \setminus H(\bar{a})$. Its orbit C is disjoint from P , and its entire type block is not small.

If $\kappa > \omega$, the saturation $G \cdot P$ has size below κ . The large block of b contains at least κ countable components, so one can interchange C with an isomorphic component avoiding $G \cdot P \cup \ker(y)$. This atom permutation commutes with all the names, fixes $\bar{a}$, and sends b outside the old kernel of y . Its lift contradicts uniqueness. In this argument one does *not* protect all of D_κ pointwise: it may be large, and it is unnecessary to protect it.

Suppose instead that $\kappa = \omega$, so P and $\ker(y)$ are finite. If C is infinite, its transitive abelian action is G/K . Translation by an element of G/K commutes with the entire named action and can send b to any prescribed point of C . Choose a point outside the finite set $\ker(y)$, translate on C , and act identically on every other component. This fixes $\bar{a}$ and changes y , again a contradiction.

If C is finite, its type block must be infinite because $b \notin D_\omega$. Infinitely many copies of its fixed finite type are available. Swap C with a copy avoiding the finitely many components meeting P or $\ker(y)$. This supplies the same contradiction.

Every atom permutation used above fixes all pure objects. Therefore arbitrary additional pure parameters leave the upper bound unchanged, and the lower inclusion already needed none. This proves the final assertion. $\square$

[write] Question 10.1 for commuting permutations, 4 October 2026, batch 93. For commuting permutations the component closure $\text{cl}_p(P)$ of (5.4) is the orbit saturation $G \cdot P$, so [theorem 17.1](#) says that the bound of Part I's Question 10.1 ([Question 10.1](#)), component closure plus D_κ , is exactly the atom kernel of definable closure, at every cutoff. A dated note after [Question 10.1](#) records this and what stays open.

Corollary 17.2 (A single definable bound when the hull is small). *The class $H(\bar{a})$ has a first-order definition from $\bar{a}$. If $|H(\bar{a})| < \kappa$, it is itself one internal definable set of atoms, containing the kernels of all objects definable from $\bar{a}$.*

For $\kappa > \omega$, expanded Replacement holds exactly when the parameter-free definable kernel is smaller than κ .

Proof. The first term of $H(\bar{a})$ is defined by finite-word reachability from the atom kernels of the parameters. To define D_κ uniformly, let a finite integer matrix code a subgroup of G , and ask whether there is an internal set containing exactly the atoms with that stabilizer. An atom belongs to D_κ exactly when it belongs to one of these complete internal blocks. This quantifies over matrices as pure sets and over the individual block witnesses; it does not assume that all witnesses have already been collected. If the resulting class has size below κ , its external extension is an internal set and is uniquely defined by this formula.

With no parameters, [theorem 17.1](#) gives precisely D_κ . The uncountable-cutoff equivalence is now [theorem 14.2](#). For a nonempty parameter tuple, the orbit saturation of its kernel is always small at an uncountable cutoff, so Replacement also makes its full hull small. $\square$

At the finite cutoff the last equivalence requires an additional clause. A single regular infinite $\mathbb{Z}'$ -orbit has empty D_ω and no parameter-free definable atom kernel, yet Replacement fails once an atom is admitted as a parameter and all its iterates are collected. The orbit-finiteness condition in [theorem 14.2](#) is therefore indispensable.

Example 17.3 (Definable kernels can be large without a collecting set). In the countable-cofinality cycle example, each B_n is parameter-free definable. Their union has size κ and cannot be an internal set. Formula (17.1) records exactly this external kernel, while making no assertion that the model contains a set of all parameter-free definable objects. The failed Replacement instance is what would otherwise perform the forbidden collection.

Example 17.4 (Why injections require a correction at ω). Let A be a single countably infinite ray and let one injection be its successor map. At $\kappa = \omega$, its entire ray block is infinite, so $D_\omega = \emptyset$. Nevertheless the root is the unique atom without a predecessor and is parameter-free definable. Every finite successor of the root is then definable. Thus the parameter-free definable kernel is all of A . This disproves the unchanged version of (17.1) for injections and explains the permutation hypothesis in the theorem.

18 A group-theoretic threshold for named automorphisms

Commutation is one possible relation among named automorphisms. A fixed finitely generated group provides a common framework for arbitrary such relations. Let $G = \langle s_1, \dots, s_r \rangle$ and let $\rho : G \rightarrow \text{Sym}(A)$ be an external action. Name the canonical lifts of $\rho(s_1), \dots, \rho(s_r)$ in $M_\kappa(A)$. Relations in G hold for these lifts because canonical lifting respects composition. Neither faithfulness of the action nor finite presentation of G is assumed.

Write

$$\tau(G) = |\{H : H \leq G\} / \text{conjugacy}|. \quad (18.1)$$

This counts conjugacy classes of subgroups, rather than conjugacy classes of elements. It measures how many different transitive atom components the named action can distinguish.

Lemma 18.1 (Transitive component types). *The possible connected-component types for the named generators of G are exactly the transitive G -sets G/H , with H taken up to conjugacy.*

Proof. The component containing a is its orbit Ga . If $H = \{g \in G : ga = a\}$, the map $gH \mapsto ga$ is an equivariant bijection $G/H \rightarrow Ga$. Changing the basepoint conjugates its stabilizer. Conversely, an equivariant bijection $G/H \rightarrow G/K$ taking H to gK identifies stabilizers and therefore gives $H = gKg^{-1}$. A bijection preserving the named generators preserves their inverses and every word in them, so it is exactly a G -equivariant bijection. $\square$

Because G is finitely generated, it is countable. All its transitive components are consequently finite or countably infinite. There is no definability assumption hidden in the use of subgroup types. The pure set of codes for connected structures in the finite generator language, with domain a positive finite ordinal or $\mathbb{N}$, exists in $M_\kappa(A)$, as does its quotient by isomorphism. One can restrict that pure set to the codes satisfying the relations of G . The group, its multiplication table, and a choice of words representing its elements are pure set parameters if needed.

For $\kappa > \omega$, a component code is recognized at an atom by an internal enumeration graph preserving every generator, whose image contains that atom and is closed under incoming and outgoing labelled edges. Such a graph has at most countably many atoms in its kernel. Finite components have finite enumeration graphs even when $\kappa = \omega$. Thus the type-block formula underlying Theorem 14.2 applies without using expanded Replacement to construct its witnesses. This point matters: simply counting abstract subgroup types would not, by itself, supply an internal failed instance of an axiom scheme.

Theorem 18.2 (Exact group robustness threshold). *Let G be a finitely generated group, with a fixed finite generating tuple. Every action of G on A preserves Replacement after the canonical lifts of those generators are named if and only if*

$$\begin{cases} G \text{ is finite,} & \kappa = \omega, \\ \text{cf}(\kappa) > \tau(G), & \kappa > \omega. \end{cases} \quad (18.2)$$

In particular, the answer is independent of the finite generating tuple.

Proof. Suppose first $\kappa > \omega$ and $\text{cf}(\kappa) > \tau(G)$. An action has at most $\tau(G)$ type blocks. The union of its blocks of cardinality below κ therefore has cardinality below κ , so Theorem 14.2 proves preservation.

For the converse put $\delta = \text{cf}(\kappa)$ and assume $\delta \leq \tau(G)$. Choose pairwise nonconjugate subgroups H_i for $i < \delta$, omitting G itself. Omitting one type does not reduce the available infinite cardinality. Choose infinite cardinals $\lambda_i < \kappa$ with

$$\sum_{i < \delta} \lambda_i = \kappa.$$

For regular κ , take $\delta = \kappa$ and every $\lambda_i = \aleph_0$; for singular κ , take a cofinal family of infinite cardinals below κ .

On a subset of A of cardinality κ , place λ_i disjoint copies of G/H_i for each $i < \delta$. Fix the remaining atoms. The selected type block B_i has size λ_i , since each G/H_i is nonempty and countable, and the selected blocks together have size κ . The remainder has trivial type and cannot enlarge any B_i . Thus the small-type union violates the criterion in Theorem 14.2.

Explicitly, a pure sequence of the selected type codes is an internal parameter. The formula sending i to the set of all atoms having the i th type has a unique internal value B_i for every $i < \delta$. Replacement for this function would produce a range set whose kernel contains their union of size κ . This is the required failed instance, on the pure internal domain δ .

At $\kappa = \omega$, a finite group has only finitely many transitive types, all finite. Its union of finite type blocks is finite, so the finite-cutoff clause of Theorem 14.2 applies. If G is infinite, place one regular G -orbit on a countable atom subset and fix the remainder. From an atom in that orbit, endpoints of finite words in the generators and inverses are uniquely definable. Every finite path graph exists internally. Replacement on the pure set of words would collect the entire infinite orbit into an internal atom set, impossible at the finite-kernel cutoff. $\square$

[write] **The finite cutoff in the repository, 4 October 2026, batch 93.** The clause for $\kappa = \omega$ also follows from Part VIII of birthday-cutoffs-and-hereditary-sets, written after this manuscript's pin: by its Proposition 62.1 (hset:ns:prop:finitewords) the generator names define the uniform action, and its Theorem 58.1 (hset:ns:thm:full) then requires finite orbits and a finite exceptional set, which every action has exactly when G is finite. The clause for $\kappa > \omega$, the threshold $\text{cf}(\kappa) > \tau(G)$, is new here.

18.1 The three possible type counts

Proposition 18.3 (Cardinality trichotomy). *For a countable group G , the cardinal $\tau(G)$ is finite, $\aleph_0$, or $\mathfrak{c}$.*

Proof. Represent a subgroup by its characteristic function in 2^G . The subgroup conditions are closed: failure to contain the identity or to be closed under products and inverses has a finite witness. Thus the space of subgroups is a closed subspace of the compact metrizable space 2^G . If uncountable, it

has a nonempty perfect subset and therefore cardinality $\mathfrak{c}$. For clarity, the perfect-set step follows by repeatedly splitting basic clopen neighborhoods around distinct condensation points; the resulting binary tree embeds $2^{\mathbb{N}}$. Only countably many points can fail to be condensation points, since the space has a countable base. Every conjugacy class of subgroups is countable, being an orbit of G . Consequently an uncountable subgroup space has $\mathfrak{c}$ conjugacy classes as well. The remaining possibilities are finite or countably infinite. $\square$

For an infinite finitely generated abelian group the count is $\aleph_0$: every subgroup is finitely generated, so a countable collection of finite tuples lists them, and infinitely many finite-index subgroups give distinct types. The same upper-bound argument applies whenever every subgroup of a countable group is finitely generated. At the other extreme, the free group $F(a, b)$ has $\mathfrak{c}$ types. The elements $a^n b a^{-n}$, $n \in \mathbb{Z}$, freely generate a subgroup with a countable free basis. Indeed, after combining adjacent equal basis indices, expansion of a nonempty reduced word leaves nonzero powers of a between consecutive nonzero powers of b , so it cannot reduce to the identity. Distinct subsets of this basis generate distinct subgroups, giving $\mathfrak{c}$ subgroups; quotienting by the countable conjugacy classes preserves that cardinality.

Remark 18.4 (Infinite groups with finitely many types). The finite alternative cannot be identified with finiteness of G . Smith records finitely generated infinite simple groups constructed by Ivanov and Ol'shanskii in which all proper nontrivial subgroups are conjugate [11, p. 69]. Such a group has $\tau(G) = 3$. We use this published existence statement and do not reproduce that construction. Theorem 18.2 then says that every named generator action preserves Replacement at every uncountable cutoff, while a regular orbit still defeats it at $\kappa = \omega$. This is why the finite-cutoff clause must be stated separately.

[write] An imported existence theorem, 4 October 2026, batch 93. This remark is not proved here: it imports the existence of a finitely generated infinite group all of whose proper nontrivial subgroups are conjugate, a construction of Ivanov and Ol'shanskii [12] that the manuscript read only through Smith [11]. At the write, the first page of [11], freely visible on the publisher's site, was checked: it attributes to Ivanov and Ol'shanskii finitely generated infinite simple groups with all proper nontrivial subgroups conjugate. The primary source (paywalled) was not consulted, by the manuscript or at the write. Given the existence statement, $\tau(G) = 3$ is immediate: the classes are $\{1\}$, G , and the single class of proper nontrivial subgroups, which is nonempty because an infinite simple group is not cyclic. What depends on the import is only the claim that the two clauses of theorem 18.2 genuinely differ, that is, that an infinite finitely generated group can have finite $\tau(G)$; theorem 18.2 and proposition 18.3 are proved without it. Question 23.13 records the missing primary check.

The topology of subgroup spaces has a much deeper established theory, including a complete analysis for countable abelian groups in [10]. Only the elementary cardinality fact above is needed here.

These results translate group structure into preservation properties of a set-theoretic scheme. They do not classify groups with countably many subgroups or claim to solve the existing classification problems about subgroup spaces. Their role is to identify exactly which invariant of a fixed finitely generated action controls this family of kernel models.

19 Uniformly evaluating an infinite family of automorphisms

There are two different ways to name an action of a countable group. In the first, the language has a separate unary symbol J_g for each $g \in G$. Every first-order formula still contains only finitely

many of these symbols. In the second, one binary symbol E uniformly evaluates the action:

$$E(g, x) = j_{\rho(g)}(x) \quad (g \in G). \quad (19.1)$$

Set $E(g, x) = x$ when $g \notin G$ to make it total, and allow the pure set G as an internal parameter. Now a single formula can quantify over all $g \in G$. Although the individual maps in the two descriptions are identical, their Replacement schemes need not agree.

Proposition 19.1 (Separate names for a locally finite group). *If G is locally finite, every action of G preserves Replacement in the language of separate unary names at every infinite cutoff. For countable G , the converse holds for universal preservation at $\kappa = \omega$.*

Proof. A formula uses finitely many names. They generate a finite subgroup when G is locally finite, so Theorem 18.2 applies to that formula. Conversely, if a finite subset generates an infinite subgroup K , a regular G -orbit on countably many atoms restricts to an infinite K -orbit. The finite-word endpoint instance using those finitely many names fails at $\kappa = \omega$. $\square$

Proposition 19.2 (Uniform group profile). *For a countable group acting with a uniform evaluation name, the component-profile criterion of Theorem 14.2 holds with components equal to full G -orbits and types equal to conjugacy classes of stabilizers. Consequently the universal robustness criterion (18.2) holds for every countable group in this language, without a finite-generation assumption.*

Proof. Use pure codes for actions of the pure set G on a positive finite ordinal or $\mathbb{N}$. An internal enumeration graph h recognizes a component by requiring $h(g \cdot d) = E(g, h(d))$ for every $g \in G$ and every code element d . Transitivity of the code and inclusion of the chosen atom identify the image with its whole orbit. This is one formula precisely because evaluation is uniform. The graph is internal at uncountable cutoffs, and for finite orbits at the finite cutoff.

The necessity argument therefore collects the small type blocks just as before. At $\kappa = \omega$, the uniformly defined function $g \mapsto E(g, a)$ first forces every orbit to be finite. For sufficiency, saturate the kernels of the input set and parameters under G , and adjoin the small-type part. This protected union has size below κ : countable saturation preserves that bound when $\kappa > \omega$, while saturation of finitely many atoms is finite when all orbits are finite. A fresh equivariant component swap fixes the protected union and changes any purported unique output using an atom outside it. Its lift preserves E , so uniqueness confines all output kernels to the protected union. External Replacement then produces an internal range. Finally the sharp type-block construction in Theorem 18.2 uses only countability of G and applies unchanged. $\square$

[write] Prior in the repository at the finite cutoff, 4 October 2026, batch 93. At $\kappa = \omega$ both propositions are in Part VIII of birthday-cutoffs-and-hereditary-sets, written after this manuscript's pin: [proposition 19.1](#) is its finite-symbol principle, Proposition 62.2 (`hset:ns:prop:language`), together with Corollary 58.2 (`hset:ns:cor:finitigroup`); [proposition 19.2](#) is its Theorem 58.1 (`hset:ns:thm:full`), stated there for a uniformly named action of an arbitrary pure set-sized group, its exceptional set X_{Act} being D_ω once all orbits are finite. The proofs are independent. For $\kappa > \omega$ both propositions are new here; [proposition 19.2](#) answers in part that Part's Question 65.2 (`hset:ns:q:kernelbounds`), whose dated note lists "actions of arbitrary pure groups at uncountable κ " as open: for countable groups and full Replacement the criterion is the profile criterion with full orbits, and the universal threshold is $\text{cf}(\kappa) > \tau(G)$. Uncountable groups and the bounded levels $\lambda < \kappa$ of that question remain open.

19.1 A separation using only two-atom orbits

Take

$$G = \bigoplus_{n < \omega} C_2,$$

coded as the pure set of finite subsets of $\mathbb{N}$ with symmetric difference. This group is locally finite. For each nonempty $S \subseteq \mathbb{N}$, its nonzero character

$$\chi_S(F) = |F \cap S| \pmod{2} \quad (19.2)$$

defines a transitive action on $\{0, 1\}$ by $F \cdot \varepsilon = \varepsilon + \chi_S(F) \pmod{2}$. Distinct characters have distinct kernels: an index-two subgroup has a unique nonzero quotient character to C_2 . Because G is abelian, these kernels are never identified by conjugacy. There are therefore $\mathfrak{c}$ different two-point transitive types.

Theorem 19.3 (An amenable uniform name can change Replacement). *For every infinite κ with $\text{cf}(\kappa) \leq \mathfrak{c}$, there is an action of $G = \bigoplus_{n < \omega} C_2$ on A , all of whose orbits have size at most two, such that:*

1. *naming every individual lift separately preserves Replacement;*
2. *naming the uniform evaluation function E violates Replacement;*
3. *E is amenable: its restriction graph on every internal set of input pairs belongs to $M_\kappa(A)$.*

For $\text{cf}(\kappa) > \mathfrak{c}$, every uniformly named action of this group preserves Replacement.

Proof. At $\kappa > \omega$, let $\delta = \text{cf}(\kappa)$ and choose distinct nonempty codes S_i , $i < \delta$. Choose small infinite multiplicities λ_i with sum κ . Put λ_i copies of the two-point action of χ_{S_i} in a block B_i , and fix the remaining atoms. Then $|B_i| = \lambda_i < \kappa$, whereas the union of the B_i has size κ . At $\kappa = \omega$, take one copy of each of countably many distinct nonzero characters; each B_i then has size two.

In the uniform language the formula

$$\text{At}(a) \wedge \forall g \in G (E(g, a) = a \iff \chi_S(g) = 0) \quad (19.3)$$

recognizes the type of χ_S . Its arithmetic character computation is pure and uniformly definable. A pure sequence of the selected S_i therefore defines the unique internal block B_i from i . Any Replacement range for these values would have a kernel reaching the cutoff. This proves failure even though no orbit is infinite. Preservation for the separate-name expansion follows from Proposition 19.1.

For amenability, let w be an internal set of input pairs and put $P = \ker(w)$. The kernel of any value $E(g, x)$ appearing on w is contained in the saturation GP . Since every orbit has at most two atoms, $|GP| \leq 2|P| < \kappa$, with the evident finite interpretation. The externally constructed restriction graph has kernel contained in GP and is therefore internal. Finally $\tau(G) = \mathfrak{c}$ by the two-point types and the general continuum upper bound, so Proposition 19.2 gives the asserted positive threshold. $\square$

The witness here does more than expose an unavailable infinite orbit. All individual named maps are commuting involutions, every full orbit has at most two atoms, and the uniform evaluation class is itself amenable. The failed instance uses the ability of one formula to test all group indices and distinguish their stabilizers. Replacing infinitely many separate names by a uniform evaluator is thus a substantive change to an axiom scheme.

[write] **Relation to Part VIII of the surreal report, 4 October 2026, batch 93.** Theorem 62.3 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:thm:evaluator`), written after this manuscript's pin, makes the same separation for the same group at $\kappa = \omega$: countably many separately named involutions keep Replacement, and one uniform evaluator $T(i, a, b)$ destroys it. Its witness is the regular action of $\bigoplus_{\omega} C_2$ on atoms labelled by the group, whose orbit is infinite; the evaluator with one atom parameter then collects infinitely many single atoms, so already R_1 (Replacement for outputs with at most one atom) fails.

[Theorem 19.3](#) differs in four ways: every orbit has at most two atoms, so the failure comes from distinguishing stabilizers rather than from an infinite orbit; it holds at every κ with $\text{cf}(\kappa) \leq \mathfrak{c}$, not only at ω ; the evaluator is amenable; and the converse for $\text{cf}(\kappa) > \mathfrak{c}$ is included. At $\kappa = \omega$ the two witnesses are also separated by the graded levels of that Part: for the witness above, every orbit is finite, and at most one atom (a fixed atom, when it is the only fixed one) has an orbit of size at most one under the centralizer of the action, because the transposition of a two-atom orbit commutes with the action; so by its Theorem 59.1 (`hset:ns:thm:graded`; its language is interdefinable with the evaluator E by its Proposition 62.4) R_1 holds, while R_2 fails, since each of the infinitely many two-atom orbits is the only orbit of its type and so has centralizer orbits of size two. This last observation is the write's, by that theorem, and is not in either manuscript.

Proposition 19.4 (Uniform evaluation for a finitely generated group). *If G is finitely generated, its uniform evaluation function is definable from the finitely many generator names and a pure parameter. Adjoining that evaluator therefore does not change whether Replacement holds.*

Proof. Choose a pure function assigning to every $g \in G$ a finite word in the generators and their inverses representing g . To define $E(g, x)$, ask for the endpoint of a finite path starting at x and following that word. A positive letter applies its named lift; a negative letter requires the named lift of the next value to equal the previous value. These steps are unique because the lifts are automorphisms.

Every such finite path graph exists internally. Its finitely many values have kernels which are images of $\ker(x)$ under group elements, so their union has size below κ , even when κ is singular or equals ω . The defining formula thus uses only the finite generator language and the pure word-selector parameter; it assumes no expanded Replacement. Substituting this definition into an evaluation-language Replacement instance translates it to an instance in the generator language. Conversely each generator is an evaluation at one fixed pure group element. $\square$

This proposition specifies the boundary of the preceding counterexample. The issue is not merely that an index variable occurs in the notation. For finitely generated actions a finite path can evaluate every index using fixed named operations, and its existence is already guaranteed by the kernel bound. For $\bigoplus_{n < \omega} C_2$, no finite family of group elements generates the full action; a formula in separate names has access only to a finite subgroup, while the uniform predicate (19.3) compares all its indices at once.

[write] **Added 4 October 2026, batch 93.** At $\kappa = \omega$, [proposition 19.4](#) is Proposition 62.1 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:prop:finitewords`), stated there for the words of a pure free group on finitely many named permutations and written after this manuscript's pin; here it holds at every cutoff.

19.2 Finite orbits and the profinite completion

There is an exact version of this phenomenon for any countable group. Define

$$\tau_{\text{fin}}(G) = |\{H \leq G : [G : H] < \omega\} / \text{conjugacy}|.$$

Corollary 19.5 (Universal preservation for finite-orbit actions). *Restrict to actions of a countable group G whose every orbit is finite, and name their uniform evaluation function. Every such action preserves Replacement exactly under the conditions*

$$\begin{cases} \tau_{\text{fin}}(G) \text{ is finite,} & \kappa = \omega, \\ \text{cf}(\kappa) > \tau_{\text{fin}}(G), & \kappa > \omega. \end{cases}$$

At the finite cutoff the condition is equivalent to finiteness of the profinite completion $\widehat{G}$.

Proof. The uniform profile proof now uses only finite transitive types. At uncountable cutoffs its upper bound and sharp block construction apply with $\tau_{\text{fin}}(G)$. At the finite cutoff, finitely many types give a finite union of finite blocks. Infinitely many types allow one copy of each of countably many distinct nontrivial types, whose finite blocks have infinite union.

A finite-index subgroup has finitely many conjugates, since its normalizer contains it. Hence finite $\tau_{\text{fin}}(G)$ is equivalent to having finitely many finite-index subgroups. Intersect their normal cores to obtain a finite-index normal subgroup R contained in every such subgroup. Every finite quotient factors through G/R , so $\widehat{G} = G/R$ is finite. Conversely, each finite-index subgroup is the inverse image of an open subgroup of $\widehat{G}$: its finite coset action extends to the profinite completion, and one takes the stabilizer of its distinguished coset. If $\widehat{G}$ is finite there are only finitely many such subgroups. $\square$

For a finitely generated group there are only finitely many actions on each fixed finite labelled set: the images of finitely many generators determine the action. Thus $\tau_{\text{fin}}(G)$ is at most countable. The continuum-sized family of two-point types above relies on infinitely many generators and their uniform evaluation. This explains precisely why its language distinction has no counterpart for a fixed finite generating tuple.

20 Collection and reflection in the commuting cases

The preceding results concern Replacement. The relation to Glazer and Yao's research becomes clearer when Collection and reflection are kept distinct. In the present external-choice models, the general Collection and full-reflection criteria were already established in [7, Theorems 7.5 and 7.7]. We record and prove the consequence for the countable-type commuting signatures.

Theorem 20.1 (Collection and reflection for at most one injection). *Fix a finite nonempty commuting tuple of injections in which all but at most one map are permutations. In its canonical expansion of $M_\kappa(A)$, the following are equivalent:*

- (i) *full expanded Collection;*
- (ii) *for every finite family Φ of expanded formulas and every internal set w , there is an internal transitive set t with $w \subseteq t$, closed under the named functions, such that every formula in Φ is absolute between t and the full model for all parameter tuples from t ;*
- (iii) *κ is an uncountable successor cardinal.*

This is a consequence of the prior Collection/reflection criterion and the type classification proved here.

Proof. Failure at a limit cutoff. Suppose κ is a limit cardinal, with ω included. Choose a pure set w of ordinals below κ whose cardinalities are unbounded below κ . For ω , take $w = \omega$. For each $\alpha \in w$ there is an internal graph of an injection from α into the atoms: it uses only $|\alpha| < \kappa$ atoms. A set collecting such graphs for all $\alpha \in w$ would have a kernel of size at least $|\alpha|$ for every $\alpha \in w$, contrary to the cutoff. This is a failed base-language Collection instance, so adding names cannot repair it.

A common reservoir at a successor cutoff. Write $\kappa = \nu^+$, with ν infinite. There are at most countably many component types, and every component is countable. Close the kernels of an internal input domain w and its parameters under components; this uses at most ν atoms. Add every entire type block of size at most ν . There are only countably many such blocks, so the resulting component-closed set P^* still has size at most ν .

For each remaining realized type, choose ν components disjoint from P^* . Such a type has at least ν^+ components. Their union with P^* is a fixed component-closed reservoir H of size at most ν . If y is any internal witness, its kernel meets at most ν components outside P^* . For each type, match those components injectively into its reservoir. Because the full component family has size at least ν^+ , the partial matching extends to a permutation of that family; choose the corresponding component isomorphisms. The resulting atom permutation commutes with all names and fixes P^* pointwise. It carries y to a witness whose kernel lies in H while fixing every input and parameter.

For a fixed total relation on w , external Collection and Choice now supply witnesses all supported on the same H . Their collected set is internal. This proves full expanded Collection independently of the cardinality of the pure part of w .

Reflection. For a finite family of formulas, include its subformulas, expressing universal quantifiers through negation and existence. Construct externally increasing sets

$$X_n = V_{\alpha_n}(H_n) \quad (n < \omega),$$

where H_n is component-closed of size at most ν , the α_n increase, and the initial stage contains the desired parameters and elements of the prescribed internal set. Given X_n , the reservoir construction over H_n compresses all needed witnesses while fixing their parameter tuples. There are only set-many tuples in X_n and finitely many existential subformulas. External Collection gives a common rank bound, producing H_{n+1}, α_{n+1} such that each true existential instance over X_n has a witness in X_{n+1} .

The union $t = \bigcup_n X_n$ is transitive, is closed under each named function, and has kernel of size at most ν , so it is internal. Induction on the finite collection of formulas proves absoluteness between the full model and t . The existential step is precisely the witness property; the atomic step uses transitivity and closure under the named functions.

Conversely full reflection implies Collection by reflecting a totality assertion and its matrix. For a domain w and parameter tuple p , prescribe an internal set containing w itself and all the entries of p , so that these objects belong to the reflecting set t . Transitivity then puts every element of w in t . The witnesses supplied in t are genuine witnesses by absoluteness, and t itself is an internal collecting set. This completes the equivalence. $\square$

Remark 20.2 (The reservoir must grow). Reflection uses increasing atom reservoirs as well as increasing rank bounds. A single $V_\alpha(H)$ can contain its whole atom supply H as one of its elements, whereas the full kernel model has no internal set of all atoms. The growing-reservoir proof avoids incorrectly treating one fixed atom supply as a reflecting universe.

These results give several distinct possibilities. At an uncountable successor cutoff, every commuting-permutation expansion has full Replacement, Collection, and the specified full reflec-

tion. At a limit cardinal of uncountable cofinality, every such expansion still has Replacement, but Collection and full reflection fail. At a limit cardinal of countable cofinality, Replacement depends on the specific action's small-block profile, and Collection always fails.

This does not recover or strengthen the choiceless separation theorem of Glazer and Yao. Their work varies choice and reflection principles in a broader collection of models. Here the ambient construction uses Choice, and the new classification varies the named action and the atom cutoff. The connection is the common role of atom symmetries in deciding whether witnesses can be confined to one internal set.

21 Forcing stability for a fixed finite action

The universal thresholds quantify over all available actions, so they can change when a forcing extension supplies new actions. For an action already present in the ground universe, the profile theorem yields a stronger stability statement. This answers the fixed-action part of the forcing question proposed in the repository [7, Question 10.10] under an explicit hypothesis on the cutoff.

Theorem 21.1 (Preserving the cutoff preserves the verdict). *Let $V \subseteq W = V[\mathcal{G}]$ be a set-forcing extension of the ambient universe. Suppose A , an infinite cardinal κ , and a fixed finite tuple f of injections on A belong to V , with $|A|^V \geq \kappa$. Assume that the ground ordinal κ remains a cardinal in W . Form the kernel model and the canonical lifts separately in V and in W . Then the ground expansion satisfies expanded Replacement if and only if the rebuilt expansion in W does.*

Proof. The atom set and the named maps have the same elements and values in both universes. A component is determined by finite paths through these values, so each component is the same ground set in V and W . It is countable in V .

We next show that the partition into component types does not change. Fix two ground components and ground enumerations of them. Their labelled isomorphisms correspond to infinite branches through a ground tree of finite partial isomorphisms, requiring at the n th stage that the first n enumerated elements on both sides be covered. A ground isomorphism remains one in W . If there is no ground isomorphism, this tree is well-founded in V and has an ordinal-valued rank function there. The same rank function remains valid in W and precludes a new infinite branch. Thus no two previously different component types become isomorphic. Indexing types by the actual partition of components, rather than by all possible pure codes, gives the same blocks B_i in both universes.

For any ground set X , the assertion $|X| < \kappa$ is absolute under the stated cardinal hypothesis. A ground injection of X into an ordinal below κ persists and proves the forward direction. If $|X|^V \geq \kappa$, a ground injection of κ into X persists and, because κ is still a cardinal, prevents $|X|^W < \kappa$.

Consequently the family of small blocks B_i is unchanged, its union D_κ is the same ground set, and the smallness of that union is unchanged. At $\kappa = \omega$, the finiteness of every component and of D_ω is also absolute. Apply [theorem 14.2](#) in the two universes. The models have different pure sets in general, but the criterion applies to all formulas and all internal parameters in each rebuilt model, which is exactly the required equivalence. $\square$

[write] **Question 10.10, fixed-action part, 4 October 2026, batch 93.** “Question 10.10” of [7] is [Question 10.10](#) of Part I; a dated note there records this answer and what remains open ([Question 23.11](#)).

Remark 21.2 (What can still change). The theorem permits a change of $\text{cf}(\kappa)$ as long as κ remains a cardinal. If such a change crosses a universal threshold, counterexample actions available in the extension need not be ground actions. The theorem does not assert that the old proper-class model is unchanged, that each old formula has the same truth value, or that Collection is absolute under arbitrary cardinal collapses below κ . For the countable-type commuting signatures of [theorem 20.1](#), preserving whether κ is a successor cardinal additionally preserves Collection and the stated reflection scheme; forcing that preserves all cardinals is sufficient.

22 Formalization, verification, and proof dependencies

22.1 How the argument fits the existing ProveIt formalization

The inspected repository contains a formula-indexed ZF development at `SetTheory/ZF/Lean/ZF/Zf.lean`. Its declarations include `Sep_form`, `Func_form`, `Image_form`, `Repl_form`, `ZFax`, and `ZFprov` [8]. They provide a natural starting point for distinguishing a base formula language from the enlarged language used by a scheme. They do not already formalize the urelement class models, component profiles, or theorems of this article.

A useful formalization should make the signature an explicit input. A proof of Replacement for formulas in $\{\in, \text{At}\}$ must not automatically become a proof for formulas in $\{\in, \text{At}, J_1, \dots, J_r\}$. Similarly, an interface for amenability should assert the existence of individual restriction graphs and should not quietly include expanded Replacement.

Milestone	Mathematical content to certify
Syntax and interpretation	Separate atom and set predicates; finite function signatures; relativization of each schema instance.
The kernel model	Tagged universe, transitive closure, kernel operations, closure under subsets, and the fresh-atom proof of base Replacement.
Finite paths and lifts	Recursive lift, preservation of kernels, extension of a small partial injection to an atom permutation, and internal finite path graphs.
Component types	Pure countable codes, labelled isomorphisms, uniform recognition, and the fresh-component swap.
Commuting normal forms	Subgroups of $\mathbb{Z}'$, the fiber quotient for one injection, holonomy on cycles, and rigidity of the coded staircases.
Schema certificates	An explicit input domain, functional formula, and cutoff-violating kernel for each negative example.

The core mathematical proof is amenable to modular formalization: finite combinatorics establishes the type families, elementary cardinal arithmetic gives their possible unions, and the kernel model converts those unions into schema instances. The hard formal boundary is semantic bookkeeping for a definable proper-class structure. An eventual formalization may instead work over externally quantified set models with explicit satisfaction and transport each fixed instance. Either route must state its metatheory rather than conceal a truth predicate.

22.2 The supplied computational checks

The file `verify_examples.py` performs deterministic finite checks using integer arithmetic and finite sets only. Its recorded run passes 347,278 explicit checks. These cover all 256 eight-bit staircase prefixes continued by zeros, boundary decoding, coordinate closure and commutation, translated finite boundary shadows, all 15 subgroups of $(\mathbb{Z}/4\mathbb{Z})^2$, quotient-action stabilizers and equivariant maps, and all 31 nonzero characters of $(\mathbb{F}_2)^5$.

The file `make_figure.py` regenerates [figure 1](#). The picture is an exact integer-lattice visualization of the defining inequalities; it is not generated imagery. Both the vector figure and a high-resolution preview are included.

The role of these checks is limited and useful. They test the explicit finite formulas underlying the constructions and make arithmetic or indexing errors reproducible. They do not prove that all infinite staircase codes are distinct, do not verify any infinite-cardinal argument, and do not certify any instance of set-theoretic Replacement. The mathematical proofs of those claims are supplied in the text. No Lean, Coq, or other proof-assistant verification is claimed.

[write] Shipped layout and replay, 4 October 2026, batch 93. In ProveIt the two programs are `code/02-commuting-actions-verify_examples.py` and `code/02-commuting-actions-make_figure.py`, the recorded run is `data/02-commuting-actions-verification_output.txt`, and the figure is `figures/02-commuting-actions-staircases.pdf` with the preview `figures/02-commuting-actions-staircases.png`. Both programs were rerun on scratch copies at the write (Section 12.4); the report’s README gives the commands, which never write into this directory.

22.3 Dependency audit

Conclusion	Essential input	Boundary to preserve
Commuting threshold	Finite descriptions of subgroups of $\mathbb{Z}^r$; type profile	Countability of types is used with cofinality, not mere size of κ .
One-injection threshold	Injective map on permutation orbits; untwisting of fiber maps	Induced fiber maps are bijections even though the global map may not be.
Two-injection lower bound	Rigid staircase boundary; block multiplicities	Complete infinite labelled types are distinct; finite shadows are only checks.
Exact definable kernel	Parameter-free subgroup definitions and transitive translations	Kernels of definable objects differ from individually definable atoms.
General group threshold	Coset-action classification and uniformly recognized codes	Subgroups are counted up to conjugacy; the finite-cutoff case is separate.
Uniformity separation	Infinitely many index-two stabilizers; finite reducts	A binary evaluator permits index quantification unavailable to a fixed finite list of unary names.

Conclusion	Essential input	Boundary to preserve
Collection and reflection	Compression into growing component reservoirs	Arbitrary witnesses require a reservoir argument; uniqueness alone gives only Replacement.

23 Further research questions and topics

The questions below specify where the proved classifications end. They are proposed research tasks, not a claim that every formulation is a longstanding published open problem. The commuting-permutation question itself is no longer included as an open task in the setting of this article.

Question 23.1 (Atom-valued definable closure for nonabelian actions). For a finitely generated nonabelian group G , determine the atoms that are individually definable from specified parameters, as distinct from the kernel of definable objects. The equivariant automorphism group of G/H is controlled by $N_G(H)/H$. How does that normalizer quotient interact with the internal availability of complete orbit blocks? A useful first target is the infinite dihedral group, where orbit and stabilizer data remain explicit.

[write] **Answered with pure real parameters; re-scoped, 4 October 2026, batch 96.** Part III (source 03 there) answers this question with pure real parameters. For a finitely generated group G named by generators, an atom of type G/H with multiplicity $\mu_{[H]}$ is definable from pure real parameters alone exactly when $\mu_{[H]} = 1$ and $N_G(H) = H$; from finitely many atom parameters F and pure reals, exactly when it lies in GF or its residual multiplicity outside GF is 1 and $N_G(H) = H$ (corollary 28.1 and theorem 27.3); for arbitrary set parameters the same criterion is exact outside the saturation of their kernels. The interaction with complete orbit blocks is the formula $D_\kappa^{\text{pt}} = \bigcup \{A_{[H]} : \mu_{[H]}[N_G(H) : H] < \kappa\}$ for the atoms occurring in objects definable with pure reals. The infinite dihedral first target is worked out in Section 28.1. At $\kappa = \omega$, theorem 27.5 characterizes all objects definable from arbitrary parameters and arbitrary pure parameters as those fixed by every centralizer element fixing the parameters. Open: ordinary definability, without pure parameters (Question 37.4), and arbitrary set parameters inside the saturation of their kernels at $\kappa > \omega$ (Question 37.5).

Question 23.2 (Exact finite-kernel formulas for several injections). Develop the analogue of theorem 17.1 when two or more commuting injections are named and $\kappa = \omega$. The ray in example 17.4 shows that boundary points create definable atoms absent from the small-block term. Which finite layers or distinguished boundary configurations account for all such atoms in the staircase families? Can the result be stated through a finite-boundary invariant of a cancellative monoid action?

[write] **Answered with pure real parameters; open for ordinary definability, 4 October 2026, batch 96.** Part III (source 03 there) proves the analogue of theorem 17.1 for every finite tuple of injections, commuting or not, at every infinite cutoff including $\kappa = \omega$, with pure real parameters: the atom kernel of the objects definable from $\bar{a}$ and finitely many pure reals is $\text{Sat}(B) \cup D_\kappa^{\text{pt}}$ (theorem 27.1). At $\kappa = \omega$ a component type contributes to D_ω^{pt} exactly when it has finitely many copies and a finite automorphism group, even if the component is infinite (corollary 27.2). For a cancellative commutative monoid action this is the finite-boundary invariant asked for: the automorphism group of a component $S \subseteq G/H$ is its period group $\text{Per}(S)$ (corollary 29.4); for $f^m = g^n$ it is C_δ , read off the boundary word (theorem 33.4); the staircases of lemma 16.2 are rigid, so in the staircase families every atom of a type with finitely many copies is in the kernel, not only boundary points, and an atom is individually definable with pure reals exactly when its

type occurs once ([theorem 27.3](#)). For *ordinary* definability, as in [theorem 17.1](#), [proposition 27.4](#) reduces the question at $\kappa = \omega$ to the definability of rooted codes, which is open ([Question 37.4](#)). The question stays open in that form.

Question 23.3 (Relations among noninvertible generators). Classify the connected-type cardinality for injective actions of specified finitely presented commutative monoids. The free case $\mathbb{Z}^s \times \mathbb{N}^t$ is settled here. Relations such as $f^m = g^n$ can collapse two apparent directions. Determine when such relations force countably many types and when a staircase or another real-coded family survives.

[write] **Answered, 4 October 2026, batch 96.** Part III answers this question in full, by two independent proofs (its sources 03 and 04). For a finitely generated commutative monoid M acting by injections, with group completion G , cancellative image P , units $U = P \cap (-P)$ and $d = \text{rank}(G/U)$, the number of connected labelled types is $|\text{Sub}(G)|$ when G is finite, $\aleph_0$ when G is infinite and $d \leq 1$, and $\mathfrak{c}$ when $d \geq 2$, in which case a rigid real-coded family survives ([theorems 30.4 and 30.6](#)). The integer d is computable from a finite presentation ([theorem 32.1](#)). A relation $f^m = g^n$ always gives $d = 1$, hence countably many types, which are enumerated exactly by divisor and necklace sums ([theorem 33.1](#)); relations can also collapse directions outright ($f^2g = f$ forces $fg = \text{id}$, and $fh = gh$ forces $f = g$; [table 6](#)). The universal Replacement thresholds are: none for G finite, $\text{cf}(\kappa) > \aleph_0$ for G infinite and $d \leq 1$, $\text{cf}(\kappa) > \mathfrak{c}$ for $d \geq 2$ ([theorem 34.3](#)); Collection and reflection in [corollary 34.5](#). AI-assisted and unrefereed, like the rest of the report.

Question 23.4 (Using finer subgroup-space invariants). The theorem uses only $\tau(G)$, but subgroup spaces have much finer topology and descriptive complexity; for countable abelian groups see [10]. Can Cantor–Bendixson rank or a natural complexity bound on stabilizer codes control the syntactic complexity of failed Replacement instances? The first goal should be a precise theorem for a class of groups with already classified subgroup spaces.

Question 23.5 (Decidability from a finite presentation). For a finitely presented group G , what is the complexity of determining which of the finite, countable, and continuum type-count cases occurs? More realistically, find effective sufficient certificates for a countable bound, a perfect family of subgroup types, or a selected bad multiplicity profile. Any undecidability claim should be proved by an explicit reduction rather than inferred from the general difficulty of group theory.

[write] **Not answered; the monoid analogue is decided, 4 October 2026, batch 96.** [Theorem 32.1](#) of Part III (both of its sources) decides, from a finite commutative *monoid* presentation, which of the three cases holds for the number of connected injective action types: the commutative-monoid analogue of this question. Source 04 there calls this “a commutative answer” to the question. It is not the commutative case of the question as posed: for a finitely generated abelian group given by a commutative presentation, $\tau(G) = |\text{Sub}(G)|$ is finite when G is finite and $\aleph_0$ otherwise, which a Smith normal form decides, and this was immediate before ([Section 24.2](#)). The question for finitely presented groups beyond the abelian ones stays open; [Question 37.13](#) asks for the bit complexity of the monoid test.

Question 23.6 (Separate names for groups that are not locally finite). Classify universal Replacement preservation for countable groups in the language of separate names. Local finiteness gives unconditional preservation, and finite generation is settled by [theorem 18.2](#). Between these cases, every formula sees a finitely generated subgroup, but only actions that extend to the whole group are available. Determine whether an invariant of extendible subgroup actions gives a sharp general criterion.

Question 23.7 (Infinitary formulas). Replace ordinary first-order syntax by a precisely specified infinitary logic and re-evaluate the separate-name/uniform-evaluation distinction. For example, can a countable conjunction uniformly distinguish the index-two stabilizers used in [theorem 19.3](#)? The treatment must separately specify formula codes, satisfaction, the allowed parameters, and which Replacement or Collection scheme is being asserted.

Question 23.8 (How much ambient Choice is necessary?). The present proofs use external Choice for cardinal arithmetic, families of isomorphisms, and witness selection after external Collection. Isolate these uses in the commuting case. Rooted finitely generated orbits have explicit word enumerations, but selecting and matching components still needs analysis. Establish the weakest natural choice hypotheses under which the exact threshold and the definable-kernel formula survive.

Question 23.9 (Atom ideals beyond cardinal cutoffs). Replace the condition $|\ker(x)| < \kappa$ by membership in an atom ideal I . Seek conditions on I and the action which are both necessary and sufficient for expanded Replacement. Closure under orbit saturation, closure under unions of recognizable small blocks, and the ability to transport witnesses into fresh copies are distinct requirements and should not be merged into one unexamined symmetry assumption.

Question 23.10 (Complexity of a failed instance). Find the least formula complexity needed to witness the negative examples. The type selector quantifies over whole blocks, while the staircase construction has a directly readable boundary. Can a lower-complexity formula uniformly recognize each coded block? The answer should distinguish a fixed code as parameter, a pure sequence of codes, and a parameter-free formula.

Question 23.11 (Forcing beyond preservation of the cutoff). [Theorem 21.1](#) settles the fixed-action question when the original cutoff remains a cardinal. Analyze what happens when that ordinal is collapsed and the model is rebuilt using its new cardinality as the cutoff. Find exact examples in which Replacement changes, and determine which changes of block sizes cause the transition. Separately, when cofinality changes but the cutoff remains a cardinal, identify how new actions witness any change in a universal threshold.

Question 23.12 (Formal schema independence certificates). Build a reusable ProveIt interface certifying a definable model, its valid scheme instances, and one explicit failed instance in an expanded signature. Begin with the distinct-period example, then certify the two-point uniformity separation. This would provide a concrete test that the formal infrastructure keeps elementary lifts, amenability, base Replacement, and expanded Replacement separate.

[write] **A question added at the write, 4 October 2026, batch 93.** ProveIt’s standing rule for unproved claims (docs/incoming/README.md, Section 4, item 11) moves the one imported, unproved premise of this Part here.

Question 23.13 (A primary check of the Ivanov–Ol’shanskii groups). *Source:* [remark 18.4](#), which credits Ivanov and Ol’shanskii [12] through Smith [11, p. 69]. *Claim:* there is a finitely generated infinite group G all of whose proper nontrivial subgroups are conjugate; then $\tau(G) = 3$, so the finite-cutoff clause “ G finite” of [theorem 18.2](#) and the clause “ $\text{cf}(\kappa) > \tau(G)$ ” read at $\kappa = \omega$ (that is, $\tau(G)$ finite) differ. *Argument available:* none beyond Smith’s attribution (checked on the first page of [11]); the title of [12] names its method, graded diagrams in combinatorial group theory. *Missing:* a check of the construction in the primary source, or in a textbook account of it, confirming finite generation, infinitude and conjugacy of all proper nontrivial subgroups; neither the manuscript nor the write consulted [12]. Short of that, is there a more accessible infinite finitely generated group with finitely many conjugacy classes of subgroups?

[write] **Status of the questions, 4 October 2026, batch 93.** None of these thirteen questions is answered in ProveIt. Several continue questions of Part I and of Part VIII of birthday-cutoffs-and-hereditary-sets (hset:ns:): [Questions 23.1](#) and [23.2](#) continue [Question 10.1](#) beyond commuting permutations; [Question 23.7](#)

meets Question 10.6 (uncountably many names and infinitary syntax); Question 23.8 is Question 10.8 (removing ambient choice) for the commuting case; Question 23.9 is Question 10.7 and Part VIII's Question 65.1 (hset:ns:q:ideals); Question 23.10 is Question 10.9 and Part VIII's Question 65.6 ($\text{hset:ns:q:complexity}$); Question 23.11 is the remainder of Question 10.10; Question 23.12 is Question 10.12 and Part VIII's Question 65.12 ($\text{hset:ns:q:formalization}$); Questions 23.6 and 23.7 meet Part VIII's Questions 65.4 and 65.8 ($\text{hset:ns:q:generators}$, $\text{hset:ns:q:evaluators}$), which this Part answers in part (the notes after proposition 19.2 and theorem 19.3). [Added 4 October 2026, batch 96: the first sentence of this note is no longer true. Part III answers Question 23.3 in full, and Questions 23.1 and 23.2 with pure real parameters only (dated notes after the three questions); Question 23.5 is not answered, although one source of Part III says so of its commutative case (the note after it). Part III's questions (Section 37) continue Questions 23.1, 23.2 and 23.8 to 23.12.]

D Conventions and construction data

Notation	Meaning and location
A	An external set of atoms with $ A \geq \kappa$; it is not an internal set of $M_\kappa(A)$.
$\ker(x)$	Atoms occurring in $\text{TC}(\{x\})$, including x itself when x is an atom.
$M_\kappa(A)$	All objects over A whose atom kernels have cardinality below κ . Pure ranks are unrestricted.
j_f	The recursively lifted atom injection f , fixing every pure object.
B_t	All atoms in all components of type t , not a single component.
D_κ	Union of the entire type blocks that are internal atom sets. It is always an external set but need not be internal.
$\tau(G)$	Number of conjugacy classes of all subgroups of G .
$\tau_{\text{fin}}(G)$	Number of conjugacy classes of finite-index subgroups of G .
$E(g, x)$	One binary evaluator for the whole group action, totalized by the identity outside the pure set G .
$\mathcal{K}(\text{dcl}(\bar{a}))$	Scheme notation for kernels of uniquely definable objects, as explained before theorem 17.1; not an added object-language truth predicate.

D.1 An explicit singular-cutoff witness

At $\kappa = \aleph_\omega$, choose disjoint atom blocks B_n of size $\aleph_n$. Partition B_n into cycles of length $n + 2$. Let p rotate each cycle and fix any unused atoms. The single formula

$$“y \text{ is the set of all atoms of exact } p\text{-period } n + 2”$$

has the unique internal value B_n at each $n \in \omega$. The statement of exact period uses finite path graphs and the requirement that no shorter positive iterate returns to the atom. Its range cannot be an internal set because its kernel would contain $\bigcup_n B_n$, of size $\aleph_\omega$. Adding finitely many identity names gives commuting counterexamples of every finite positive arity.

D.2 An explicit regular-cutoff witness

At $\kappa = \aleph_1$, choose distinct staircase codes S_α for $\alpha < \omega_1$ and use countably many copies of each U_{S_α} . Such a family of codes exists by Choice and $\aleph_1 \leq \mathfrak{c}$. Each entire type block is countable. Their union has size $\aleph_1$.

The pure sequence $\langle S_\alpha : \alpha < \omega_1 \rangle$ is an internal parameter, even though it has uncountable size. Its type-block selector is functional on the internal pure domain ω_1 and has no internal range. Each unary coordinate map has just the ray type and preserves Replacement. This explicitly illustrates why large pure input sets cannot be ignored in a model that restricts only atom kernels.

D.3 The finite-cutoff uniformity witness

Take $G = \bigoplus_\omega C_2$ and nonzero characters $\chi_{\{n\}}$ for $n < \omega$. Give each one exactly one two-atom orbit. Separate unary names preserve Replacement, since any finite set of them acts through a finite subgroup. With uniform evaluation, the formula comparing an atom's stabilizer with $\ker \chi_{\{n\}}$ defines the complete two-atom block. Collecting those blocks would require infinitely many atoms, which is forbidden at $\kappa = \omega$. The evaluator's restriction to any internal set still uses only finitely many atoms, so it remains amenable.

E Source and verification record

The following record makes the research boundary reproducible without relying on an unpinned repository branch.

- The target question and prior profile theorem were read in the full source of *Naming Elementary Embeddings* at commit a9ab9a69872c2fb7464097af32fd7b76c75cb030, file SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings/article.tex. Its Git blob is 5426fc8da5f470efc448c3dff041751d0d7dc4a9. The question is Question 10.4, and the profile theorem is Theorem 5.1.
- The group-action connection to Glazer was checked against the full arXiv HTML of Glazer–Yao, version 1, 25 February 2026, especially its kernel-model discussion. LinkedIn identifies the researcher but is not used as the mathematical authority for the results.
- Smith's published 1995 PDF was read directly for the existence of an infinite finitely generated group with three conjugacy classes of subgroups. The construction itself is not reproduced. The original Ivanov–Ol'shanskii chapter is included bibliographically; its publisher metadata was checked, but its paywalled proof was not used.
- The main mathematical arguments received an independent review of the group types, mixed normal forms, strict staircase rigidity, the finite-cutoff definable-kernel argument, and the uniform-action construction. This is an additional review step within preparation of the manuscript, not external peer review.
- The exact finite checks are recorded in `verification_output.txt`. The source and scripts are supplied so that arithmetic and figure generation can be reproduced. The semantic and infinite-cardinal conclusions remain mathematical proofs in the manuscript.

The literature checks support a direct research intersection and a concrete answer to the repository question. They do not establish that every proposed refinement is absent from all unpublished notes or the wider literature. Any submission for publication should retain the prior-work distinctions in [section 12](#) and undergo an additional specialist review.

Part III

Relations among injections: definable atoms, connected types, and Replacement

A finite/countable/continuum classification, exact kernels from rooted components, exact enumeration, and Replacement

An investigation of ProveIt Questions 23.1–23.3 at the intersection with Elliot Glazer’s research

Two research manuscripts, merged: *Relations, Definable Atoms, and Replacement* (source 03) and *Relations among Commuting Injections: Connected Actions, Exact Enumeration, and Replacement* (source 04, subtitled *An effective rank classification at the Glazer–ProveIt research intersection*), both 4 October 2026

Status (source 03). Complete mathematical proofs are supplied for the stated results. The component-count classification and the definability refinements are developed here; their standard localization, group-action, and polyhedral ingredients are credited. The results answer the specified repository questions with the parameter conventions stated in the article. Historical publication priority has not been established. The accompanying computations check finite examples; these proofs have not been checked by a proof assistant. Elliot Glazer and Bokai Yao are cited researchers, not authors or endorsers of this manuscript.

Abstract (source 03). Let a finitely generated commutative monoid act by injective maps. Write G for its Grothendieck group, P for its image in G , $U = P \cap (-P)$ for the group of units, and $d = \text{rank}(G/U)$. We prove that the number of nonempty connected labelled action types is finite when G is finite, countably infinite when G is infinite and $d \leq 1$, and exactly $2^{\aleph_0}$ when $d \geq 2$. For a finite presentation, d is the dimension of the span of an explicitly given nonnegative rational solution cone, so the trichotomy is decidable by exact linear algebra. In particular, commuting injections satisfying $f^m = g^n$ have only countably many types. If m, n are coprime, precisely $1 + \binom{m+n}{m} / (m+n)$ of those types are infinite. A necklace formula classifies and counts all finite and infinite types without a coprimality assumption, and determines the automorphism group of every proper component.

We apply this classification to $M_\kappa(A)$, the universe of sets involving fewer than κ atoms. It gives sharp universal thresholds for preserving Replacement after the canonical lifts of the generators are named. A second theorem, valid without commutativity, identifies the exact atom kernel of definable closure with pure real parameters: it is the component saturation of the parameter kernels together with the atoms whose orbits under the commuting automorphism group have size below κ . Finite path codes make this result valid even at $\kappa = \omega$ and for infinite components. For group actions the criterion is expressed through subgroup normalizers; for commutative actions it is expressed through translation periods. We distinguish kernels of definable objects from individually definable atoms, give exact parameter-sensitive formulas, explain the relation to Glazer–Yao small-kernel methods, and propose further research directions.

Abstract (source 04). We classify the number of connected injective actions of every finitely generated commutative monoid. After cancellativization, let P be the monoid inside its finitely generated abelian group completion G , let $U = P \cap (-P)$, and put $d = \text{rank}(G/U)$. There are finitely many connected types when G is finite, countably infinitely many when G is infinite and $d \leq 1$, and continuum many when $d \geq 2$. The last case admits an explicit Cantor family of rigid components. For a fixed completion stabilizer the alternatives sharpen to finite or continuum.

The invariant is computable by exact rational linear algebra from a finite commutative presentation. For two distinguished commuting injections satisfying $f^m = g^n$, we give a complete enumeration: finite connected types are counted by a divisor sum, and infinite connected types by a finite sum of binary-necklace numbers. The coprime boundary case recovers the classical rational Catalan number.

In cardinal-kernel models with urelements, the type classification yields sharp universal thresholds for expanded Replacement: no restriction on the infinite cutoff for finite effective monoids, cofinality above $\aleph_0$ for infinite monoids of positive rank at most one, and cofinality above $2^{\aleph_0}$ in higher positive rank. Explicit counterexamples witness every negative case. These results answer Provelt’s Question 23.3 in the inspected revision and give a commutative answer to its related decidability question.

[write] Part III of this report since 4 October 2026: batch-96 manuscripts 05 and 06 of Provelt’s incoming-reports intake, two independent answers to one assignment, *merged* (every result, proof, example, remark, question and non-claim of both is printed; a result proved by both is printed once, with the other proof as a marked second route). In this report they are *source 03* (manuscript 05, *Relations, Definable Atoms, and Replacement*, labels `nee:aa:`, the base of the merge) and *source 04* (manuscript 06, *Relations among Commuting Injections*, labels `nee:ci:`); Parts I and II are sources 01 and 02. Both answer Question 23.3 of Part II; source 03 also answers Questions 23.1 and 23.2 of Part II and Question 10.1 of Part I *with pure real parameters*, which the write re-scopes rather than closes (Section 24.2). The sections are renumbered from 24 and the appendices from F; a line “Sources” after each numbered item names the item of each manuscript that it prints. The second sentence of source 04’s abstract uses source 03’s letters (source 04 swaps M and P ; Section 24.6); its last sentence overstates what is decided about Question 23.5 (the note in Section 24.2). Provenance, the merge choices and the notation are in Sections 24.5 and 24.6.

24 The research intersection and the results

24.1 Why this is a Glazer–ProveIt topic

Glazer and Yao study the interaction of Collection, reflection, and choice in set theory with urelements, including models in which the atoms form an internal proper class [1]. Their small-kernel constructions use permutations to control which atoms can occur in a uniquely specified output. Glazer’s thesis places these questions in a broader program involving choice, forcing, and measure theory [16]. These primary mathematical sources provide the research-interest match; the LinkedIn address supplied with the research request did not provide readable contents during this investigation.

The inspected ProveIt report [13] develops canonical elementary lifts of injections on atoms and asks what changes when such lifts become named operations. Its Part II ends with explicit questions about nonabelian definable closure, finite kernels for several injections, and relations among commuting noninvertible generators. We use the repository snapshot

715a716a3002b1e9f28daa2d81c009a392851b94.

The mathematical source is

SetTheory/Cardinals/docs/reports/ordinals-and-order-types/
naming-elementary-embeddings/article.tex.

The connection is precise: equations among named injections constrain their connected components, which constrain available automorphisms, which in turn control definable atom sets and the Replacement scheme.

Source 04's account of the same intersection. Glazer and Yao's *Reflection Principles in ZFU* studies how Collection and reflection diverge in set theory with urelements [1]. Its relevance here is the role of atom symmetries and of the availability of sets collecting witnesses. ProveIt's report *Naming Elementary Embeddings* develops cardinal-kernel models and asks how naming atom maps changes the full Replacement scheme [13]. Its Part II settles free commuting signatures and explicitly asks, in Question 23.3, what relations among noninvertible generators do to the number of connected action types.

We answer that repository question for all finitely generated commutative monoids, including those initially presented without cancellation. The answer is controlled by the rank of the group completion after removing the directions already invertible in the effective monoid. We then compute considerably more than cardinality for the two-generator relation $f^m = g^n$.

The external ambient theory throughout is ordinary set theory with Choice; for the kernel-model application we use its usual version with an external set of urelements. The conclusions concern explicit expansions of those models. They do not strengthen the choiceless separation theorem of Glazer and Yao.

24.2 Questions answered and conventions

Repository question	Mathematical issue	Result in this Part (source 03)
23.1	Individually definable atoms for nonabelian actions	A normalizer formula with residual component multiplicity for finite atom parameters; an exact formula outside the support saturation for arbitrary parameters. At the finite cutoff, a general orbit-definability theorem also treats arbitrary parameters.
23.2	Finite definable kernels for several injections	An exact rooted-component formula for every infinite cutoff, with pure real parameters. At the finite cutoff, finite multiplicity and a finite local automorphism group are decisive.
23.3	Relations in finitely presented commutative monoids	A complete finite/countable/continuum classification, a decision procedure, sharp axiom thresholds, and an explicit power-relation classification.

Pure-parameter definability means that each definition uses only finitely many pure parameters; it does not add a constant symbol for every pure object to a single formula. The distinction from

parameter-free definability is essential. An infinite union of rigid components can have many more atoms than there are parameter-free formulas. We give the corresponding counterexample below.

The inspected repository question (source 04). The audit used commit

715a716a3002b1e9f28daa2d81c009a392851b94

of VladimirReshetnikov/ProveIt. The relevant file is

SetTheory/Cardinals/docs/reports/ordinals-and-order-types/
naming-elementary-embeddings/article.tex.

Part II's Question 23.3 asks for a classification for finitely presented commutative monoids, with $f^m = g^n$ as a motivating example. Its Question 23.5 asks about decidability, primarily for general finitely presented groups. Our algorithm settles the commutative presentation case, without making a decidability claim for arbitrary groups.

The source is a research report, not a blanket certificate of formal verification. We use the mathematical arguments explicitly stated below; no unexamined repository claim is a premise of the main classification. [Section 36](#) records the proof boundaries and the supplied reproducibility material.

What is answered, as re-scoped at the write

[write] Added 4 October 2026, batch 96. “Question 23.*k*” above is Question 23.*k* of Part II: 23.1 is [Question 23.1](#), 23.2 is [Question 23.2](#), 23.3 is [Question 23.3](#) and 23.5 is [Question 23.5](#); the sources' numbers are this report's. ProveIt's standing rule for unproved claims (docs/incoming/README.md, Section 4, item 11) asks that a question only partly answered be re-scoped, not closed. Read against the questions as posed, the two sources settle the following.

- *Question 23.3 is answered* in full, by both sources independently: the type count is finite, $\aleph_0$ or $2^{\aleph_0}$ according to G and $d = \text{rank}(G/U)$ ([theorem 30.4](#)), decidably from a finite presentation ([theorem 32.1](#)); relations $f^m = g^n$ are enumerated exactly ([theorem 33.1](#)), and the universal Replacement, Collection and reflection thresholds follow ([theorem 34.3](#) and [corollary 34.5](#)). A dated note after the question records this.
- *Question 23.2 is answered with pure real parameters only.* [Theorem 27.1](#) gives, at every infinite cutoff and for every finite tuple of injections (commuting or not), the exact atom kernel of objects definable from the parameters *together with finitely many pure reals*. The question asks for the analogue of [theorem 17.1](#), which is a statement about ordinary definability; for ordinary definability source 03 reduces the problem, at $\kappa = \omega$, to the definability of rooted codes ([proposition 27.4](#)) and leaves that open ([Question 37.4](#)). The question stays open in its ordinary form.
- *Question 10.1 of Part I ([Question 10.1](#)) is answered* in the same pure-real form, for every finite tuple of injections: at $\kappa > \omega$ the bound “component closure plus $D_\kappa(f)$ ” of [theorem 5.1](#)'s proof is exactly the atom kernel of objects definable with pure real parameters ([theorem 27.1](#) and [corollary 27.2](#)); at $\kappa = \omega$ the small-type part must be replaced by D_ω^{pt} , the atoms whose centralizer orbits are finite. Ordinary definability stays open.
- *Question 23.1 is answered with pure real parameters for atom parameters, and at $\kappa = \omega$ for arbitrary parameters with arbitrary pure parameters.* [Theorem 27.3](#) and [corollary 28.1](#) give the individually definable atoms from finitely many named atoms and pure reals, through $N_G(H)/H$ and the residual multiplicity, including the infinite dihedral “first target” ([Section 28.1](#)); for arbitrary set parameters the criterion is exact only outside the saturation $\text{Sat}(B)$ of their kernels; at $\kappa = \omega$, [theorem 27.5](#) characterizes all objects definable from arbitrary parameters and arbitrary pure parameters. Open: ordinary definability, and arbitrary set parameters inside $\text{Sat}(B)$ at $\kappa > \omega$ ([Questions 37.4](#) and [37.5](#)).

- *Question 23.5 is not answered.* Source 04’s abstract says that its results “give a commutative answer to” it, and its README that they give “an effective answer to the related commutative case”. What [theorem 32.1](#) decides is the *commutative-monoid analogue*: from a finite commutative monoid presentation, which of the three cases holds for the number $\tau(M)$ of connected injective action types. Question 23.5 asks about $\tau(G)$, the number of conjugacy classes of subgroups, for finitely presented *groups*. Its commutative case, finitely generated abelian groups given by commutative presentations, was immediate before source 04: every subgroup of such a group is finitely generated, so $\tau(G) = |\text{Sub}(G)|$ is finite when G is finite and $\aleph_0$ otherwise (an infinite one has infinitely many finite-index subgroups), and finiteness of G is read off a Smith normal form. (Presenting such a group as a commutative monoid with inverse generators, [theorem 30.4](#) returns the same answer: $P = G$, so $d = 0$.) Question 23.5 stays open beyond abelian groups; source 04’s own text in [Section 24](#) says as much (“without making a decidability claim for arbitrary groups”). A dated note after the question records this.

24.3 Main conclusions of source 04

Source 04 opens by stating its two main theorems in summary form, its Theorem 1.1 (the type classification) and Theorem 1.2 (universal Replacement). Each is proved by both sources and is printed once in this Part: Theorem 1.1 as [theorem 30.4](#) with source 04’s Cantor-family clause, Theorem 1.2 as [theorem 34.3](#). In source 04’s words: the theorem is proved in [Section 30](#), after the normal form in [section 29](#). Its effective version in [section 32](#) determines the answer directly from a finite commutative presentation. The complete two-generator enumeration is [theorem 33.1](#). The main foundational consequence, proved in [section 34](#), is the universal Replacement table of [theorem 34.3](#).

Here a statement about *every* action is essential. At a cutoff below the threshold, particular actions can still preserve Replacement. We give the exact criterion for each individual action in [theorem 34.1](#).

24.4 What is inherited, and the contribution

The small-kernel universe itself is established groundwork [3, 1]. The fixed-action Replacement and Collection criteria are proved in the inspected repository report [13]. A related report already classifies Replacement with a fixed finite bound on output kernels for uniformly named group actions [14, Part VIII, Theorem 59.1]. We do not claim those classifications as new. The passage from a cancellative action to an action of its group of fractions is standard [17, Section 6]; so are the normalizer formula for automorphisms of a transitive group action [18] and the geometry of units in affine monoids [19].

The contributions here are the complete type-count trichotomy for finitely generated commutative actions, its explicit pure-code construction and effective form, and the rooted-path definability refinements that remain valid for infinite components with finite atom kernels. Some proofs recombine standard ingredients. The claim of progress is tied to the precise statements proved, rather than to an assertion of a field-wide breakthrough or of settled publication priority.

Contribution and status (source 04). The new extension relative to the inspected ProvelIt report is the general monoid rank classification, its effective presentation test, the explicit two-generator enumeration, and their resulting uniform foundational classification. Localization of commutative monoid actions, rational-cone facts, necklaces, and rational Catalan enumeration are classical ingredients. The component-block criterion is prior repository work and is reproved with attribution. The proofs have received multiple internal reviews during AI-assisted preparation and exact finite checks, but are not proof-assistant certified or externally refereed. A claim of historical priority or of an established mathematical breakthrough is not made.

24.5 Provenance and place in this report

[write] Added 4 October 2026, batch 96. *Sources.* This Part is built from two manuscripts of batch 96 of ProvelIt’s incoming-reports intake, both in the arrival commit 26e036956 and placed in commit 27f200305:

- *source 03*, manuscript 05, *Relations, Definable Atoms, and Replacement: A finite/countable/continuum classification and exact kernels from rooted components*, delivered as `atom_actions_research.zip` (594,987 bytes; inner directory `atom_actions/`, main file `atom_actions.tex`, 1,761 lines, 45 labels, 10 bibliography entries, 12 questions numbered 1–12, a 28-page US Letter PDF); author line “Research manuscript prepared with ChatGPT”, 4 October 2026; files prefixed `03-atom-actions-`, labels `nee:aa;`
- *source 04*, manuscript 06, *Relations among Commuting Injections: Connected actions, exact enumeration, and Replacement*, delivered as `commuting_injections_research.zip` (309,489 bytes; inner directory `commuting_injections/`, main file `commuting_injections.tex`, 1,887 lines, 51 labels, 5 bibliography entries, 12 questions, a 24-page US Letter PDF); author line “Research manuscript prepared for Vladimir Reshetnikov”, 4 October 2026 (the PDF metadata reads “Research manuscript prepared with ChatGPT”); files prefixed `04-commuting-injections-`, labels `nee:ci:`

The source numbers continue those of the report’s README (Part I is source 01, Part II source 02) and match the file prefixes. Both Parts’ sources are AI-assisted; this Part is unrefereed and not formalized. The two manuscripts answer one assignment independently (their shared 8-grams, about 1% of each, are mostly bibliography) and both answer Question 23.3 of Part II, so the placement made them one addition to this report, which they continue, rather than a new report or an addition to Part VIII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets.

How they were merged. Union, not selection: every statement, proof, example, remark, figure, table, question and non-claim of both manuscripts is printed. Source 03 is the base: its kernel theorems need no commutativity, and it answers three repository questions to source 04’s one. Its section order is kept, and source 04’s sections are inserted where their subject is: its Section 1 is merged into Section 24, its Sections 2–3 (localization, cones, the trichotomy) are merged into Sections 29–30, its Section 4 is Section 31, its Section 5 is merged into Section 32, its Section 6 into Section 33, its Sections 7–10 into Sections 25 and 34, its Section 11.1 is Section 35, its Section 11.2 is merged into Section 30.3, its Sections 12–13 into Sections 37 and 36. After each numbered item a line “Sources” names the manuscript items it prints. Where both sources prove a result, it is printed once, in a statement containing every clause of either, with source 03’s proof first and source 04’s as “Second proof (source 04)” whenever the arguments differ: [proposition 25.3](#), [lemma 29.1](#), [theorem 29.2](#), [lemma 30.1](#), [theorem 30.4](#), [theorem 32.1](#), [theorem 33.1](#) (i) and (iii), [theorem 34.1](#), [theorem 34.3](#), [proposition 34.4](#) and [corollary 34.5](#). Where the merge had to choose: (i) source 04’s two summary theorems (its Theorems 1.1–1.2) are printed once, as [theorems 30.4](#) and [34.3](#), carrying their labels; (ii) source 03’s Remark 7.2 and the same torsion example in source 04’s Section 2.3 are printed once ([remark 30.2](#)), as are source 03’s Section 7.3 and source 04’s Example 11.2 ([example 30.7](#)); (iii) both figures are kept ([figures 2](#) and [3](#)); the two tables of presentations are merged into one with a source column ([table 6](#)), and the two tables of power-relation counts, whose columns differ, are both kept ([tables 7](#) and [8](#)); (iv) both abstracts and both status statements are printed; (v) the 24 questions are merged into 16, with one added at the write (Section 37); (vi) the notation is unified on source 03’s letters (Section 24.6). Section, theorem and equation

numbers of both manuscripts change; the “Sources” lines give the correspondence. Each source’s equations, numbered consecutively in source 03 and within sections in source 04, are numbered within sections here.

Pin and repository claims. Both manuscripts are pinned to 715a716a3002b1e9f28daa2d81c009a392851b94 (“Construct a non-dyadic outer family with actual compiler constants fixed”, 4 October 2026), an ancestor of the placement. This report’s `article.tex` at the pin is the blob `af39ad7c6`, identical to Parts I–II as printed here before this Part was added, so every number the sources quote (Questions 23.1–23.3 and 23.5; Theorems 5.1, 7.5 and 7.7 of Part I) is this report’s, and birthday-cutoffs-and-hereditary-sets is also unchanged since the pin. The sources’ repository statements are true at the pin and at this write: Part II ends with the questions they quote, under the labels source 03 names (`nee:ca:q:nonabelian`, `nee:ca:q:injections`, `nee:ca:q:monoids`); it “settles free commuting signatures” (theorem 16.3, $\mathbb{Z}^s \times \mathbb{N}^t$); the fixed-action Replacement and Collection criteria are theorems 5.1, 7.5, 7.7 and 14.2, the reservoir argument is Part I’s Section 7 (lemma 7.4); Part VIII of birthday-cutoffs-and-hereditary-sets, Theorem 59.1 (`hset:ns:thm:graded`), classifies Replacement with a finite bound on output kernels for uniformly named group actions, at $\kappa = \omega$; and the repository’s root README says that generated data and exploratory computations enter the trusted boundary only through proved checkers. No repository statement is contradicted, and nothing needed retraction.

Formal status. No statement of this Part is formalized in Lean or Rocq; the formal-status note of Section 1.4 applies unchanged, and the formalization orders of Section 36 are proposals.

Relation to Parts I and II. Sections 25 and 34 reprove, with attribution, the foundations and the fixed-action criteria of Part I (proposition 25.2 is theorem 2.4, proposition 25.3 is part of theorem 3.2, lemma 26.1 is lemma 4.2, theorem 34.1 is theorem 5.1, and proposition 34.4 is theorems 7.5 and 7.7); example 34.6 is the commuting-injection counterpart of example 7.10. New, and consistent with Parts I–II: theorem 27.1 generalizes theorem 17.1 from commuting permutations to every finite injective tuple, with pure real parameters (the note after corollary 27.2); theorem 30.4 generalizes lemmas 15.1, 16.1 and 16.2 to every finitely generated commutative monoid, and theorem 34.3 restricted to $\mathbb{Z}^s \times \mathbb{N}^t$ is theorem 16.3 ($d = t$ there), restricted to $r = 1$ the one-name case of theorem 6.1, and restricted to finitely generated abelian groups the abelian case of theorem 18.2 and corollary 15.4; theorem 35.1 complements theorem 21.1.

Prior in the repository: Part VIII of the surreal report. Source 03 cites Part VIII of birthday-cutoffs-and-hereditary-sets (labels `hset:ns:`) for its graded classification. Its lemma 26.3 generalizes Part VIII’s Lemma 56.3 (`hset:ns:lem:centralizer`) from finite orbits of group actions to injections and infinite components (the note after it), and its two-involution example (Section 28.2) bears on Part VIII’s Theorem 61.3 and Question 65.7 (`hset:ns:q:joining`). Source 04 does not cite Part VIII.

Imported facts. The normalizer formula $\text{Aut}_G(G/H) \cong N_G(H)/H$ [18] is standard and its proof is sketched in corollary 28.1; the localization of cancellative actions [17, 21] and the cone facts [19] are proved in full; Anderson, Gorsky–Mazin and Johnson [20, 22, 23] are cited for the classical rational Catalan enumeration, which is also derived here. No unproved premise is imported.

The standing rule. No claim of either source was found to be wrong. Moved to Section 37: source 03’s countability of the parameter-free definable kernel, a metatheoretic argument (Question 37.17), and the residues of Questions 23.1, 23.2 and 10.1 (Questions 37.4 and 37.5). Source 04’s statement about Question 23.5 is corrected in scope above, not refuted.

The bibliography. The sources’ entries for Glazer–Yao (GY, GlazerYao) and Yao (Yao) are [1] and [3], printed once (source 03 adds “Especially Definition 18 and Theorem 19” to Glazer–Yao and dates Yao’s arXiv version only as “2023”; source 04 gives “2026”). Their entries for this report (RepoNaming, Repo), pinned at 715a716a3, are cited as [13], so that Part II’s entry [7] (pinned at a9ab9a698) stays as it was; source 03’s RepoBirthday and RepoRoot are [14, 15]. The entries [16, 17, 18, 19, 20, 21, 22, 23] are new.

Non-claims of the delivery READMEs (not shipped), beyond those in the text: source 03’s “separate ChatGPT agents reviewed the written proofs” and source 04’s “multiple internal proof reviews” are process statements, not external review; neither manuscript claims proof-assistant verification, external peer review or publication priority; source 03’s principal type-count theorem assumes a finitely generated commutative monoid, its all-object fixed-point theorem is proved at the finite cutoff only, and it distinguishes ordinary definability from definability with pure real or arbitrary pure parameters; its questions are extensions of the

present results, not longstanding published problems; source 04's finite checks do not substitute for the infinite and foundational proofs; Glazer and Yao are cited researchers, not authors or endorsers.

Checks at the write. The placement (27f200305) read the main proofs of both sources in full and checked, by an independent program, the automorphism spectra of proper types for $m, n \leq 8$ (source 03's drop sequences against source 04's gap words and [theorem 33.5](#)), the subgroup counts $\sigma_1(\gcd(k, g_0))$, the labelled commuting pairs of (33.8) for $N \leq 5$, the rational Catalan counts up to (5, 7), and every table entry; it found no wrong claim. At this write both verification programs were rerun on scratch copies, and the figure of source 03 was regenerated (Section 36, the note there).

24.6 Notation of Part III

[write] Added 4 October 2026, batch 96. The objects of Parts I–II keep their meaning: $M_\kappa(A) = M_\kappa(A)$ of (2.1), $\ker, j_f, \mathcal{N}_{\kappa,f}, \mathcal{T}_r, D_\kappa(f)$ (written D_κ), $I(f), V_\alpha(C), \mathfrak{c} = 2^{\aleph_0}$, Replacement, Collection, RP, $\mathcal{K}(\text{dcl}(\cdot))$ and $\mathcal{Z}$ (the centralizer, as in Part VIII of birthday-cutoffs-and-hereditary-sets). The two sources' letters are unified on source 03's. Renamed from the sources, with no change of normalization:

- source 03's atom kernel $\text{ak}(x)$ is printed $\ker(x)$, its expanded structure $\mathcal{M}$ is printed $\mathcal{N}_{\kappa,f}$, and its $\mathcal{N}(a, b)$ (rotation classes of weak compositions) is printed $\text{Neck}(a, b)$, source 04's name, so that it is not read as the expanded model $\mathcal{N}_{\kappa,f}$;
- source 04 *swaps* M and P : its P is the presented monoid and its M the cancellative image; here, as in source 03, M is the presented monoid and P its image in G , and source 04's $\bar{M}$ (also written N) is $\bar{P}$. Its type count $\theta(P)$ is $\tau(M)$; its realized-type set I is Part I's $I(f)$, whose cardinality source 03 writes θ ; its pure set of types $\mathcal{T}$ is $\mathcal{T}_r$; its cumulative universe $V(A)$ is $U(A)$;
- for $f^m = g^n$, source 04's $\delta = \gcd(m, n)$ is source 03's g_0 ; its divisor count $\tau(s)$ is $\text{div}(s)$; the order of the localization stabilizer, source 04's h , is source 03's e ; the repetition number of a boundary word and order of the automorphism group, source 04's e , is source 03's δ ; its Möbius function μ is $\mu_{\text{Möb}}$;
- source 04's Cantor family $B_r, S_r, r \in 2^{\mathbb{N}}$ is source 03's $A_R, S_R, R \subseteq \mathbb{N}$ (a subset and its characteristic function are identified); source 04's relation matrix A_0 is source 03's B , its cone $W \subseteq Q'$ is $\mathcal{W}_B^Q = \mathcal{W}_B \cap Q'$, and its column submatrix $(A_0)_J$ is B_J ;
- $2^{\aleph_0}$ is printed $\mathfrak{c}$, as in Parts I–II; source 04's At is At; $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are as in both sources.

Several letters still carry more than one meaning:

Symbol	Meaning in Part III (and what it is not)
M, P, G	from Section 29 on: a presented finitely generated commutative monoid, its cancellative image, its group completion. <i>Not</i> the model $M_\kappa(A)$; in Section 28 G is a finitely generated group. In Part II, G is a group and P a union of parameter kernels; in Part I's (5.4), P is a set of atoms
U, Q, d	the units $P \cap (-P)$, the quotient G/U , its rank. <i>Not</i> the universe $U(A)$, the staircases U_S of Part II or the unlabelled counts $U_N(m, n)$ of corollary 33.7; in Section 33, Q_e is a quotient group. d is <i>not</i> Part VIII's level d or the order $d_t = \text{Aut} $ of birthday-cutoffs-and-hereditary-sets (that order is δ_t here)
$\tau(M)$	the number of connected labelled injective action types of M . For a finite abelian group it is $ \text{Sub}(G) $, which is Part II's $\tau(G)$ (conjugacy classes of subgroups) for abelian G . <i>Not</i> a divisor count
θ	$ I(f) $, the number of realized types (proposition 34.4)
$A, A_t, A_{[H]}, A_R$	the atom set; the union of all copies of the type t (Part I's block B_t) or of G/H ; the height-zero antichain of (30.2)
B, B_w, B_t	the union of the parameter kernels (Section 27) and its image under the word w ; in Section 32, B is the relation matrix; in source 04's proof of theorem 34.1, B_t is Part I's type block
$\mu_t, \mu, \mu_{[H]}, \mu_t^F$	multiplicities (numbers of copies); $\mu_{\text{Möb}}$ is the Möbius function
δ_t, δ	the order of the automorphism group of a component; in Part I and Part II, $\delta = \text{cf}(\kappa)$, a usage source 04 shares in its second proof of theorem 34.3 (written η in source 03's proof)
e, g_0	the order of the localization stabilizer of an infinite component of $f^m = g^n$; $\text{gcd}(m, n)$
H	a subgroup (Sections 28–33); a protected set of atoms in source 04's proof of theorem 34.1; reservoirs H_n in proposition 34.4, as in lemma 7.4. <i>Never</i> H_κ
$\mathcal{R}, \text{Pure}$	the pure reals $\mathcal{P}(\mathbb{N})$; the class of pure objects
$D_\kappa^{\text{pt}}, D_\kappa$	the atoms with centralizer orbit of size below κ ; Part I's union of small type blocks. They agree for $\kappa > \omega$ (corollary 27.2)
$W, \mathcal{W}_B$	the pure set of signed words; the nonnegative real kernel cone of the relation matrix
ℓ, v	a positive integral height and an infinite-order element of its kernel. In Section 33, source 04 writes u, v for the generators of $G_{m,n}$; in Part II, v is a holonomy
c_i	drops of a boundary function (source 03, length a , sum b); source 04's gaps c_j have length b and sum a ((33.7))
$C_k(m, n), C_\delta$	the number of connected types on k points; a cyclic group
$L_n, L_N(m, n)$	layers of the dihedral example; labelled finite actions

The same type count appears under three names in the sources: source 03's $\tau(M)$, source 04's $\theta(P)$ and, for a finite abelian group, Part II's $\tau(G)$. The same necklace number appears as source 03's $\mathcal{N}(a, b)$, the rotation classes of weak compositions of b into a parts (drop sequences), and source 04's $\text{Neck}(a, b)$, those of a into b parts (gap words); the formula (33.1) is symmetric in a and b , so the two are the same number.

25 The ambient universe and named injections

25.1 Sets with a restricted atom kernel

Work externally in a well-founded universe of sets with Choice and a set A of atoms. Let κ be an infinite cardinal and assume $|A| \geq \kappa$. For any object x , define its *atom kernel*

$$\ker(x) = A \cap \text{TC}(\{x\}).$$

In particular $\ker(a) = \{a\}$ for an atom a . Let

$$M_\kappa(A) = \{x : |\ker(x)| < \kappa\}.$$

There is no bound on pure rank. Every pure set belongs to $M_\kappa(A)$, but A itself does not. Thus A is an external set and an internal proper class. The predicates “set” and “atom” are distinguished; Extensionality applies to sets.

Lemma 25.1 (Kernel facts). *If $y \in x$, then $\ker(y) \subseteq \ker(x)$. Moreover*

$$\ker(\{x, y\}) = \ker(x) \cup \ker(y), \quad \ker(\mathcal{P}(x)) \subseteq \ker(x), \quad \ker(Y) = \bigcup_{y \in Y} \ker(y)$$

for every external set Y of objects. Consequently $M_\kappa(A)$ is transitive and is closed under subsets of internal sets, Pairing, Union, and Powerset.

Sources: source 03, Lemma 2.1. It is [lemma 2.2](#) of Part I.

Proof. An atom in a transitive closure is witnessed by a finite membership path. Concatenating such paths proves the inclusions and identities. A finite union of cardinals below an infinite κ remains below κ , including when κ is singular. The closure assertions follow. $\square$

Every permutation π of A has a canonical recursive lift to all objects, with $\hat{\pi}(x) = \{\hat{\pi}(y) : y \in x\}$ on sets. It is an automorphism of the membership structure on $M_\kappa(A)$ and fixes every pure object. If it fixes $\ker(x)$ pointwise, it fixes x .

Proposition 25.2 (The classical reduct). *The structure $(M_\kappa(A), \in, \text{At})$ satisfies ZF with atoms, Replacement, and Choice. Its internal cardinalities of internal sets agree with their external cardinalities.*

Sources: source 03, Proposition 2.2; source 04, Section 7.1 (second route below). It is [theorem 2.4](#) of Part I and [proposition 13.3](#) of Part II.

Proof. Only Replacement needs a substantive argument. Suppose a base-language formula uniquely specifies y from internal parameters x, p . If $b \in \ker(y) \setminus (\ker(x) \cup \ker(p))$, choose $c \in A \setminus (\ker(x) \cup \ker(p) \cup \ker(y))$. Such c exists by the kernel bound. Swapping b, c fixes x, p and changes y , contradicting uniqueness. Thus a unique output introduces no atoms beyond the input and parameter kernels. The externally constructed range on an internal domain w has kernel contained in $\ker(w) \cup \ker(p)$, so it is internal. This proves each Replacement instance. The remaining structural axioms follow from [lemma 25.1](#); Separation uses external Separation. An external well-order of an internal set x , and the graph of its enumeration by an ordinal, use only atoms from $\ker(x)$. They are internal, proving Choice and cardinality agreement. $\square$

Source 04's framework (its Section 7.1). Work in a well-founded ambient universe with urelements and Choice. Fix an external set A of atoms and an infinite cardinal $\kappa \leq |A|$. Let $U(A)$ be the cumulative universe over A . For an object x , let $\ker(x)$ be the set of atoms in $\text{TC}(\{x\})$, and define the class

$$M_\kappa(A) = \{x \in U(A) : |\ker(x)| < \kappa\}.$$

This is a definable class structure. No set of all its objects or internal set of all its atoms is asserted. Every pure set lies in $M_\kappa(A)$; an atom set is internal exactly when it has size below κ .

Subsets of an internal set are internal. Pairing, Union, and Powerset preserve the kernel bound, as do the usual other ZFU axioms. An external well-order of an internal set has its graph supported on the same atom kernel, so is internal. The base-language Replacement assertion follows, for example, from the component argument below with no named maps: there is one singleton component type with a large atom block. Thus the underlying model satisfies ZFU with Choice and Replacement. This does not assert Collection at every cutoff. (That is, [theorem 34.1](#) below, applied to the empty tuple of named maps, or to the identity, gives a second proof of base Replacement.)

25.2 Lifts are elementary before they are named

For an injection $f : A \rightarrow A$, let j_f be the recursively defined map

$$j_f(a) = f(a), \quad j_f(x) = \{j_f(y) : y \in x\} \quad (x \text{ a set}).$$

It satisfies $\ker(j_f(x)) = f[\ker(x)]$.

Proposition 25.3 (Canonical lifts). *Every j_f is an elementary self-embedding of the classical reduct and is amenable: the graph of its restriction to any internal set is internal. Naming several such maps need not preserve expanded Replacement.*

Sources: source 03, Proposition 2.3; source 04, Proposition 7.1 (elementarity). It is part of [theorem 3.2](#) and [proposition 13.4](#).

Proof. For a finite tuple of parameters, the union B of their kernels has size below κ . The partial bijection $f|B$ extends to a permutation of A , since both complements have size $|A|$. Its lift agrees with j_f on the tuple, proving elementarity one formula at a time. The restriction graph on an internal x has kernel contained in $\ker(x) \cup f[\ker(x)]$, proving amenability. The last assertion will follow from the explicit failed instances in [section 34](#); elementarity of a map in the original language is a different assertion from Replacement for formulas that can mention that map. $\square$

Source 04's setting for the lifts. Let the effective monoid P act on A by injections. For a chosen finite generating tuple $p_1, \dots, p_r$, let $f_i(a) = p_i \cdot a$, and define its recursive lift by

$$J_i(a) = f_i(a) \quad (a \in A), \quad J_i(x) = \{J_i(y) : y \in x\} \quad (x \text{ a set}).$$

These class maps preserve $M_\kappa(A)$, since $\ker(J_i(x)) = f_i[\ker(x)]$. The expanded language is

$$\{\in, \text{At}, J_1, \dots, J_r\}.$$

Expanded Replacement concerns *every formula* in this language, not only the existence of the individual images $J_i[w]$.

Second proof of elementarity (source 04). For any finite parameter tuple, its combined atom kernel B has size below κ . The injection $f_i|B$ extends to a permutation of A , because both complements have cardinality $|A|$. The recursive lift of that permutation is an automorphism of the base structure and agrees with J_i on the chosen parameters. Every base-language formula therefore has the same truth value before and after applying J_i . $\square$

Thus the new scheme obstruction can coexist with elementary self-embeddings of the reduct. Introducing a symbol permits formulas to use the component structure of that map, which was unavailable in the base language. This is the specific foundational issue studied by the repository report. (The two proofs are the same argument, that of [theorem 3.2](#), written independently.)

Fix a finite tuple of injections $f = (f_1, \dots, f_r)$ and let $\mathcal{N}_{\kappa, f} = (M_\kappa(A), \in, \text{At}, j_{f_1}, \dots, j_{f_r})$. Unless explicitly stated, they need not commute. The relevant atom automorphism group is

$$\mathcal{Z} = \text{Aut}(A; f) = \{\pi \in \text{Sym}(A) : \pi f_i = f_i \pi \text{ for all } i\}.$$

Its lifts are automorphisms of $\mathcal{N}_{\kappa, f}$.

Remark 25.4 (Schemes and external semantics). Assertions about a formula in $\mathcal{N}_{\kappa, f}$ refer to the external relativization of that fixed formula to the definable class $M_\kappa(A)$. Replacement, reflection, and definable closure are interpreted formula by formula. No internal satisfaction predicate for a proper-class structure is assumed. All collection of external witnesses occurs in the ambient theory and is followed by a separate atom-kernel bound. This convention prevents a concealed use of the very expanded Replacement scheme being investigated.

Sources: source 03, Remark 2.4; source 04, Section 7.2 (below). Compare [remarks 2.1](#) and [13.1](#).

External arguments versus internal assertions (source 04, Section 7.2). All uses of satisfaction below concern one fixed formula, whose quantifiers can be restricted to the definable class $M_\kappa(A)$ in the ambient theory. There is no uniform truth predicate for the universe. Type codes and sets of type codes are pure sets, hence internal; an isomorphism from a countable pure code to an atom component need not be internal at the finite cutoff. We treat that cutoff separately.

26 Rooted components encoded by finite paths

Give A the labelled directed edges $a \xrightarrow{i} f_i(a)$. A component is connected after directions are forgotten. Since each label has at most one incoming and one outgoing edge at a vertex, every component is countable. Let W be the pure countable set of finite words in $1^+, 1^-, \dots, r^+, r^-$. A positive letter follows f_i ; a negative letter follows its partial inverse.

Lemma 26.1 (Internal finite evaluation). *There is a fixed first-order formula $\text{Eval}(w, a, b)$ saying that w has a feasible path from atom a to atom b . It has its intended external meaning in $\mathcal{N}_{\kappa, f}$ at every infinite κ . For fixed w, a , the endpoint is unique when it exists.*

Sources: source 03, Lemma 3.1. It is [lemma 4.2](#) of Part I.

Proof. The formula asks for a finite sequence of atoms of length $|w| + 1$ with the specified endpoints. For i^+ require $j_{f_i}(a_k) = a_{k+1}$; for i^- require $j_{f_i}(a_{k+1}) = a_k$. Every actual finite path has a finite kernel and therefore has an internal sequence graph. Conversely an internal such graph is a genuine path. Injectivity provides uniqueness for negative steps; positive steps are already functional. $\square$

For an atom a define

$$E_a = \{(u, v) \in W^2 : \exists b (\text{Eval}(u, a, b) \wedge \text{Eval}(v, a, b))\}.$$

Using a fixed injective natural-number coding of W^2 , regard E_a as a pure real $c_a \subseteq \mathbb{N}$; numbers outside the coding image are excluded. The diagonal records feasible words; all pairs record equality of endpoints.

Theorem 26.2 (Exact rooted codes). *There is one first-order formula $\text{Code}(c, a)$ defining $c = c_a$. For atoms a, b , the following are equivalent:*

- (i) $c_a = c_b$;
- (ii) *their rooted labelled components are isomorphic;*
- (iii) $b \in \mathcal{Z}a$.

The result includes infinite components when $\kappa = \omega$.

Sources: source 03, Theorem 3.2.

Proof. The defining formula for Code requires $c \subseteq \mathbb{N}$, excludes numbers outside the coding image, and states, for every coded word pair, exactly the displayed condition defining E_a . The pure real c_a exists externally and is internal because it contains no atoms. This does not require an internal graph of all endpoints.

If the codes agree, send the endpoint of w at a to the endpoint of w at b . Equality data makes this well defined and injective, and feasibility gives surjectivity. Appending a letter shows preservation of each labelled edge. Every point of a component is reached by a finite signed word, so this is a rooted isomorphism. The converse follows by transporting paths.

A rooted isomorphism between distinct components extends to a permutation of A by interchanging the components through the isomorphism and its inverse. Within a single component, extend the corresponding automorphism by the identity on other components. These extensions commute with every f_i . Finally a commuting permutation preserves every path code. This proves the three-way equivalence. $\square$

[write] **Rooted codes and Part I's recognition lemma, 4 October 2026, batch 96.** Lemma 4.3 of Part I recognizes a whole component type by an enumeration graph of the component, which is internal only when $\kappa > \omega$ or the component is finite. The rooted code c_a uses only finite path graphs, one formula at a time, so theorem 26.2 holds at $\kappa = \omega$ for infinite components too; this is what lets theorem 27.1 cover the finite cutoff.

Lemma 26.3 (Multiplicity and local symmetry). *Let X_t represent a nonempty connected labelled type t , let μ_t be its multiplicity in A , and put $\delta_t = |\text{Aut}(X_t)|$. Its local automorphism group acts freely on X_t , has cardinality at most $\aleph_0$, and for every atom a of that type,*

$$|\mathcal{Z}a| = \mu_t \delta_t.$$

Sources: source 03, Lemma 3.3.

Proof. An automorphism fixing one vertex fixes all finite signed paths from it and hence the entire component. It is determined by the image of one vertex, so there are at most countably many

such automorphisms. In each target copy of the type, the possible images of a form a free orbit of cardinality δ_t . Every component isomorphism extends globally as in [theorem 26.2](#). Counting these disjoint target-copy contributions gives the formula. $\square$

[write] A generalization of a lemma of Part VIII of the surreal report, 4 October 2026, batch 96. For a uniformly named action of a pure group Γ at the finite cutoff, Lemma 56.3 of Part VIII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (hset:ns:lem:centralizer) proves that $\mathcal{Z}a$ is infinite when the type t has infinitely many copies, has $m_t d_t$ elements when it has finitely many ($d_t = |\text{Aut}_\Gamma(O_t)|$), and that $\text{Aut}_\Gamma(O_t)$ acts freely on the orbit O_t ; there every orbit is a transitive Γ -set and, in the applications, finite. [Lemma 26.3](#) is the same statement for a finite tuple of *injections* (not necessarily permutations, so a component need not be an orbit of a group) and for *infinite* components, as one cardinal identity $|\mathcal{Z}a| = \mu_t \delta_t$ in which both factors may be infinite (with $\delta_t \leq \aleph_0$). The proofs are the same argument (an automorphism fixing one point of a connected structure is trivial; component isomorphisms extend to commuting permutations), written independently; source 03 cites Part VIII only for its Theorem 59.1. Part VIII's m_t, d_t are μ_t, δ_t here; its divisibility $d_t \mid n_t$ for a finite orbit has no counterpart for infinite components. A reciprocal note for Part VIII is proposed separately.

27 The exact atom kernel of definable closure

Let $\mathcal{R} = \mathcal{P}(\mathbb{N})$ denote the pure reals. For a finite tuple of internal parameters $\bar{a}$, write

$$B = \bigcup_i \ker(a_i), \quad \text{Sat}(B) = \text{the union of components meeting } B, \quad D_\kappa^{\text{pt}} = \{b \in A : |\mathcal{Z}b| < \kappa\}.$$

Use $\mathcal{K}(\text{dcl}(\bar{a}, \mathcal{R}))$ for the external union of atom kernels of objects uniquely definable using $\bar{a}$ and finitely many pure real parameters. This is scheme notation as in [remark 25.4](#).

Theorem 27.1 (Exact kernel formula). *For every infinite κ , every finite tuple of injective maps, and every finite internal parameter tuple,*

$$\mathcal{K}(\text{dcl}(\bar{a}, \mathcal{R})) = \text{Sat}(B) \cup D_\kappa^{\text{pt}}. \quad (27.1)$$

Allowing arbitrary pure parameters instead of pure reals does not enlarge this atom kernel.

Sources: source 03, Theorem 4.1 and its equation (1).

Proof. The set B is definable from the parameters using the ordinary kernel operation. Each fixed signed word w induces a partial bijection on atoms. Its image

$$B_w = \{b : \exists a \in B \text{ Eval}(w, a, b)\}$$

has size at most $|B| < \kappa$ and is an internal atom set. Its exact defining formula gives a unique object from $\bar{a}$. The sets B_w cover $\text{Sat}(B)$. This uses neither individually definable elements of B nor expanded Replacement to collect the family of all B_w .

For $b \in D_\kappa^{\text{pt}}$, its rooted code c_b defines exactly the fiber $\mathcal{Z}b$ by [theorem 26.2](#). This fiber is an internal atom set, since it has size below κ . It is uniquely specified from the pure real c_b , and its kernel contains b . This proves the lower inclusion.

For the upper inclusion, a unique output y from the given parameters and any pure parameters is fixed by every member of $\mathcal{Z}$ fixing B pointwise. Its kernel is consequently invariant under

$\mathcal{Z}_{(B)}$. Suppose b lies outside the right side of (27.1). Its component does not meet B . Write μ, δ for the multiplicity and local automorphism cardinality of its type. Then $\mu\delta \geq \kappa$. Fewer than κ components meet B . If $\delta \geq \kappa$, the automorphisms within the untouched component already give at least κ images of b . Otherwise cardinal arithmetic gives $\mu \geq \kappa$; deleting fewer than κ touched copies leaves at least κ target copies. In either case $|\mathcal{Z}_{(B)}b| \geq \kappa$. An invariant kernel of size below κ cannot contain b . This proves the upper inclusion, and also proves that additional pure parameters provide no larger kernel. $\square$

Corollary 27.2 (The finite-cutoff correction). *Let A_t be the union of all copies of component type t . Then*

$$D_\kappa^{\text{pt}} = \bigcup \{A_t : \mu_t \delta_t < \kappa\}.$$

At $\kappa = \omega$, a type contributes exactly when it has finite multiplicity and a finite local automorphism group. Its component may still be infinite. At $\kappa > \omega$,

$$D_\kappa^{\text{pt}} = \bigcup \{A_t : |A_t| < \kappa\}.$$

Sources: source 03, Corollary 4.2.

Proof. Use lemma 26.3. At an uncountable cutoff both $|X_t|$ and δ_t are at most countable, and both are nonzero. Thus $\mu_t \delta_t < \kappa$ if and only if $\mu_t |X_t| < \kappa$. At the finite cutoff the two conditions differ when X_t is infinite and δ_t is finite. $\square$

[write] Relation to Parts I and II, 4 October 2026, batch 96. A_t is Part I's type block B_t , so for $\kappa > \omega$ the corollary says $D_\kappa^{\text{pt}} = D_\kappa(f)$, and theorem 27.1 says that the bound $\text{cl}_f(P) \cup D_\kappa(f)$ of (5.4), for parameters with kernel union P , is exactly the atom kernel of the objects definable from them *with pure real parameters*: the pure-real form of Question 10.1, for every finite tuple of injections. Theorem 17.1 of Part II proves, for commuting permutations, the same formula for *ordinary* definability, at every cutoff. The two agree where both apply; the write checked this. For commuting permutations an infinite component is a transitive $\mathbb{Z}^r$ -set $\mathbb{Z}^r/K$, whose translations by $\mathbb{Z}^r/K$ are automorphisms, so δ_t is infinite and the type does not contribute to D_ω^{pt} ; a finite component has $\delta_t = |\mathbb{Z}^r/K|$ finite, so it contributes exactly when μ_t is finite, that is, when its block is finite. Hence $D_\omega^{\text{pt}} = D_\omega(p)$ for commuting permutations, matching (17.1). For a single ray at $\kappa = \omega$ (example 17.4), $\mu_t = \delta_t = 1$, so $D_\omega^{\text{pt}} = A$ although $D_\omega = \emptyset$, which is the correction that example calls for. Whether ordinary definability gives the same kernel for injections is Question 37.4.

27.1 Individually definable atoms and parameters

Theorem 27.3 (Exact atom closure for atom parameters). *If the impure parameters are a finite tuple of individually named atoms $F = \{a_1, \dots, a_n\}$, let μ_t^F count the components of type t disjoint from $\text{Sat}(F)$. Then*

$$A \cap \text{dcl}(F, \mathcal{R}) = \text{Sat}(F) \cup \bigcup_{\mu_t^F \delta_t = 1} \{\text{atoms in untouched copies of type } t\}.$$

In particular, without impure parameters an atom is definable from a pure real exactly when its type occurs once and its component is rigid.

Sources: source 03, Theorem 4.3.

Proof. Every atom in $\text{Sat}(F)$ is the unique endpoint of a fixed feasible word from a named atom. Every member of $\mathcal{Z}_{(F)}$ fixes these components pointwise. On untouched components, all component swaps and local automorphisms remain available. Thus an untouched atom has stabilizer-orbit size $\mu_t^F \delta_t$. This orbit is defined by its rooted code together with the requirement of being outside $\text{Sat}(F)$; saturation is expressible by existence of a finite path. A singleton orbit supplies a definition. A larger orbit supplies an automorphism fixing all parameters and moving the atom, precluding a definition. $\square$

For arbitrary set parameters $\bar{a}$, the same singleton criterion is exact *outside* $\text{Sat}(B)$, with μ_t^B counting untouched components. Inside $\text{Sat}(B)$, an unordered set parameter need not individually name its members. For example, two interchangeable atoms form a definable two-element set without either being individually definable. The full kernel formula (27.1) intentionally retains this distinction.

27.2 Why pure parameters cannot be omitted

Consider a family of $\mathfrak{c}$ pairwise nonisomorphic rigid countable components, one copy of each, for two commuting injections. Such a family is constructed explicitly in [theorem 30.4](#). Its global automorphism group is trivial, so $D_\omega^{\text{pt}} = A$ has cardinality $\mathfrak{c}$. Every atom is definable from its pure real rooted code. Yet there are only countably many parameter-free formulas. Each uniquely defined internal object has a finite atom kernel, so the parameter-free definable kernel is at most countable. This counting observation is external semantic reasoning, for example after placing the construction in a set-sized ambient model whose atom set is uncountable in a larger metatheory with access to its satisfaction relation; it is not an internal enumeration of definitions. It shows that the pure-parameter qualification is mathematically necessary, even with a rigid underlying atom structure.

[write] **A metatheoretic argument, kept as a remark, 4 October 2026, batch 96.** The preceding paragraph is not a theorem of the framework of [remark 25.4](#), and source 03 does not state it as one. There, “uniquely definable” is read one fixed formula at a time; the union over *all* formulas of the kernels of parameter-free definable objects is not an object that the scheme reading defines, so “that union is at most countable” is not one assertion of the ambient theory. What the framework does prove, for each fixed finite set of formulas, is that the kernels of the objects they define without parameters have a finite union. The counting argument becomes a proof once the ambient universe is a set-sized model W inside a larger universe that has its satisfaction relation: then the parameter-free definable objects of the expanded model built in W are countably many, each with a finite kernel. The intended contrast with $D_\omega^{\text{pt}} = A$, “of cardinality $\mathfrak{c}$ ”, needs in addition that A be uncountable in the larger universe (for a countable W it is countable there), and it needs W to satisfy enough of the ambient theory for [theorems 26.2](#) and [30.4](#) to hold inside it. Neither source states or proves such a version, and the write does not supply one. The claim is recorded, with what is missing, as [Question 37.17](#). Nothing else in this Part depends on it: [theorems 27.1](#) and [27.3](#) are stated with pure real parameters precisely so as not to need it.

An infinite component still violates atom-valued Replacement at the finite cutoff when an atom parameter is allowed: send each feasible signed word to its endpoint and every infeasible word to the starting atom. The domain W is pure, the outputs are single atoms, and the range is the infinite component. Definable finite objects inside a component do not imply that their total family can be collected internally.

27.3 An exact reduction for ordinary finite-kernel definability

The pure-parameter theorem also yields an exact statement when no extra parameters are automatically supplied. This statement identifies the remaining issue as definability of pure codes.

Proposition 27.4 (Ordinary definability at the finite cutoff). *Assume $\kappa = \omega$, and put $\mathcal{D}_{\bar{a}} = \mathcal{R} \cap \text{dcl}(\bar{a})$, with ordinary definable closure. For a code c , let $F_c = \{b \in A : \text{Code}(c, b)\}$. Then*

$$\mathcal{K}(\text{dcl}(\bar{a})) = \text{Sat}(B) \cup \bigcup \{F_c : c \in \mathcal{D}_{\bar{a}}, |F_c| < \omega\}.$$

Sources: source 03, Proposition 4.4.

Proof. The saturation term is obtained by fixed-word images of B , as before. If c is definable from $\bar{a}$ and F_c is finite, the exact fiber is an internal set definable from $\bar{a}$.

Conversely, let y be definable from $\bar{a}$ and let $K = \ker(y)$, a definable finite atom set. The set of rooted codes of members of K is a finite pure set, exists internally, and is uniquely specified from $\bar{a}$. Order its distinct reals lexicographically, using the least natural number at which two codes differ. Each member of this finite ordered set is individually definable by its finite position. Thus each code appearing in K belongs to $\mathcal{D}_{\bar{a}}$. For an atom of K outside $\text{Sat}(B)$, the invariance argument in [theorem 27.1](#) forces its global fiber to be finite. This proves the upper inclusion. $\square$

27.4 All objects at the finite cutoff

Theorem 27.5 (Fixed objects and pure parameters). *At $\kappa = \omega$, for every finite tuple $\bar{a}$ of arbitrary internal parameters,*

$$\text{dcl}(\bar{a}, \text{Pure}) = \{x : \hat{\pi}(x) = x \text{ for all } \pi \in \mathcal{Z} \text{ satisfying } \hat{\pi}(\bar{a}) = \bar{a}\}.$$

This concerns all internal objects, not just atoms or their kernels.

Sources: source 03, Theorem 4.5.

Proof. Automorphism invariance gives the forward inclusion. For the reverse, fix such an x and enumerate the finite set $\ker(x) \cup \bigcup_i \ker(a_i)$ as $b_1, \dots, b_s$. Code the complete signed-word equality diagram of this tuple by a pure real: for all i, j, u, v , record whether paths u from b_i and v from b_j exist and have the same endpoint. Its diagonal includes feasibility. A tuple $b'_1, \dots, b'_s$ realizes the same diagram exactly when the word map gives an isomorphism between the two saturated unions of components, carrying each b_i to b'_i .

This isomorphism extends to a member of $\mathcal{Z}$. It matches equal finite numbers of touched components of every type. Removing equal finite numbers from the same component multiplicity leaves equal complementary multiplicities, whether finite or infinite. Choose component isomorphisms on the remaining copies and extend the given partial isomorphism. Infinite touched components cause no difficulty; the matching is external.

Take also a pure code for the well-founded membership structure on $\text{TC}(\{x, a_1, \dots, a_n\})$. Concretely, enumerate this external set by an ordinal, record its membership relation, mark its atom nodes by $1, \dots, s$, and mark the distinguished roots. All this information is pure. Given atom labels $b'_1, \dots, b'_s$, a realization is a function on the pure index set satisfying the recorded membership equations, with these labels at the atom nodes and a set at every unmarked node. Its graph has

a finite atom kernel and belongs to the model. Existence follows by well-founded recursion and uniqueness by well-founded induction on the coded relation.

Define y by requiring such a labelled realization whose parameter roots equal the actual $\bar{a}$ and whose distinguished x -root equals y , with the labels realizing the signed-word diagram. Every realization is obtained from the original one by a commuting atom permutation. Equality of the parameter roots forces that permutation to fix $\bar{a}$, and the hypothesis then forces $y = x$. The original realization proves existence. The definition uses only $\bar{a}$ and finitely many pure codes. $\square$

The membership code in this theorem can be large and need not be a real. Accordingly the theorem states Pure, not $\mathcal{R}$. It is an explicit coding theorem in this particular well-founded setting; it does not assume that automorphism-invariance implies definability in arbitrary structures.

28 Nonabelian actions and subgroup normalizers

Suppose the named permutations generate an action of a finitely generated group G . A transitive component is G/H , where H is a point stabilizer. The type depends on the conjugacy class $[H]$. The standard formula [18, Theorem 8] is

$$\text{Aut}_G(G/H) \cong N_G(H)/H,$$

with the harmless opposite-group convention if a different choice of left or right action is used. In particular its cardinality is $[N_G(H) : H]$.

Corollary 28.1 (Normalizer formulas). *If $\mu_{[H]}$ is the multiplicity of type G/H , then*

$$D_\kappa^{\text{pt}} = \bigcup_{\mu_{[H]}[N_G(H):H] < \kappa} A_{[H]}.$$

An atom is definable from pure real parameters alone exactly when

$$\mu_{[H]} = 1 \quad \text{and} \quad N_G(H) = H.$$

For finite atom parameters F , replace $\mu_{[H]}$ by the residual multiplicity $\mu_{[H]}^F$ outside GF , and include every atom of GF . For arbitrary parameter tuples the same residual criterion is exact outside the saturation of their kernels; [theorem 27.5](#) gives the full fixed-point characterization at $\kappa = \omega$.

Sources: source 03, Corollary 5.1.

Proof. An equivariant map of G/H is determined by the image gH of H . It is an automorphism exactly when the image has stabilizer H , equivalently $gHg^{-1} = H$. The possible images are therefore the cosets of H in its normalizer. Apply [lemma 26.3](#) and [theorems 27.1](#) and [27.3](#). $\square$

The normalizer index cannot be replaced by the size $[G : H]$ of the component. Nor can the residual multiplicity be replaced by the original multiplicity: if there are exactly two fixed atoms and one is individually named, the other is the unique remaining fixed atom and becomes definable.

28.1 Infinite dihedral layers

Let $D_\infty = \langle r, t : t^2 = 1, trt = r^{-1} \rangle$, and act by $D_\infty \times C_m$ on $A = \mathbb{Z} \times C_m$ through

$$r(n, j) = (n + 1, j), \quad t(n, j) = (-n, j), \quad z(n, j) = (n, j + 1).$$

The stabilizer of $(0, 0)$ is $H = \langle t \rangle \times \{0\}$, and direct calculation gives

$$N_G(H) = \langle t \rangle \times C_m, \quad [N_G(H) : H] = m, \quad [G : H] = \aleph_0.$$

There is one infinite component, but its commuting-automorphism orbits are the finite layers $L_n = \{n\} \times C_m$. The layer L_0 is the set of atoms fixed by t , and $L_n = r^n L_0$. Thus every atom belongs to the kernel of a parameter-free definable finite set in $M_\omega(A)$. For $m > 1$ no individual atom is definable from pure parameters, because cyclic rotation moves it. For $m = 1$ every atom is individually parameter-free definable.

With q identical copies, each pointed fiber has size qm . Finite nonzero q preserves the finite-kernel phenomenon; infinite q removes every atom from the pure-parameter definable kernel. This gives explicit instances of all three factors: component infinitude, internal symmetry, and multiplicity.

[write] **Question 23.1, 4 October 2026, batch 96.** This is the “useful first target” of [Question 23.1](#), the infinite dihedral group, worked out (here as $D_\infty \times C_m$, with $m = 1$ giving D_∞ itself). With [theorem 27.3](#) and [corollary 28.1](#) it answers that question for finitely many atom parameters with pure real parameters, and with [theorem 27.5](#) at $\kappa = \omega$ for arbitrary parameters with arbitrary pure parameters; the dated note after the question records what stays open ([Section 24.2](#)).

28.2 Two involutions can jointly change Replacement

On atoms indexed by $\mathbb{Z}$, put $s(n) = 1 - n$ and $t(n) = -n$. The expansion of $M_\omega(A)$ by either single involution preserves Replacement: each has only finite cycles and finitely many types; for t include its unique fixed atom in the finite kernel reservoir. Yet jointly, t defines atom 0 and $(s \circ t)(n) = n + 1$. The parameter-free formula

$$k \mapsto (s \circ t)^k(0), \quad k \in \mathbb{N},$$

defined by finite paths, has an infinite atom range. It is a failed Replacement instance. The symbols $0, k$ in the formula mean, respectively, the uniquely specified t -fixed atom and a pure natural number; they are different sorts of objects.

[write] **Relation to earlier joint failures, 4 October 2026, batch 96.** The phenomenon is that of [corollary 7.9](#) of Part I (two permutations each safe alone, jointly failing Replacement), here at the finite cutoff and with involutions; there the cutoff is $\aleph_1$ and the joint failure comes from continuum many component types. Theorem 61.3 of Part VIII of birthday-cutoffs-and-hereditary-sets gives, at $\kappa = \omega$, two involutions each harmless alone *with all joint orbits finite* whose joint expansion keeps exactly the levels $R_k, k < d$. In the example above the joint orbit is infinite (s and t generate an infinite dihedral group acting transitively on $\mathbb{Z}$), so Replacement fails already for single-atom outputs; the example is not one of the finite-orbit pairs asked about in Part VIII’s Question 65.7 (hset:ns:q:joining), and it answers no question.

29 Commutative relations and localization

From this point let M be a finitely generated *commutative* monoid with fixed labelled generators $x_1, \dots, x_r$. An action by injective maps means that each generator acts injectively. Surjectivity is

allowed; it is not required. Isomorphisms preserve each operation label. We avoid the phrase “injective act,” which in the monoid literature usually denotes a different, categorical extension property.

Let $G = M^{\text{gp}}$ be the Grothendieck group and $P \subseteq G$ the image of M . Thus P is the universal cancellative image of the original monoid. The original presentation need not be cancellative.

Source 04's conventions (its Section 2.1). We write monoids additively, with identity 0. An M -action on a set X is *injective* if every translation $x \mapsto p \cdot x$ is injective. It need not be faithful, and an injective translation need not be surjective. Choose a finite generating list for M . The action is *connected* if the undirected graph joining x to $p_i \cdot x$ is connected. This property is independent of the chosen finite generating list.

The Grothendieck group is constructed from pairs of monoid elements; the images of $p, q \in M$ agree precisely when

$$\exists r \in M \quad p + r = q + r.$$

Consequently $P = \text{im}(M \rightarrow G)$ is a cancellative commutative monoid embedded in G , and $G = P - P$.

Lemma 29.1 (Cancellation forced by injectivity). *Actions of M by injective maps are exactly actions of P by injective maps, pulled back along $M \rightarrow P$. This identification includes their equivariant maps and connected components.*

Sources: source 03, Lemma 6.1; source 04, Lemma 2.1 (“Injective actions see only cancellativization”; the second sentence is its addition).

Proof. Two elements $a, b \in M$ have the same group-completion image exactly when $a + c = b + c$ for some $c \in M$. Their action maps satisfy $T_c \circ T_a = T_c \circ T_b$. Since T_c is injective, $T_a = T_b$. The action therefore factors through P . The converse is immediate. $\square$

Second proof (source 04). If $p + r = q + r$, then

$$r \cdot (p \cdot x) = r \cdot (q \cdot x).$$

Injectivity of the action of r gives $p \cdot x = q \cdot x$. Thus the action factors uniquely through P . The factored maps remain injective, and their images contain the original generators. Equivariance and the component relation are unchanged. $\square$

For example, a relation $f^2 g = f$ between commuting injections forces $fg = \text{id}$. Both maps are then bijections. The number of generators that look noninvertible in a presentation is consequently an unreliable invariant.

Source 04's example: this elementary observation prevents a significant ambiguity. The relation $f^2 = f$ does not make f the identity in the original idempotent monoid, but it forces $f = \text{id}$ in every injective action. All unit groups in our statements refer to the effective monoid P .

A direct construction of the localization (source 04, Section 2.2). For an injective P -set X , form pairs $(x, p) \in X \times P$ and put

$$(x, p) \sim (y, q) \iff q \cdot x = p \cdot y.$$

Transitivity uses cancellation of an acting element: from $q \cdot x = p \cdot y$ and $r \cdot y = q \cdot z$, applying r and p and cancelling q gives $r \cdot x = p \cdot z$. Denote the quotient by X^{loc} , and write $[x, p]$ for a class. For $a, b \in P$, define

$$(a - b) \cdot [x, p] = [a \cdot x, p + b].$$

The displayed equivalence and injectivity verify independence of all representatives. The resulting action is a G -action. The map $x \mapsto [x, 0]$ is an injective equivariant map.

This construction is the usual localization mechanism for injective commutative monoid actions; [21, §2.1] provides related background. We give the construction because its uniqueness properties are needed in the counting argument.

Theorem 29.2 (Localized component classification). *Every nonempty connected action of P by injective maps is isomorphic to translation on a nonempty subset $S \subseteq G/H$, for some subgroup $H \leq G$, such that $S + P \subseteq S$. Conversely every such subset is connected under the labelled generator maps. Two descriptions (H, S) and (H', S') give isomorphic labelled actions exactly when $H = H'$ and $S' = S + q$ for some $q \in G/H$.*

Sources: source 03, Theorem 6.2; source 04, Theorem 2.2 (“Translation normal form”).

Proof. For an action on X , form pairs $(p, x) \in P \times X$ and set

$$(p, x) \sim (q, y) \iff T_q x = T_p y.$$

This is an equivalence relation. For transitivity, apply T_s and T_p to $T_q x = T_p y$ and $T_s y = T_q z$, then cancel the injective T_q . The map $x \mapsto [0, x]$ is injective. On the quotient, $T_m[p, x] = [p, T_m x]$ becomes invertible, with inverse $[p, x] \mapsto [p + m, x]$. Hence the action extends to G .

Two original points are in the same connected component exactly when they have a common forward image $T_a x = T_b y$: this relation is an equivalence relation containing the generator edges, and every such equality supplies an undirected path. Therefore a connected component localizes to one transitive G -set, necessarily G/H . Its embedded image is P -stable.

Conversely, for $s, t \in S$ write $t - s = a - b$ modulo H , with $a, b \in P$. Then $s + a = t + b$ is a common forward point in S , proving connectedness. Translation gives injective maps.

An isomorphism extends to the localized transitive actions. As G is abelian, its stabilizer is independent of the basepoint, so the stabilizers agree. An equivariant isomorphism between these coset actions is translation. Conversely translation taking S onto S' preserves all labels. $\square$

Second proof (source 04). Localizing a connected action makes all generator steps invertible, so its localization is a transitive G -set. Choose a base point; its stabilizer H identifies the localization with G/H . The image of X is the required P -invariant subset.

For the converse, take $s, t \in S$. Since $G = P - P$, there are $a, b \in P$ with $t - s = a - b$ modulo H . Then

$$s + a = t + b.$$

Both paths to this common upper point stay in S , establishing connectedness.

An equivariant isomorphism $S \rightarrow S'$ extends uniquely to their localizations by $[x, p] \mapsto [\varphi(x), p]$. Equivariant isomorphisms between transitive abelian G -sets preserve stabilizers exactly, so an isomorphism requires $H = H'$. After this identification it is translation by the image of one base point. Restricting such a translation proves the reverse direction. $\square$

This explicit construction is the commutative specialization of the standard act-of-quotients construction [17, Section 6]. We need both the embedding and the isomorphism criterion, since merely bounding component cardinalities would not count their types.

Remark 29.3 (Labels and point stabilizers). The isomorphisms preserve the action of each monoid element. Permuting generators or applying an automorphism of G is not allowed. The subgroup H is the stabilizer of a *point* in the localization. It is different from the group of translations preserving the subset S as a whole, which is the automorphism group of the component.

Sources: source 04, Remark 2.3.

Corollary 29.4 (Periods are local automorphisms). *For a component $S \subseteq G/H$,*

$$\text{Aut}_P(S) \cong \text{Per}(S) := \{q \in G/H : S + q = S\}.$$

At the finite cutoff, its type contributes to the pure-parameter definable kernel exactly when its multiplicity and $\text{Per}(S)$ are both finite. For a torsion-free G/H , a finite period group is necessarily trivial.

Sources: source 03, Corollary 6.3 (stated without a separate proof: the first assertion is the isomorphism criterion of theorem 29.2 with $S' = S$, the second is corollary 27.2).

Removing unit directions (source 04, Section 2.3). Let

$$U = P \cap (-P).$$

This is a subgroup of G , equal to the unit group of P . Every P -invariant subset of G/H is invariant under the image of U , in both directions. It is therefore the full inverse image of a subset of

$$G/(H + U).$$

This simple quotient accounts for the effective number of one-way directions. A finitely generated abelian group has countably many subgroups: each subgroup is finitely generated, and there are only countably many finite tuples. The quotient G/U may have torsion (remark 30.2). Throughout, rank means the dimension after tensoring with $\mathbb{Q}$; we never silently discard torsion from the action classification.

30 The finite/countable/continuum theorem

30.1 Units and the dimension that matters

Put

$$U = P \cap (-P), \quad Q = G/U, \quad \bar{P} = \text{im}(P \rightarrow Q), \quad d = \text{rank}(Q).$$

The elements of U are precisely those monoid motions with monoid inverses. The integer d counts independent directions remaining after these units are removed. It can differ from both r and $\text{rank}(G)$.

Lemma 30.1 (Cone, units, and positive height). *Let $V = G \otimes_{\mathbb{Z}} \mathbb{R}$, let $C \subseteq V$ be the cone generated by the images of the monoid generators, and let $L = C \cap (-C)$. Then*

$$\text{span}_{\mathbb{R}}(U) = L, \quad d = \dim V - \dim L.$$

The monoid $\bar{P}$ has no nonzero units and no nonzero torsion elements. If $d > 0$, there is a homomorphism $\ell : Q \rightarrow \mathbb{Z}$ with

$$\ell(p) > 0 \quad (p \in \bar{P} \setminus \{0\}).$$

When $\bar{P} = \{0\}$, take $\ell = 0$. If $d = 0$, then $P = G$.

Sources: source 03, Lemma 7.1; source 04, Lemma 3.1 (“Unit directions are exactly the lineality directions”, which states $U \otimes_{\mathbb{Z}} \mathbb{R} = L$ and the positive height, including the convention $\ell = 0$).

Proof. Write p_i for generator images in G and v_i for their images in V . A generator is a unit exactly when $-v_i \in C$. The nontrivial implication uses a nonnegative rational expression for $-v_i$. Clear denominators and then multiply by the exponent of the finite torsion subgroup of G to obtain

$$Np_i + \sum_j c_j p_j = 0, \quad N \geq 1, c_j \geq 0.$$

It follows that $-p_i = (N-1)p_i + \sum_j c_j p_j \in P$.

Every unit is a sum of unit generators: if u and $-u$ are expressed using generators, their sum is a nonnegative relation equalling zero; each generator occurring in it has its inverse obtained by rearrangement. Also L is spanned by the v_i lying in L : express $v \in L$ and $-v$ in C and add the expressions. Every generator with positive coefficient in the resulting zero relation has its negative in C . These observations prove the span identity.

If $p, q \in P$ and $p + q \in U$, then $-p = q - (p + q) \in P$. Hence any unit in $\bar{P}$ is zero. A torsion element of a monoid inside a group is a unit, so $\bar{P}$ contains no nonzero torsion. Applying the span identity to $\bar{P}$ shows that its rational cone is pointed. Choose a rational functional strictly positive on this cone away from zero and clear denominators on the free quotient of Q . This gives ℓ ; finite torsion is killed by it. These standard cone facts are also developed in [19, Lemmas 3.4.4–3.4.5].

If $d = 0$, the finitely generated Q is finite. Its submonoid $\bar{P}$ is then a group, but has no nonzero units, so is zero. It generates Q , forcing $Q = 0$ and $P = U = G$. $\square$

Second proof of the span identity and the positive height (source 04). Units clearly map into L . If a generator p maps into L , then $-p$, in V , is a nonnegative rational combination of generator images. Clear denominators, and then multiply by the order of the remaining torsion discrepancy in G . This gives $-Np \in P$ for a positive integer N . Hence

$$-p = (N-1)p + (-Np) \in P,$$

so p is a unit.

Every vector of L is a nonnegative combination of generator images lying in L . Indeed, if $v = \sum a_i p_i \in L$ with $a_i > 0$, then $-a_i p_i = -v + \sum_{j \neq i} a_j p_j \in C$, whence $p_i \in L$. Thus the unit generators span L .

The quotient cone in V/L is pointed and rational. A rational linear functional can be chosen strictly positive on its nonzero generating rays. For completeness, normalize those finitely many nonzero rational generators to points on a sphere. Their convex hull does not contain zero, because a nontrivial nonnegative dependence summing to zero would give a line in the cone. Separation from zero gives a strictly positive functional; a rational approximation preserves the finitely many strict inequalities. Clearing denominators gives an integral homomorphism on the lattice G/U modulo torsion.

If an element $a \in \bar{P}$ had zero height, a representation as a sum of generators could use only unit generators. Hence $a = 0$. This establishes the stated positivity including the possible torsion in G/U . $\square$

Remark 30.2 (Torsion must be retained). Q need not be torsion-free. For example, the submonoid generated by $(1,0), (1,1)$ in $\mathbb{Z} \oplus C_2$ has $U = 0$. Its positive height is the first coordinate. The rank-one argument below uses finitely many cyclic cosets, so it includes this case.

Sources: source 03, Remark 7.2; source 04, Section 2.3 (the same example: “one has $U = 0$, so the torsion survives”).

Lemma 30.3 (Finite height levels). *For the positive height above, each set $\{a \in \bar{P} : \ell(a) = h\}$ is finite.*

Sources: source 04, Lemma 3.2.

Proof. Discard zero generators of $\bar{P}$. All remaining generators have positive integer heights. At a fixed nonnegative height, each coefficient in a nonnegative generator expression is bounded. There are only finitely many such expressions. Negative levels are empty. $\square$

30.2 Exact type count

Let $\tau(M)$ be the number of isomorphism types of nonempty connected labelled actions by injective maps.

Theorem 30.4 (Type-count trichotomy). *For every finitely generated commutative monoid,*

$$\tau(M) = \begin{cases} |\{H : H \leq G\}| = |\text{Sub}(G)|, & G \text{ finite,} \\ \aleph_0, & G \text{ infinite and } d \leq 1, \\ \mathfrak{c}, & d \geq 2. \end{cases} \quad (30.1)$$

Isomorphisms preserve the named generators. In the last case, already the rigid connected components have exactly $\mathfrak{c}$ types, all with the same localization stabilizer U , and there is a continuously coded Cantor family of pairwise nonisomorphic rigid components (theorem 30.6).

Sources: source 03, Theorem 7.3 and its equation (2); source 04, Theorem 1.1 (“The type classification, summarized”, with the Cantor-family clause), proved in its Section 3.4 (second proof below).

Proof. Every subgroup of a finitely generated abelian group is finitely generated. Thus G has at most countably many subgroups. The localization theorem reduces the problem to invariant subsets of the countable coset groups, up to translations. This also supplies the general upper bound $\mathfrak{c}$.

If G is finite, its submonoid P is all of G . A nonempty invariant subset of a transitive coset action is the whole action, giving exactly one type for each subgroup. If $d = 0$ with G infinite, the same argument gives at most countably many types.

Suppose $d = 1$. Pull back a P -invariant subset of G/H to G . It is invariant under both u and $-u$ for $u \in U$, and so is the inverse image of a $\bar{P}$ -invariant subset of Q . Choose nonzero $p \in \bar{P}$. It has infinite order, and $\mathbb{Z}p$ has finite index in Q . On each of the finitely many cyclic cosets $q_i + \mathbb{Z}p$, a subset invariant under $+p$ is empty, the whole coset, or

$$\{q_i + np : n \geq b_i\} \quad (b_i \in \mathbb{Z}).$$

It is therefore encoded by finitely many cutoffs in $\mathbb{Z} \cup \{-\infty, +\infty\}$. There are only countably many choices. This bounds the possibilities for every H , and hence the total type count. An infinite finitely generated abelian G has a quotient $\mathbb{Z}$, and therefore transitive actions of every finite size through $\mathbb{Z}/k\mathbb{Z}$. The image of P generates each finite quotient also as a monoid. These different cardinalities give infinitely many types, proving the countably infinite case.

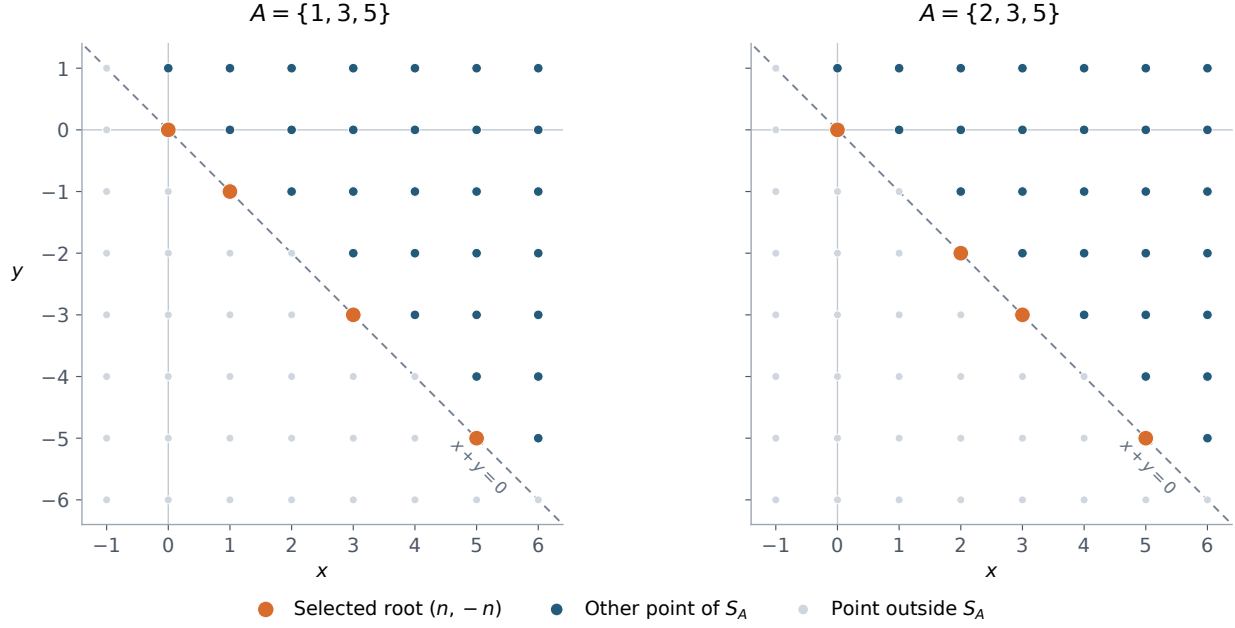


Figure 2: Finite windows of the construction in $\mathbb{Z}^2$, with $\bar{P} = \mathbb{N}^2$, $\ell(x, y) = x + y$, and $v = (1, -1)$. The orange zero-height points recover the chosen subset of positive multiples of v ; adding the quadrant gives the blue points. In the theorem the chosen subset may be infinite. The drawing illustrates the construction, rather than serving as evidence for its infinite cardinality. (Source 03, Figure 1.)

Finally assume $d \geq 2$. Use [lemma 30.1](#) to choose ℓ , and choose an infinite-order $v \in \ker \ell$, whose rank is $d - 1 \geq 1$. For each $R \subseteq \mathbb{N}$ set

$$A_R = \{0\} \cup \{(n+1)v : n \in R\}, \quad S_R = A_R + \bar{P} \subseteq Q. \quad (30.2)$$

It is a nonempty stable subset and hence a connected action. Positivity gives

$$S_R \cap \ker \ell = A_R, \quad \min \ell[S_R] = 0.$$

An isomorphism $S_R \rightarrow S_T$ is translation by $q \in Q$. Equality of minimum heights gives $\ell(q) = 0$, so $A_T = A_R + q$. Since 0 lies in both sets, $q \in A_T$ and $-q \in A_R$. Both are nonnegative multiples of the same infinite-order v , forcing $q = 0$. Thus $R = T$. The same argument for $R = T$ shows rigidity. This is an explicit injection of $\mathcal{P}(\mathbb{N})$ into the type set; no choice of representatives of translation classes is needed. The stabilizer of the localized G -action on Q is U . The upper bound already proved completes the trichotomy. $\square$

Source 04's route to the trichotomy (its Sections 3.2–3.4).

Proposition 30.5 (Countability in rank at most one). *If $d = \text{rank}(G/U) \leq 1$, then there are at most countably many connected injective P -action types.*

Sources: source 04, Proposition 3.3; the same argument as the case $d = 1$ in source 03's proof of [theorem 30.4](#).

Proof. Fix $H \leq G$. By the normal form and unit invariance, it suffices to count invariant subsets of $Q = G/(H + U)$. This group is finite or has rank one.

In the rank-one case the image of some $p \in P$ has infinite order, since the monoid image generates Q as a group. The cyclic subgroup $\mathbb{Z}p$ has finite index in Q . On each coset $r + \mathbb{Z}p$, closure under $+p$ means that the subset is empty, the whole coset, or a tail

$$\{r + np : n \geq k\} \quad (k \in \mathbb{Z}).$$

Thus finitely many values in $\mathbb{Z} \cup \{-\infty, +\infty\}$ determine the subset. There are at most countably many choices. The finite case is immediate. Finally, G has countably many subgroups H . $\square$

Theorem 30.6 (A Cantor family of rigid components). *If $d \geq 2$, there is a continuous map from $2^{\mathbb{N}}$ to the space of subsets of G/U whose images are connected injective action carriers, are pairwise nonisomorphic, and have trivial equivariant automorphism groups.*

Sources: source 04, Theorem 3.4; its family is source 03's (30.2), with continuity added.

Proof. Put $Q = G/U$. Choose the positive integral height ℓ . Its kernel has positive rank, so choose an element $v \in \ker \ell$ of infinite order. For $R \in 2^{\mathbb{N}}$, put

$$A_R = \{0\} \cup \{(n+1)v : n \in R\}, \quad S_R = A_R + \bar{P}.$$

Then S_R is nonempty and $\bar{P}$ -invariant, and hence connected. Positivity gives

$$\min \ell[S_R] = 0, \quad S_R \cap \ker \ell = A_R.$$

If $S_R + t = S_T$, their minimum heights force $\ell(t) = 0$. Their zero-height traces satisfy $A_R + t = A_T$. Since both traces lie in $\mathbb{N}v$ and contain zero, first $t \in \mathbb{Z}v$, and then the minimum integer coefficient in each trace forces $t = 0$. It follows that $R = T$. The same argument with $R = T$ proves rigidity, using [theorem 29.2](#).

For a fixed $a \in Q$, membership in S_R asks whether $a = b + n$ for $b \in A_R$ and $n \in \bar{P}$. Necessarily $\ell(n) = \ell(a)$, and there are only finitely many such n by [lemma 30.3](#). Membership therefore depends on finitely many bits of R . This is precisely continuity into the product space $2^{\mathbb{Q}}$. $\square$

Second proof of theorem 30.4 (source 04). If G is finite, its submonoid P is a group: positive powers of an element include its inverse. Since P generates G , one has $P = G$. A nonempty invariant subset of G/H is the whole quotient, so there is exactly one type for each subgroup.

If G is infinite, it has a quotient $\mathbb{Z}$. Reducing this quotient modulo each positive integer n supplies connected actions of cardinality n , and hence infinitely many types. [Proposition 30.5](#) therefore yields exactly $\aleph_0$ when $d \leq 1$.

When $d \geq 2$, [theorem 30.6](#) is the continuum lower bound. For the upper bound, there are countably many subgroups and at most continuum many subsets of each countable group G/H . This completes all cases. $\square$

30.3 Why finite generation matters

Example 30.7 (Rank one without finite generation). The additive monoid of all nonnegative rational numbers has group completion Q and rational rank one, but the invariant cuts $S_\alpha = \{q \in Q : q > \alpha\}$, for $\alpha \in \mathbb{R}$, have continuum many types modulo rational translation. Each translation class contains only countably many cuts. Thus rational rank alone does not extend [theorem 30.4](#)

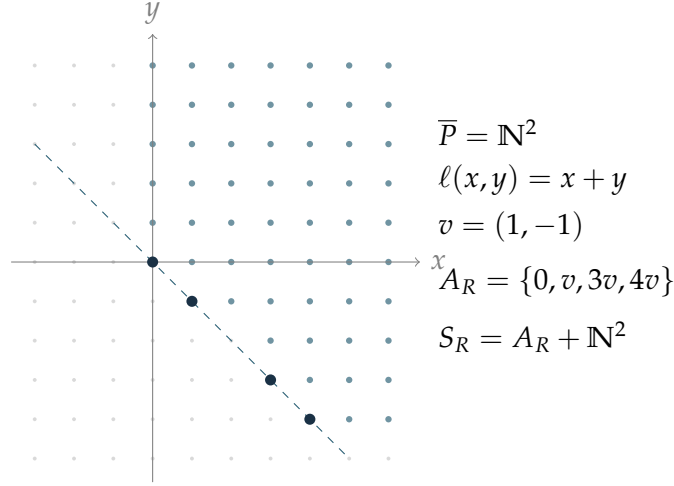


Figure 3: A finite window of a component from the higher-rank construction. The dark points on height zero encode the chosen bits; each generates its northeast quadrant. The proof concerns the entire infinite subset, not this finite window. (Source 04, Figure 1, drawn in TikZ; its labels M , B , S are printed $\bar{P}$, A_R , S_R .)

to infinitely generated monoids. The finite-index cyclic subgroup used in the countable case is indispensable.

In source 04's words: let $P = \mathbb{Q}_{\geq 0}$ inside $G = \mathbb{Q}$. Its unit group is zero and its rational rank is one. For $\alpha \in \mathbb{R}$, let $S_\alpha = \{q \in \mathbb{Q} : q > \alpha\}$. These are nonempty connected invariant subsets. They are translation-isomorphic exactly when $\alpha - \beta \in \mathbb{Q}$. Each isomorphism class contains at most countably many parameters, so there are continuum many types, all rigid. This violates both low-rank finiteness conclusions once finite generation is omitted.

Sources: source 03, Section 7.3; source 04, Example 11.2 (its rigidity clause, and “both low-rank finiteness conclusions”, those of [theorem 30.4](#) and [corollary 31.2](#), are its additions).

The failed step is precise: a positive cyclic subgroup of $\mathbb{Q}$ has infinite index, so finitely many integer thresholds no longer suffice. The example is only an algebraic limitation. It makes no claim that a language with infinitely many separate function symbols has the same Replacement behavior as a finite signature or as one uniform action evaluator. (Source 04, Section 11.2.)

31 A finer classification at a fixed stabilizer

The global countably infinite case hides a useful finiteness theorem. For a fixed subgroup $H \leq G$, let $Q_H = G/H$, let N_H be the image of P in Q_H , and let $V_H = N_H \cap (-N_H)$. Define

$$d_H = \text{rank}(Q_H/V_H).$$

The new unit group V_H can be larger than the image of U : passing to a quotient can make additional translations invertible.

Theorem 31.1 (Finite or continuum at fixed isotropy). *Among connected action types whose localization stabilizer is H , the number is one if $d_H = 0$, finite if $d_H = 1$, and continuum if $d_H \geq 2$.*

Sources: source 04, Theorem 4.1.

Proof. Every invariant subset is the inverse image of one in $Q = Q_H/V_H$, invariant under the sharp monoid N induced by N_H . Translation classes correspond exactly under this quotient. If $d_H = 0$, Q is finite. Its generating submonoid is a group, and sharpness forces $Q = 0$. There is only the full carrier. The construction of [theorem 30.6](#) in the quotient gives the continuum alternative when $d_H \geq 2$.

Suppose $d_H = 1$. Choose a positive height $\ell : Q \rightarrow \mathbb{Z}$ and an infinite-order $p \in N$. The quotient $Q/\mathbb{Z}p$ is finite. The image of N in this finite group is the whole group: it generates the group, and a submonoid of a finite group is a subgroup. Choose $n_r \in N$ in each coset. Every sufficiently high element of that coset is $n_r + kp$ with $k \geq 0$, so belongs to N . Thus

$$F = \{x \in Q : \ell(x) \geq 0\} \setminus N$$

is finite, since height fibers in a rank-one finitely generated group are finite.

A nonempty proper invariant subset S cannot be unbounded below in height. Otherwise one of the finitely many $\mathbb{Z}p$ -cosets contains points of S arbitrarily far below; $+p$ invariance fills that coset, and addition by the n_r 's fills all cosets. Translate a point of minimum height in S to zero. The normalized carrier satisfies

$$N \subseteq S \subseteq \{x : \ell(x) \geq 0\}.$$

It is $N \cup E$ for some $E \subseteq F$, leaving only finitely many possibilities. Include the full carrier as one further type. $\square$

This proof gives a finite method for a fixed rank-one quotient: enumerate subsets of F , retain the generator-invariant ones, and identify translations. The global infinity in low rank can come from countably many stabilizers instead of complicated boundaries inside a single quotient.

Corollary 31.2 (Rigid-component dichotomy). *There are finitely many rigid connected action types when $\text{rank}(G/U) \leq 1$, and continuum many when $\text{rank}(G/U) \geq 2$.*

Sources: source 04, Corollary 4.2.

Proof. Rigidity of a carrier in G/H forces $U \subseteq H$, because unit translations preserve the carrier. If G/U is finite, the action is transitive with automorphism group G/H , so only the singleton is rigid. If G/U has rank one, either H/U is finite or it has rank one. There are finitely many finite subgroups, all in the finite torsion subgroup; each contributes finitely many types by [theorem 31.1](#). In the other case G/H is finite, again leaving only the singleton. The higher-rank conclusion is [theorem 30.6](#). $\square$

32 A decision procedure from the relations

Suppose the input is a finite commutative presentation

$$M = \langle x_1, \dots, x_r \mid x^{u_j} = x^{v_j}, 1 \leq j \leq s \rangle, \quad u_j, v_j \in \mathbb{N}^r.$$

Let B be the integer matrix with rows $u_j - v_j$. Then $G = \mathbb{Z}^r / \langle u_j - v_j \rangle_{\mathbb{Z}} = \mathbb{Z}^r / \text{row}_{\mathbb{Z}}(B)$ and its free rank is $g = r - \text{rank}_{\mathbb{Q}}(B)$; $P = \text{im}(\mathbb{N}^r \rightarrow G)$. Define the rational polyhedral cone

$$\mathcal{W}_B = \{w \in \mathbb{R}^r : Bw = 0, w_i \geq 0 \text{ for all } i\}.$$

Commutativity is part of the input; this is not an algorithm for arbitrary noncommutative presentations (source 04). Following source 04, put

$$\mathcal{W}_B^{\mathbb{Q}} = \mathcal{W}_B \cap \mathbb{Q}^r, \quad J = \{j : \exists w \in \mathcal{W}_B^{\mathbb{Q}}, w_j > 0\}, \quad K = \{1, \dots, r\} \setminus J,$$

and write B_J for the submatrix retaining the columns in J .

Theorem 32.1 (An effective relation invariant; exact presentation test). (i) *The integer d of [theorem 30.4](#) is*

$$d = \dim \operatorname{span}_{\mathbb{R}}(\mathcal{W}_B).$$

(ii) *The effective unit group U is generated by the images of the K -generators, and*

$$G/U \cong \mathbb{Z}^I / \operatorname{row}_{\mathbb{Z}}(B_J).$$

In particular,

$$d = |J| - \operatorname{rank}_{\mathbb{Q}}(B_J) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \mathcal{W}_B^{\mathbb{Q}}. \quad (32.1)$$

(iii) *G is finite exactly when $r - \operatorname{rank}_{\mathbb{Q}}(B) = 0$.*

Consequently a finite presentation decides whether its connected action types are finite, countably infinite, or continuum many. No enumeration of infinite components is required.

Sources: source 03, Theorem 8.1 (“An effective relation invariant”: (i) and the decision); source 04, Theorem 5.1 (“Exact presentation test”: (ii), (iii), the decision, and a second proof of (i)) and its equation (5.1).

Proof (source 03). The dual of $V = \mathbb{R}^r / \operatorname{rowspan}(B)$ consists of the vectors w with $Bw = 0$. Such a functional is nonnegative on the image cone of the generators exactly when all $w_i \geq 0$. Thus $\mathcal{W}_B$ is its dual cone. For a full-spanning closed polyhedral cone $C \subseteq V$, the annihilator of the span of its dual is $C \cap (-C)$. Indeed, every dual functional vanishes on that lineality space; conversely, if all dual functionals vanish on v , both v and $-v$ belong to C by polyhedral separation. Taking dimensions and applying [lemma 30.1](#) gives the formula.

The group rank and finite torsion invariants are computable by rational elimination and Smith normal form. The span of a rational polyhedral cone is computable by finite exact linear methods. If $g = 0$, G is finite; otherwise compare d with one. The trichotomy then supplies the answer. $\square$

Proof (source 04); a second proof of (i). Each $w \in \mathcal{W}_B^{\mathbb{Q}}$ defines a homomorphism on G , nonnegative on P , which vanishes on units. Thus a generator in J is not a unit. Conversely, the rational-cone argument of [lemma 30.1](#) identifies the generators outside J with the lineality generators and makes them units.

There is a useful exact certificate. Infeasibility of

$$Bw = 0, \quad w \geq 0, \quad w_j = 1$$

gives by rational linear separation a vector λ with $B^T \lambda \geq e_j$. Clearing denominators gives an integer z and $v \in \mathbb{N}^r$ with

$$B^T z = e_j + v. \quad (32.2)$$

This certifies $[e_j] + [v] = 0$ in G . Conversely, pairing such an identity with $w \in \mathcal{W}_B^{\mathbb{Q}}$ forces $w_j = 0$. The separation used here is the finite rational linear-inequality alternative, equivalently the dual cone description of lineality in [lemma 30.1](#).

If a unit $u = \sum n_j[e_j]$ had a representation using an index $j \in J$, a nonnegative witness with $w_j > 0$ would make $w(u) > 0$. Thus units use only the K -generators. Killing those generators gives the stated presentation and rank.

For the span identity, add one witness for each index in J to obtain $w^* \in \mathcal{W}_B^{\mathbb{Q}}$ positive exactly on J . For any $z \in \ker_{\mathbb{Q}}(B_J)$, sufficiently small positive rational ε makes $w^* \pm \varepsilon z$ nonnegative. Their difference spans z , proving the identity. $\square$

Smith normal form can recover the torsion in G/U . The algorithm decides the algebraic branch of the foundational classification; it does not compute the continuum or decide an arbitrary set-theoretic cofinality inequality. (Source 04.)

[write] **Question 23.5, 4 October 2026, batch 96.** [Theorem 32.1](#) decides, for a finite commutative *monoid* presentation, which of the three cases of [theorem 30.4](#) holds for $\tau(M)$. This is the commutative-monoid analogue of [Question 23.5](#), which asks the same about $\tau(G)$ for finitely presented groups. It is not the commutative case of that question: for finitely generated abelian groups given by commutative presentations the answer was already immediate (Section 24.2). Source 04’s abstract and README call its result “a commutative answer” to Question 23.5; read as the monoid analogue, the claim is correct, and source 04’s Section 1.3 itself makes “no decidability claim for arbitrary groups”. Question 23.5 stays open beyond abelian groups.

32.1 A small exact implementation

The accompanying program `verify_presentations.py` implements one elementary finite procedure. For each nonempty support $I \subseteq \{1, \dots, r\}$, it computes the rational nullspace of the column submatrix B_I . If this nullspace is one-dimensional and has a generator strictly positive on I , the program records its primitive integer vector, extended by zeros outside I . These are exactly the extreme rays of $\mathcal{W}_B$; their rank is d .

For completeness, an extreme nonnegative null vector must have a one-dimensional nullspace on its support. Otherwise an independent kernel direction permits small positive and negative perturbations while preserving nonnegativity, contradicting extremality. Conversely a vector with a one-dimensional support nullspace cannot split into nonproportional nonnegative kernel vectors. The cone lies in the nonnegative orthant and is pointed; its extreme rays span it. Alternatively, successive perturbations reducing support give a direct finite decomposition proof. Thus the support search is exact. It is exponential in the number of generators and is intended for small, independently inspectable certificates. More efficient polyhedral algorithms may replace it without changing the theorem.

Source 04’s finite exact algorithm (its Section 5.2). For each nonempty support $S \subseteq \{1, \dots, r\}$, compute the rational kernel of B_S . Retain S if that kernel is one-dimensional and contains a vector strictly positive on all of S . Extend this vector by zero outside S . These are the positive circuits of B .

Lemma 32.2 (Completeness of circuit enumeration). *The union of retained supports is J . It suffices to examine supports of size at most $\text{rank}(B) + 1$.*

Sources: source 04, Lemma 5.2. Source 03’s completeness argument above (extreme rays) is the same decomposition; the size bound is source 04’s.

Proof. Take nonzero $w \in \mathcal{W}_B^{\mathbb{Q}}$ with support S . If the restricted kernel has dimension one, its support is retained. Otherwise, take an independent rational kernel vector and subtract a suitable multiple

Commutative presentation	Source	$g = \text{rank } G$	d	Connected types
One unrestricted injection	03	1	1	Countably infinite
Two unrestricted injections (no relation on f, g)	03, 04	2	2	Continuum
One permutation and one injection	03	2	1	Countably infinite
$f^2 = g^3$	03, 04	1	1	Countably infinite
$f^2 = g^4$	03	1	1	Countably infinite; torsion C_2
$f^2 = g^2$	04	1	1	Countably infinite; 2-torsion survives
$f^2 = g^3 h$	03	2	2	Continuum
$fg = \text{id}$	03, 04	1	0	Countably infinite
$f^2 = g^2, fg = \text{id}$	03	0	0	Finite; completion C_4
$fh = gh$	03	2	2	Continuum; cancellation forces $f = g$
$h^2 = h$ (source 04: $f^2 = f$)	03, 04	0	0	One type; injectivity forces $h = \text{id}$
$fg = h$	04	2	2	Continuum
$f^2 g = g^2 f$	04	1	1	Countably infinite
$f^2 = g^3, h^5 = f^7$	04	1	1	Countably infinite

Table 6: Selected exactly checked relation matrices. All maps commute, and all are injective. The code of source 03 contains fourteen presentations in total; source 04’s code checks its eight rows (section 36). [write] Source 03’s Table 2 (ten rows) and source 04’s Table 1 (eight rows, “Exact tests distinguish the number of symbols from the number of noninvertible directions”, with a column rank G , which is source 03’s g) merged; the column “Source” names the tables in which a row appears.

of w to obtain a vector z with both signs on S . Moving maximally in both directions gives

$$w^+ = w + \alpha z \geq 0, \quad w^- = w - \beta z \geq 0$$

with strictly smaller supports and

$$w = \frac{\beta}{\alpha + \beta} w^+ + \frac{\alpha}{\alpha + \beta} w^-.$$

Induction on support size decomposes w into retained nonnegative vectors, so every index in J occurs. For a retained support, $|S| = \text{rank}(B_S) + 1 \leq \text{rank}(B) + 1$. $\square$

Clear denominators and divide by the gcd on each retained vector. Their sum is an integer grading positive exactly on J , with zero set exactly the effective units. A certificate consists of this grading and one identity (32.2) for each index outside J . The accompanying implementation (source 04’s `verify_results.py`) uses rational arithmetic throughout and includes singleton supports of zero columns.

32.2 Examples

For $f^2 = g^3 h$, the relation row is $(2, -3, -1)$. The two primitive nonnegative kernel rays are $(1, 0, 2)$ and $(3, 2, 0)$, proving $d = 2$ directly. For $fg = \text{id}$, the relation row is $(1, 1)$, and the nonnegative

kernel is zero: the group still has rank one, but there is no irreversible direction. These examples show why counting equations or subtracting their number from the number of generators is insufficient.

Source 04's examples: for $f^2 = g^3$, the nonnegative kernel of $(2, -3)$ is generated by $(3, 2)$. For $fg = 1$, the kernel cone of $(1, 1)$ is zero: both generators are units although $G \cong \mathbb{Z}$ is infinite. For $fg = h$, the kernel cone of $(1, 1, -1)$ has dimension two. In the final row $(15, 10, 21)$ is a positive integer grading satisfying both relations.

33 Complete enumeration for a power relation

Fix positive integers m, n and require that f, g be commuting injections with $f^m = g^n$. The two generators remain distinguished. Let $g_0 = \gcd(m, n)$ and write $\text{div}(g_0)$ for the number of positive divisors of g_0 , and $\sigma_1(s)$ for the sum of the positive divisors of s . For positive a, b , define

$$\text{Neck}(a, b) = \frac{1}{a+b} \sum_{q|\gcd(a,b)} \varphi(q) \binom{(a+b)/q}{a/q}, \quad (33.1)$$

where φ is Euler's totient function. This is the number of cyclic weak compositions of b into a parts, as proved below; by the symmetry of the formula it is also the number of cyclic weak compositions of a into b parts, and source 04 reads it as the number of fixed-content binary necklaces with a symbols of one kind and b of the other.

Theorem 33.1 (All connected types for $f^m = g^n$; exact common-power spectrum). *For the fixed labelled operations f, g :*

(i) *The number of finite connected types of cardinality $k \geq 1$ is*

$$C_k(m, n) = \sum_{q|\gcd(k, g_0)} q = \sigma_1(\gcd(g_0, k)). \quad (33.2)$$

(ii) *There are $\text{div}(g_0)$ bilateral infinite types, meaning whole infinite localized coset groups with both operations bijective.*

(iii) *The number of proper infinite types is*

$$\sum_{e|g_0} \text{Neck}(m/e, n/e),$$

and in every proper type both f and g are proper (non-surjective) injections.

Thus the exact finite number of infinite types is

$$I(m, n) = C_\infty(m, n) = \text{div}(g_0) + \sum_{e|g_0} \text{Neck}(m/e, n/e). \quad (33.3)$$

For fixed e , the proper types are explicitly indexed by weak compositions of n/e into m/e parts, modulo cyclic rotation.

Sources: source 03, Theorem 9.1 and its equations (4)–(5); source 04, Theorem 6.1 (“Exact common-power spectrum”) and its equations (6.1)–(6.3), which write $C_k(m, n)$ and $C_\infty(m, n)$ and add the last clause of (iii) (“the first term counts components where both maps are permutations; in every component in the second term both are proper injections”).

33.1 Localization stabilizers and finite components

Proof of the finite and bilateral assertions (source 03). The group completion is

$$G = \mathbb{Z}^2 / \langle (m, -n) \rangle \cong \mathbb{Z} \oplus C_{g_0}.$$

Both generator images have positive degree under the homomorphism with $\deg(f) = n/g_0$, $\deg(g) = m/g_0$. Hence $U = 0$, $d = 1$, in agreement with the countably infinite total type count from [theorem 30.4](#).

A finite component is a whole coset group G/H of index k . Write $T = C_{g_0}$ for the torsion subgroup. If $T/(H \cap T)$ has order q , then $q \mid g_0$. The image of H in the free factor is $a\mathbb{Z}$ for some $a > 0$, and $k = aq$. Conversely, for each $q \mid \gcd(k, g_0)$, the torsion intersection is uniquely determined, while a lift of $a = k/q$ has q possible torsion coordinates modulo that intersection. These choices give distinct subgroups and all index- k subgroups. This proves (i).

An infinite quotient requires H to have rank zero, so H is a subgroup of C_{g_0} . There is exactly one such subgroup H_e of each order $e \mid g_0$. Its inverse image in $\mathbb{Z}^2$ is generated by $(m/e, -n/e)$, and therefore

$$G/H_e \cong \mathbb{Z}^2 / \langle (a, -b) \rangle, \quad a = m/e, b = n/e.$$

For each e , the whole quotient is one bilateral type. Stabilizers are invariants of labelled isomorphism, so these types do not merge. $\square$

Source 04's notation for the same group. Write the group generators of f, g as u, v :

$$G_{m,n} = \mathbb{Z}u \oplus \mathbb{Z}v / \langle mu - nv \rangle \cong \mathbb{Z} \oplus C_{g_0}.$$

Its effective positive rank is one. So $G_{m,n}$ is G above, and $G_{a,b}$, for $a = m/e$, $b = n/e$, is the quotient G/H_e , called Q_e below.

Second proof of (i) (source 04, "Finite components: subgroups with a twist"). On a finite set both maps are permutations, so connected actions correspond to index- k subgroups of $\mathbb{Z} \oplus C_{g_0}$. For such a subgroup H , set

$$c = [C_{g_0} : H \cap C_{g_0}].$$

If its projection to $\mathbb{Z}$ is $r\mathbb{Z}$, the index is $rc = k$. Thus $c \mid \gcd(g_0, k)$ and $r = k/c$. The torsion intersection is uniquely determined by c . A lift of the positive projection generator r may have any of c torsion residues modulo that intersection. These give precisely c subgroups. Sum over c . $\square$

The finite connected generating function and cumulative count are (source 04)

$$\sum_{k \geq 1} C_k(m, n) z^k = \sum_{c \mid g_0} \frac{cz^c}{1 - z^c}, \quad (33.4)$$

$$\sum_{k=1}^N C_k(m, n) = \sum_{c \mid g_0} c \left\lfloor \frac{N}{c} \right\rfloor = \text{div}(g_0)N - \sum_{c \mid g_0} c \left\{ \frac{N}{c} \right\}. \quad (33.5)$$

The sequence is periodic with period dividing g_0 , and its mean over a period is exactly $\text{div}(g_0)$.

33.2 Boundary functions and cyclic compositions

Proof of the proper-component assertion (source 03). Fix e , and let S be a nonempty proper subset of $Q_e = \mathbb{Z}^2 / \langle (a, -b) \rangle$, invariant under both positive coordinate translations. Its full inverse image $T \subseteq \mathbb{Z}^2$ is upward closed in the coordinate order and invariant under both signs of the period $(a, -b)$.

Every vertical column has a point in T . Starting from $(i_0, j_0) \in T$, translate by a sufficiently large negative multiple of $(a, -b)$ to reach a column left of the desired one, then move right. Every column also has a point outside T : start outside, translate by a sufficiently large positive multiple of $(a, -b)$, and use downward closure of the complement to move left. Consequently there is a unique integer boundary function h with

$$T = \{(i, j) : j \geq h(i)\}, \quad h(i+1) \leq h(i), \quad h(i+a) = h(i) - b.$$

Its drops

$$c_i = h(i) - h(i+1)$$

are nonnegative, periodic with period a , and satisfy $c_0 + \cdots + c_{a-1} = b$. Conversely any such weak composition defines h after choosing $h(0) = 0$, and hence defines a proper component.

Translation by (u, v) sends $h(i)$ to $h(i - u) + v$. Thus it cyclically shifts the drop sequence. Conversely, once a horizontal translation aligns the drops, a vertical translation aligns the boundary values. By [theorem 29.2](#), these are all labelled isomorphisms. The proper types for e are therefore exactly the rotation classes of weak compositions of b into a parts.

Apply the orbit-counting lemma to rotation by the cyclic group C_a . A rotation of order $q \mid a$ fixes a composition only if $q \mid b$, in which case it is q repetitions of a weak composition of b/q into a/q parts. The number fixed is

$$\binom{(a+b)/q - 1}{a/q - 1}.$$

There are $\varphi(q)$ rotations of order q . Averaging the fixed counts gives

$$\frac{1}{a} \sum_{q \mid \gcd(a,b)} \varphi(q) \binom{(a+b)/q - 1}{a/q - 1} = \text{Neck}(a, b),$$

using the elementary binomial identity converting this to [\(33.1\)](#). Summing over $e \mid g_0$ proves (iii) and [\(33.3\)](#). $\square$

In this proof (u, v) is a translation vector, not source 04's pair of generators.

Source 04's route through gap words (its Section 6.3). For an infinite component the localization stabilizer H is finite and lies in C_{g_0} . There is one such subgroup of each order $e \mid g_0$; its quotient is $G_{a,b}$ with $a = m/e$, $b = n/e$, retaining the distinguished generators.

Lemma 33.2 (Threshold and gap normal forms). *Every nonempty proper invariant subset of $G_{a,b}$ has the unique form*

$$S = \bigcup_{j=0}^{b-1} \{ku + jv : k \geq t_j\}, \quad t_0 \geq t_1 \geq \cdots \geq t_{b-1} \geq t_0 - a. \quad (33.6)$$

Its translation class is the cyclic rotation class of the weak composition

$$(c_0, \dots, c_{b-1}) = (t_0 - t_1, \dots, t_{b-2} - t_{b-1}, a + t_{b-1} - t_0), \quad \sum_j c_j = a. \quad (33.7)$$

Sources: source 04, Lemma 6.2 and its equations (6.6)–(6.7). The gap word of length b and sum a is the dual description of source 03's drop sequence of length a and sum b .

Proof. The quotient modulo $\mathbb{Z}u$ is C_b . Closure under $+u$ makes each coset intersection a ray, empty, or full. Addition by v cycles through the cosets, with wraparound $bv = au$. An empty coset forces all empty, and a full coset forces all full. Thus a nonempty proper subset has a ray in every coset.

Closure under $+v$ gives exactly the displayed inequalities. Their gaps are nonnegative and sum to a , and conversely the gaps and t_0 reconstruct the thresholds. Translation by u shifts every threshold equally. Translation by v gives $(t_{b-1} + a, t_0, \dots, t_{b-2})$, rotating the gaps. Since u, v generate the group, these are exactly the translation identifications. $\square$

Lemma 33.3 (The gap-word count). *Weak compositions of a into b parts modulo cyclic rotation are counted by $\text{Neck}(a, b)$.*

Sources: source 04, Lemma 6.3; the same orbit count as in source 03's proof above, for the dual composition.

Proof. A rotation of order q fixes only compositions that repeat q times, requiring $q \mid a$. If $q \mid \gcd(a, b)$, the fixed count is $\binom{(a+b)/q-1}{b/q-1}$. There are $\varphi(q)$ such rotations. Counting fixed word-rotation pairs and dividing by b gives

$$\frac{1}{b} \sum_{q \mid \gcd(a, b)} \varphi(q) \binom{(a+b)/q-1}{b/q-1}.$$

Use $\binom{(a+b)/q-1}{b/q-1} = \frac{b}{a+b} \binom{(a+b)/q}{a/q}$ to get (33.1). Equivalently, replace each gap entry by that many symbols followed by one separator, reading the resulting binary word cyclically. $\square$

Second proof of (ii) and (iii), by source 04. Each $e \mid g_0$ contributes its full infinite quotient and $\text{Neck}(m/e, n/e)$ proper types. Different e 's have different localization stabilizers. If f is surjective, so is $f^m = g^n$, hence g is surjective. Both generators are then invertible on the carrier, which must be the full quotient. The same argument starts with g . Thus the proper types have both maps non-surjective. $\square$

33.3 Exact local symmetry of each boundary

Theorem 33.4 (Boundary periods). *For a proper component associated with e , put $a = m/e$, $b = n/e$. Let p be the least positive period of its length- a drop sequence and let $\delta = a/p$. Then $\delta \mid \gcd(a, b)$ and*

$$\text{Aut}(S) \cong C_\delta.$$

The local automorphism group of a bilateral infinite type is countably infinite. Consequently, at $\kappa = \omega$, a proper infinite type of multiplicity μ contributes to the definable atom kernel exactly when μ is finite; an individual atom in an untouched proper type is definable from pure reals exactly when the residual multiplicity and δ are both one. Bilateral infinite types contribute no atoms to the pure-parameter finite kernel.

Sources: source 03, Theorem 9.2; source 04, the first paragraph of the proof of its Theorem 6.4 (theorem 33.5), a second proof of $\text{Aut}(S) \cong C_\delta$. “The definable atom kernel” here is the pure-parameter one of theorem 27.1.

Proof. The drop sequence consists of δ repetitions of its primitive block, whose sum is b/δ . Thus δ divides b . Any translation preserving T has horizontal coordinate a multiple of p , and the boundary recurrence then forces its vertical coordinate. Precisely,

$$\text{Per}(T) = \mathbb{Z}(p, -b/\delta).$$

The relation vector killed in Q_e is $(a, -b) = \delta(p, -b/\delta)$. Therefore

$$\text{Per}(S) = \text{Per}(T)/\mathbb{Z}(a, -b) \cong C_\delta.$$

For a whole infinite quotient, all its translations are periods, giving a countably infinite group. Apply [corollary 27.2](#) and [theorem 27.3](#). $\square$

Here e and δ have different meanings. The first is the order of the subgroup identified with zero by localization; the second counts nontrivial translations of an already formed component that preserve its boundary. A shorter boundary period is an automorphism, not an additional identification of points. Confusing the two would undercount types.

Automorphism groups and exact rigidity counts (source 04, Section 6.4). With $\mu_{\text{Möb}}$ the number-theoretic Möbius function, put

$$\text{Prim}(a, b) = \frac{1}{a+b} \sum_{q|\gcd(a,b)} \mu_{\text{Möb}}(q) \binom{(a+b)/q}{a/q}.$$

Möbius inversion on repetition number counts primitive cyclic gap words by this expression.

Theorem 33.5 (Automorphism spectrum). *For each $\delta \mid g_0$, exactly $\text{Neck}(m/\delta, n/\delta)$ proper infinite connected types have equivariant automorphism group C_δ . Thus there are $\text{Neck}(m, n)$ rigid infinite types and $1 + \text{Neck}(m, n)$ rigid connected types in total.*

Sources: source 04, Theorem 6.4 (its e is δ here and its h is e).

Proof. In $G_{a,b}$, a gap word repeating exactly δ times has period b/δ and period sum a/δ . The translation

$$\eta = (b/\delta)v - (a/\delta)u$$

preserves the carrier and has exact order δ . Every preserving translation induces a preserving rotation modulo $\mathbb{Z}u$; each rotation has at most one compensating u -translation, since a proper ray has no nonzero u -period. The automorphism group is exactly C_δ .

For the quotient indexed by $e \mid g_0/\delta$, the count with exact repetition δ is $\text{Prim}(m/(e\delta), n/(e\delta))$. Summing gives

$$\sum_{e|g_0/\delta} \text{Prim}(m/(e\delta), n/(e\delta)) = \text{Neck}(m/\delta, n/\delta).$$

This is the decomposition of a necklace into its unique primitive repeated word; algebraically it uses $\sum_{e|k} e \mu_{\text{Möb}}(k/e) = \varphi(k)$. The full infinite quotients have infinite automorphism groups. A finite transitive abelian action has automorphism group its carrier group, so only the singleton is rigid. $\square$

[write] On the indices of [theorem 33.5](#), 4 October 2026, batch 96. The theorem fixes the automorphism order δ and sums over the localization quotients e in which a gap word can repeat exactly δ times; a

component of Q_e with automorphism group C_δ has $\delta \mid \gcd(m/e, n/e)$, that is, $e\delta \mid g_0$, which is the range $e \mid g_0/\delta$ of the sum. It is a statement about the distribution of automorphism orders over *all* proper types, whereas [theorem 33.4](#) computes the group of each one; together they give source 03's per-component statement a census. The placement's independent check confirmed the census for all $m, n \leq 8$ (Section 24.5).

Proposition 33.6 (Defect profile). *For a proper infinite component with localization quotient $G_{m/e, n/e}$,*

$$|S \setminus f(S)| = n/e, \quad |S \setminus g(S)| = m/e.$$

Thus its two defect sizes determine the divisor e , although they do not determine the cyclic gap word.

Sources: source 04, Proposition 6.5.

Proof. Translation by u misses exactly the initial point of each of the $b = n/e$ ray fibers. Translation by v misses the successive gaps in (33.7), whose total is $a = m/e$. $\square$

This refines the general identity $m|S \setminus f(S)| = n|S \setminus g(S)|$ whenever the two defects are finite. For any injection T , the complement of $T^r(S)$ is the disjoint union of $T^j(S \setminus T(S))$, $0 \leq j < r$; apply this to $f^m = g^n$.

33.4 Coprime powers and rational Catalan numbers

When $\gcd(m, n) = 1$, only $e = q = 1$ occur, and

$$I(m, n) = 1 + \frac{1}{m+n} \binom{m+n}{m}.$$

Identify G with $\mathbb{Z}$ by $f(x) = x + n$, $g(x) = x + m$. There is exactly one finite type on $\mathbb{Z}/k\mathbb{Z}$ for each $k \geq 1$, one bilateral type on $\mathbb{Z}$, and one proper type for each normalized subset $\Delta \subseteq \mathbb{N}$ containing zero and closed under $+m, +n$. These are the normalized semimodules of the numerical semigroup $\langle m, n \rangle$. Their rational Catalan enumeration is classical, with the related simultaneous-core enumeration due to Anderson [20]. The boundary proof above derives the count directly and also treats the noncoprime case.

To see the normalization explicitly, $\langle m, n \rangle$ contains every sufficiently large integer. A nonempty stable subset of $\mathbb{Z}$ contains a translate of it. If unbounded below, it is all of $\mathbb{Z}$; otherwise translate its minimum to zero. Distinct normalized subsets cannot be translates. Thus this familiar semigroup enumeration describes actual connected action types.

For $f^2 = g^3$, the proper normalized subsets are $\mathbb{N}$ and $\mathbb{N} \setminus \{1\}$. The three infinite types therefore have domains

$$\mathbb{Z}, \quad \mathbb{N}, \quad \mathbb{N} \setminus \{1\},$$

with $f = +3, g = +2$ in every case. This is a complete classification of the infinite components, not merely three examples.

Source 04's account of the coprime case (its Section 6.5). For $\gcd(m, n) = 1$,

$$C_k(m, n) = 1, \quad C_\infty(m, n) = 1 + \frac{1}{m+n} \binom{m+n}{m}.$$

All proper infinite components are rigid. Identifying the completion with $\mathbb{Z}$, a normalized proper carrier is a subset of $\mathbb{N}$ containing zero and closed under $+m, +n$. These are classical numerical-semigroup semimodules, counted by rational Catalan numbers. Their Dyck-path and core-partition

(m, n)	Bilateral infinite types	Proper infinite types	Total infinite types
$(2, 3)$	1	2	3
$(2, 2)$	2	3	5
$(2, 4)$	2	4	6
$(3, 3)$	2	5	7
$(3, 6)$	2	11	13
$(4, 4)$	3	13	16

Table 7: Exact consequences of [theorem 33.1](#), also checked by finite composition enumeration. Generator labels are fixed throughout. (Source 03, Table 3.)

(m, n)	Infinite types	Rigid infinite types	Finite count at size k
$(1, 1)$	2	1	1
$(2, 3)$	3	2	1
$(3, 4)$	6	5	1
$(2, 2)$	5	2	1 odd; 3 even
$(2, 4)$	6	3	1 odd; 3 even
$(3, 3)$	7	4	1 if $3 \nmid k$; 4 otherwise
$(4, 4)$	16	10	$\sigma_1(\gcd(4, k))$

Table 8: Exact component counts with both maps distinguished. The infinite count includes the full quotient components. (Source 04, Table 2.)

connections are established combinatorics [22, 23]; the present application puts them inside the full injection-action classification.

For a direct coprime count, a binary word has m steps of weight n and n of weight $-m$. Partial sums before the final step are distinct: an equality would give a proper subword with symbol counts in ratio $m : n$, impossible by coprimality. Exactly one cyclic rotation starts after the minimum and has nonnegative partial sums. Each rotation orbit has length $m + n$, yielding the rational Catalan count.

For $(m, n) = (2, 4)$, the trivial localization stabilizer gives drop sequences $(4, 0)$, $(3, 1)$, and $(2, 2)$ up to rotation. Their automorphism groups have orders 1, 1, 2. The order-two localization stabilizer adds one proper ray type; two bilateral types bring the total to six. The example exhibits the localization and boundary-symmetry distinction concretely.

33.5 All finite actions, with and without labelled vertices

Let $L_N(m, n)$ count the ordered pairs of commuting permutations on the fixed set $\{1, \dots, N\}$ satisfying $f^m = g^n$, without requiring connectedness.

Corollary 33.7 (Disconnected finite actions). *As formal power series,*

$$\sum_{N \geq 0} \frac{L_N(m, n)}{N!} z^N = \prod_{c | g_0} (1 - z^c)^{-1}, \quad (33.8)$$

$$\sum_{N \geq 0} U_N(m, n) z^N = \prod_{k \geq 1} (1 - z^k)^{-\sigma_1(\gcd(g_0, k))}, \quad (33.9)$$

where $U_N(m, n)$ counts isomorphism classes of arbitrary finite actions.

Sources: source 04, Corollary 6.6 and its equations (6.8)–(6.9).

Proof. A connected type on k points has k automorphisms, so has $(k - 1)!$ realizations on a fixed labelled set. Decomposition into unordered connected components gives

$$\sum_{N \geq 0} \frac{L_N}{N!} z^N = \exp \left(\sum_{k \geq 1} C_k \frac{z^k}{k} \right).$$

But

$$\sum_{k \geq 1} C_k \frac{z^k}{k} = \sum_{c | g_0} \sum_{j \geq 1} \frac{z^{cj}}{j} = - \sum_{c | g_0} \log(1 - z^c).$$

For unlabelled actions each connected type may occur with arbitrary finite multiplicity, giving the second product of geometric series. $\square$

An independent count uses $G_{m,n} \cong \mathbb{Z} \times C_{g_0}$: choose t with $t^{g_0} = 1$, then any permutation commuting with t . Each conjugacy class of t contributes exactly $N!$ pairs. Its cycle lengths divide g_0 , producing the same partition product. Thus finite labelled counts depend only on the gcd, although the infinite counts depend on both exponents.

34 Replacement, Collection, and reflection

34.1 The fixed-action criterion

The following criterion is inherited from the repository's component-profile analysis [13]. We give a short proof using the new rooted codes, so the application does not depend on an unproved coding assertion. For a fixed finite injective signature let

$$D_\kappa = \bigcup \{A_t : |A_t| < \kappa\},$$

where A_t is the entire unpointed component-type block.

Theorem 34.1 (Fixed-action Replacement; established criterion). *For $\kappa > \omega$, the expansion satisfies Replacement exactly when $|D_\kappa| < \kappa$. At $\kappa = \omega$, it satisfies Replacement exactly when every connected component is finite and D_ω is finite.*

Sources: source 03, Theorem 10.1; source 04, Theorem 8.1 ("Component-block criterion"). Both credit it as prior work: it is [theorem 5.1](#) of Part I and [theorem 14.2](#) of Part II. Source 03's proof goes through the rooted codes of Section 26, source 04's through Part I's enumeration graphs.

Proof. For a pure rooted code c , the formula

$$\text{Type}(c, a) \iff \exists b [\text{Code}(c, b) \wedge b \text{ is connected to } a]$$

recognizes the full unpointed type block represented by c , or the empty class if the code is not realized. Connectedness uses finite signed paths. Define a total set-valued selector on $\mathcal{R}$ by taking that exact atom class when there is an internal set containing exactly it, and taking $\emptyset$ otherwise.

Extensionality gives uniqueness, and the existence test is a first-order quantifier over model sets. This selector is definable without presuming its range exists internally. Its range kernels have union exactly D_κ . Replacement would make the range an internal set, proving $|D_\kappa| < \kappa$.

At $\kappa = \omega$, an infinite connected component also gives the atom-valued word-evaluation counterexample described after [theorem 27.3](#); so finiteness of every component is necessary.

For sufficiency, let a formula define unique outputs on an internal domain w , with parameter tuple $\bar{a}$. Put $B = \ker(w) \cup \bigcup_i \ker(a_i)$. If $\kappa > \omega$, then $|\text{Sat}(B)| \leq \max(\aleph_0, |B|) < \kappa$; the harmless empty case is immediate. By [theorem 27.1](#) and [corollary 27.2](#), all output kernels lie in $\text{Sat}(B) \cup D_\kappa$, which is small. At the finite cutoff, B is finite, all components are finite, and D_ω is finite, giving the same conclusion. The external range set consequently has a small atom kernel and is internal. This proves each Replacement instance. $\square$

[write] On the finite-cutoff step of the proof above, 4 October 2026, batch 96. At $\kappa = \omega$, [theorem 27.1](#) puts the output kernels inside $\text{Sat}(B) \cup D_\omega^{\text{pt}}$, and [corollary 27.2](#) identifies D_ω^{pt} with D_ω only for $\kappa > \omega$. When every component is finite, the two agree at $\kappa = \omega$ as well: then $\delta_t \leq |X_t| < \omega$, so $\mu_t \delta_t < \omega$ iff $\mu_t < \omega$ iff the block A_t is finite. So the step “giving the same conclusion” is correct under the hypothesis that all components are finite, which is the hypothesis in force there. (The write’s reading of the proof; source 03 leaves this step to the reader.)

Source 04’s setting (its Section 8). Every component of the named atom graph is countable: from one atom there are finitely many forward and partial inverse generator steps at each finite distance. Code connected types by actions on a positive finite ordinal or on $\mathbb{N}$, quotienting the pure set of codes by generator-preserving isomorphism. Let $\mathcal{T}_r$ be the resulting pure set of types. For $t \in \mathcal{T}_r$, define externally

$$B_t = \{a \in A : \text{the component of } a \text{ has type } t\}, \quad D_\kappa = \bigcup_{\substack{t \in \mathcal{T}_r \\ |B_t| < \kappa}} B_t.$$

The cutoff is applied to the entire block of all components of one type, not to a single component. The following criterion and its basic symmetry mechanism are prior work in ProveIt [13, Parts I–II]. We include a complete proof to make the new threshold corollaries independent of an unchecked import.

Second proof (source 04). Uniform recognition of a type. Given a pure type code t and an atom a , quantify over a representative code and an injection of its domain into the atoms whose range contains a . Require equivariance for every named generator and closure of the range under incoming edges:

$$\forall b (\text{At}(b) \wedge J_i(b) \in \text{ran}(h) \implies b \in \text{ran}(h)).$$

The range is connected and closed under both directions, so is the whole component. Conversely, an actual component enumeration witnesses these conditions. Its graph has countably many atoms and is therefore internal for $\kappa > \omega$. For finite types it is internal even at $\kappa = \omega$. This is a uniform first-order definition; no instance of expanded Replacement is used to create the enumeration.

Necessity at an uncountable cutoff. On the internal pure set $\mathcal{T}_r$, define a functional relation that sends t to the set of all atoms of its type if that whole set exists internally, and to the empty set otherwise. Extensionality gives uniqueness. The outputs are B_t for the small blocks, and empty sets for the others. Replacement followed by Union would collect D_κ internally, forcing the bound.

Necessity at the finite cutoff. Fix an atom a . On the pure set of finite words in forward and formal inverse generator symbols, evaluate the endpoint from a when the path exists, and return a

otherwise. A successful path has a finite witnessing graph; injectivity makes inverse steps unique. This is an internally definable total function whose range is the component of a . Replacement would make it an internal, hence finite, atom set. With all components now finite, the type-block selector on finite type codes also forces D_ω finite.

Sufficiency. Suppose a fixed expanded formula with parameter tuple $\bar{a}$ specifies a unique y_x for each x in an internal domain w . Let $\text{cl}(B)$ be the union of components meeting an atom set B , and put

$$H = \text{cl} \left(\ker(w) \cup \bigcup_j \ker(a_j) \right) \cup D_\kappa.$$

At an uncountable cutoff, closing a set of size $\lambda < \kappa$ under countable components has size at most $\max(\lambda, \aleph_0) < \kappa$. At the finite cutoff, closing finitely many atoms is finite because their components are finite. Thus H is small in the relevant sense and is component-closed.

Suppose $b \in \ker(y_x) \setminus H$. Its block has size at least κ . At an uncountable cutoff it therefore has at least κ components, since each component is countable. At the finite cutoff it has infinitely many components of the same finite size. Choose a component of the same type disjoint from $H \cup \ker(y_x)$. Swap it with the component containing b , through an isomorphism and its inverse, fixing all other components. This permutation commutes with every named generator. Its recursive lift is an automorphism of the expanded model fixing $x, \bar{a}$, but changing the kernel of y_x . This contradicts uniqueness.

Every output kernel is therefore contained in H . Ambient Replacement collects the range of the fixed formula; that range has atom kernel inside H , so is internal. This proves the entire expanded scheme. $\square$

Remark 34.2 (Why singular cardinals cause no hidden step). Component closure uses a *uniform countable* bound. It does not assume that every union of fewer than κ sets of size below κ is small. That false assumption would erase precisely the cofinality obstruction appearing next.

Sources: source 04, Remark 8.2.

34.2 Sharp universal thresholds from the presentation

The preceding criterion translates a supply of component types into exact thresholds. This supply argument is already present in the free-signature repository setting; the new type classification determines it for all the commutative relations considered here. (Source 04, Section 9.)

Theorem 34.3 (Universal Replacement classification). *Fix a finitely generated commutative monoid M and an infinite cardinal κ . The canonical expansions preserve Replacement for every action by injective maps on every atom set $|A| \geq \kappa$, after the recursive lifts of a finite generating tuple are named, exactly as follows:*

Algebraic case	Condition on κ
G finite	every infinite κ
G infinite, $d \leq 1$	$\text{cf}(\kappa) > \aleph_0$
$d \geq 2$	$\text{cf}(\kappa) > \mathfrak{c}$

(34.1)

Failure of the stated condition is witnessed on any prescribed atom set of cardinality at least κ . Every negative case admits an action faithful on the effective monoid P , in which an element acts bijectively precisely when it is a unit of P .

Sources: source 03, Theorem 10.2 and its equation (6); source 04, Theorem 1.2 (“Universal Replacement, summarized”) and Theorem 9.1 (“Uniform preservation and sharp witnesses”, which adds the last sentence).

Proof. If G is finite, there are finitely many finite connected types. A finite union of small blocks is small at every infinite cutoff, so [theorem 34.1](#) applies, including at ω .

Suppose G is infinite. At $\kappa = \omega$, its translation action on G is an infinite connected permitted action and violates Replacement. It can be installed on countably many atoms with fixed-point filler on the rest.

Now let $\kappa > \omega$. By [theorem 30.4](#), the number τ of possible types is $\aleph_0$ or $\mathfrak{c}$. If $\text{cf}(\kappa) > \tau$, any union of at most τ sets individually of size below κ has size below κ . Hence D_κ is small for every action.

For sharpness assume $\eta = \text{cf}(\kappa) \leq \tau$. Choose η distinct nontrivial connected types, and infinite cardinals $\lambda_i < \kappa$, $i < \eta$, with cardinal sum κ . At a regular uncountable cutoff, take κ many indices and $\lambda_i = \aleph_0$; at a singular cutoff, take a cofinal sequence of infinite cardinals. Give type i exactly λ_i copies. Its block has size $\lambda_i < \kappa$, since each component is countable. Their union has size κ . Add at least κ fixed atoms, using a separate part of the available atom set, and put any remaining atoms there. This is possible even when $|A| = \kappa$ by splitting an infinite set into two equicardinal parts. The fixed type was not selected, so it does not alter the small blocks. Now $|D_\kappa| \geq \kappa$, and the selector in [theorem 34.1](#) is a concrete failed instance. (Source 03’s proof does not address the last sentence of the theorem; source 04’s does.) $\square$

Second proof (source 04). If P is finite, it is the finite group G . There are finitely many component types, all finite. The small-block union is a finite union of small sets, so [theorem 34.1](#) applies at every infinite cutoff.

Let P be infinite and write $\tau = \tau(M)$. If $\text{cf}(\kappa) > \tau$, the cutoff is uncountable. The union of at most τ blocks, each of size below κ , has size below κ . The criterion proves universal preservation.

For the converse suppose first that $\kappa > \omega$ and $\delta = \text{cf}(\kappa) \leq \tau$. Choose δ distinct connected types t_α , omitting the singleton and regular- P types. Choose infinite cardinals $\lambda_\alpha < \kappa$ with $\sum_{\alpha < \delta} \lambda_\alpha = \kappa$. At a regular uncountable cutoff take $\lambda_\alpha = \aleph_0$; at a singular cutoff use a cofinal sequence of infinite cardinals. Realize exactly λ_α copies of t_α . Each whole block has size $\lambda_\alpha < \kappa$, but their union has size κ . On unused atoms use the trivial action. The selector in the necessity proof of [theorem 34.1](#) is an explicit failed Replacement instance.

Reserve additionally one countable component carrying the regular action of P on itself. It is faithful. Every unit acts bijectively, whereas a nonunit translation misses 0 in this component. This enforces exactly the stated invertibility behavior without altering the bad blocks, because its type was excluded from them.

At $\kappa = \omega$, the regular component is infinite; the finite-word endpoint function already witnesses failure. Finally substitute $\tau = \aleph_0$ or $\mathfrak{c}$ from [theorem 30.4](#). $\square$

For two free commuting injections, $d = 2$. At $\kappa = \aleph_1$, some actions therefore destroy Replacement. Requiring $f^2 = g^3$ changes d to one, and *every* permitted action then preserves Replacement at that same cutoff. An additional independent injection restores $d = 2$. In contrast, at $\kappa = \aleph_\omega$ the countable-type case can already fail, since its cofinality is countable. These are exact consequences of the relations and the cutoff, not consequences of the individual lifts ceasing to be elementary.

What the common-power relation changes (source 04, Section 9.1). For every positive m, n , the relation $f^m = g^n$ places the effective monoid in the rank-one case. Thus every expansion by those

two commuting injections preserves Replacement exactly when

$$\text{cf}(\kappa) > \aleph_0.$$

This includes the case where both maps are proper injections, as the sharp witness construction permits. Two unconstrained commuting injections instead have rank two and require

$$\text{cf}(\kappa) > \mathfrak{c}.$$

For instance, at $\kappa = \aleph_1$ every common-power action preserves Replacement, while some actions of two free commuting injections fail it. The latter claim uses only $\aleph_1 \leq \mathfrak{c}$, not the continuum hypothesis.

Individual preservation is more flexible (source 04, Section 9.2). Failure of the universal condition does not mean that every action fails. A trivial action preserves Replacement at every cutoff. More structurally, an action with every realized type block of size at least κ has $D_\kappa = \emptyset$, hence preserves Replacement at every uncountable cutoff. The obstruction lies in having too many individually small recognizable blocks, rather than in the mere existence of many types.

The answer is independent of finite generating lists. Every generator in one list is a finite word in another, and the recursive lifts obey the same word identities. Changing the finite list is a definitional expansion or reduct and leaves the full scheme unchanged.

[write] Consistency with Parts I and II, 4 October 2026, batch 96. Theorem 34.3 contains the earlier universal thresholds of this report as special cases, and agrees with each. One injection ($M = \mathbb{N}$, $d = 1$): $\text{cf}(\kappa) > \aleph_0$, the case $r = 1$ of theorem 6.1. Commuting s permutations and t injections ($M = \mathbb{Z}^s \times \mathbb{N}^t$, presented with a generator for the inverse of each permutation, whose lift is definable from the permutation's lift, so that naming it changes no scheme; $G = \mathbb{Z}^{s+t}$, $U = \mathbb{Z}^s$, $d = t$): $\text{cf}(\kappa) > \omega$ for $t \leq 1$ and $\text{cf}(\kappa) > \mathfrak{c}$ for $t \geq 2$, which is theorem 16.3 (with theorem 15.3 for $t = 0$). A finite abelian group named by generators (their inverses are positive powers): every infinite κ , corollary 15.4. An infinite finitely generated abelian group named by generators (with their inverses, as before): $\text{cf}(\kappa) > \aleph_0$, the abelian case of theorem 18.2, where $\tau(G) = \aleph_0$ and G is infinite. Two non-commuting permutations, the case $r \geq 2$ of theorem 6.1, lie outside theorem 34.3, which assumes commutativity.

34.3 Collection and reflection: the exact universal table

Here reflection means the following finite-family scheme. Given finitely many expanded formulas and an internal set of prescribed objects, there is an internal transitive set containing those objects, closed under the named functions, on which those formulas are absolute for parameter tuples from it. This is the full finite-family reflection convention used in [13]; it is stronger than merely finding a set that agrees on one selected sentence with no parameters.

Source 04's formulation (its Section 10). To connect directly with Glazer and Yao's subject, we distinguish Replacement from Collection and specify the reflection principle being used. Full expanded reflection here means: for every finite family of expanded formulas and every internal set of parameters, there is an internal transitive set t , closed under all named lifts, containing the prescribed parameters, such that all those formulas agree between the full structure and t on all tuples from t . This is an external scheme of finite-formula assertions. The reservoir argument below is the existing kernel-model mechanism of [13, Part I, §7]. The classification for general monoid relations results from the type theorems proved in this article.

Proposition 34.4 (Established fixed-action Collection criterion). *Let $\theta = |I(f)|$ be the number of component types actually realized by a fixed finite tuple of injections. Expanded Collection and the stated reflection scheme are equivalent. They hold exactly when*

$$\kappa = \nu^+ \text{ is an uncountable successor cardinal and } \theta < \kappa.$$

Sources: source 03, Proposition 10.3; source 04, Theorem 10.1 (“A fixed-action Collection criterion”, with the condition written $\kappa = \nu^+ > \omega$ and $|I| \leq \nu$). Both credit it as prior: it is [theorems 7.5 and 7.7](#) of Part I.

Proof. At a limit cutoff, including ω , base-language Collection already fails. On a pure domain of cardinals unbounded below κ , ask for a graph injecting each cardinal into the atoms. Each individual witness is internal; a collecting set would require a kernel of size at least every cardinal below κ , an impossibility.

If $\theta \geq \kappa$, choose pure representative rooted codes for at least κ distinct realized unpointed types. The relation asking for one atom of the specified type is total on that pure set of codes. Any collecting set requires at least one different atom per type and cannot be internal.

Conversely let $\kappa = \nu^+$, with ν infinite and $\theta \leq \nu$. Saturate the input and parameter kernels by components, using at most ν atoms. Include every whole type block of size at most ν . For each remaining type choose ν fresh components outside this set. Since a large block has at least ν^+ components, these choices are possible. The resulting component-closed reservoir has size at most ν .

Any internal witness involves at most ν components. Leave the already included small type blocks fixed. For each large type, move the components the witness uses outside the fixed parameter saturation into the fresh reservoir copies. Such partial matches extend to permutations of the complete component families, since fewer than ν^+ components are constrained. Choose the corresponding labelled component isomorphisms. The lifted automorphism fixes the inputs and parameters and moves the witness into the common reservoir. External Collection now supplies witnesses whose collected kernel lies in that one reservoir. This proves expanded Collection.

For reflection, carry out the reservoir construction repeatedly. Use full hierarchy stages $V_{\alpha_n}(B_n)$ over increasing component-closed reservoirs B_n of size at most ν , with increasing ordinal bounds α_n . Choose the first stage to contain the prescribed objects. At each successive stage include witnesses for the relevant existential subformulas over all tuples from the current stage, enlarge the atom reservoir by fresh components, and increase the rank bound. There are only set-many such tuples, so external Collection supplies a common rank bound for the compressed witnesses. The union of the countably many full hierarchy stages is transitive and closed under the named operations, which preserve rank and each B_n . Its atom kernel has size at most ν . Induction on the finite family of formulas and their subformulas proves absoluteness. The atom reservoir must grow at each stage: an automorphism fixing a reservoir pointwise cannot move outside atoms into that same fixed reservoir.

Finally reflection implies Collection by reflecting both a totality statement on a given domain and its witnessing formula, with the domain and parameters among the prescribed objects. Transitivity then puts every input in the reflecting set, which supplies genuine witnesses and is itself a collecting set. $\square$

Second proof (source 04). Limit cutoffs fail Collection in the base language. If κ is a limit cardinal, choose a pure set of ordinals below κ whose cardinalities are unbounded below κ . For $\kappa = \omega$ use the finite ordinals. Each ordinal in the domain has an internal injection into the atoms. A collecting set of

such injection graphs would have a kernel containing at least that many atoms for every size below κ , contradicting the cutoff.

Too many types fail Collection at a successor. At $\kappa = \nu^+ > \omega$, the recognition formula in the second proof of [theorem 34.1](#) gives, for every $t \in I = I(f)$, an atom of that type. Collection would gather at least one atom of each type in an internal set. Distinct types use distinct atoms, so $|I| \leq \nu$ is necessary.

Reservoir compression. Assume this bound and protect a component-closed set P_0 of at most ν atoms containing the kernels of the inputs and parameters of a Collection instance. Add every whole type block of size at most ν . There are at most ν types, so the enlarged protected set still has size at most ν . For each remaining type, reserve ν components outside the protected set. Their union with it is a fixed reservoir R of at most ν atoms.

Every internal witness meets at most ν components. For each large type, a permutation of its component family can move the witness components into the reserved ones while fixing the protected set. Such a partial matching extends to a permutation: the full family has cardinality at least ν^+ . Choose component isomorphisms to lift these permutations. The resulting expanded automorphism fixes every input and parameter and carries the witness into the reservoir. Ambient Choice and Collection now choose witnesses for the set-sized domain, all supported on R ; their collection is internal.

Growing reservoirs give reflection. First express universal quantifiers through negation and existential quantification, then close the finite formula family under subformulas. Build externally increasing stages

$$X_n = V_{\alpha_n}(H_n) \quad (n < \omega),$$

with H_n component-closed and of size at most ν . Start with all prescribed parameters included. For every true existential subformula and every tuple in X_n , use the reservoir construction fixing H_n to place a witness over a common H_{n+1} . There are only set-many such tuples; choose a rank α_{n+1} containing all the chosen witnesses. Increase ranks further if necessary for nesting and closure under the finitely many named lifts. Then $t = \bigcup_n X_n$ is transitive and internally supported on at most ν atoms. Induction on subformulas proves the required absoluteness.

Reflection implies Collection. Reflect both the matrix of a total relation and its totality assertion. Include the domain w itself and the parameters in t , so transitivity places all elements of w in t . Totality in t provides witnesses there, and absoluteness of the matrix makes them genuine. Thus t is a collecting set. $\square$

The countable union of reservoirs is harmless even when ν is singular: every reservoir has the same upper bound ν , and $\nu \cdot \aleph_0 = \nu$. The reservoir must grow; one fixed atom supply could itself occur as a parameter in a later stage and fail to supply fresh witnesses. (Source 04.)

Corollary 34.5 (Universal Collection and reflection). *For all actions of a fixed finitely generated commutative monoid, the exact conditions for Collection and the stated full reflection are:*

$$\left. \begin{array}{l} G \text{ finite, or } G \text{ infinite with } d \leq 1 \\ d \geq 2 \end{array} \right| \begin{array}{l} \kappa \text{ an uncountable successor,} \\ \kappa \text{ an uncountable successor and } \kappa > c. \end{array}$$

Sources: source 03, Corollary 10.4 (its second row reads “ κ a successor and $\kappa > 2^{\aleph_0}$ ”; such κ is uncountable); source 04, Corollary 10.2 (“Uniform Collection and reflection”). It generalizes [theorem 20.1](#) of Part II.

Proof. Combine [proposition 34.4](#) with [theorem 30.4](#). If the possible type count reaches the cutoff, an action realizing that many types supplies the necessary counterexample. All its components are countable, so it fits on an atom set of size at least κ after adding filler. $\square$

Second proof (source 04). At $\kappa = \nu^+$, every action realizes at most $\tau(M)$ types, so $\tau(M) \leq \nu$ is sufficient. If $\tau(M) > \nu$, then $\tau(M) \geq \kappa$; realize κ different countable types on a subset of A . The fixed-action theorem gives failure. All limit cutoffs already fail in the base language. Substitute the exact type count. $\square$

Example 34.6 (Replacement without Collection at a successor). Take the free two-generator monoid and $\kappa = |A| = \aleph_1$. Choose $\aleph_1$ distinct connected types from the Cantor family and give each exactly $\aleph_1$ copies. There are still only $\aleph_1$ atoms in total. All realized blocks have size κ , so $D_\kappa = \emptyset$ and Replacement holds. But $\aleph_1$ types are realized, exceeding the reservoir bound $\nu = \aleph_0$, so Collection and full reflection fail. This is an explicit distinction between the two criteria within the same successor-cutoff construction.

Sources: source 04, Example 10.3. It is [example 7.10](#) of Part I with commuting injections in place of the two permutations f, g there, which do not commute.

The new input to these axiom tables is the classification by relations. The underlying Collection/reflection equivalence is an established result for this externally choice-based family. This does not purport to settle any additional choiceless separation problem from Glazer and Yao's paper.

35 Finite codes, forcing, and new connected types

35.1 When forcing adds new connected types

The proof gives more than a cardinal count: it distinguishes finite descriptions from genuine real parameters. The following statement concerns a fixed finite commutative presentation belonging to a ground model of ZFC. Types in an extension are compared with interpretations of the ground-model carriers.

Theorem 35.1 (Type absoluteness and new reals). *If $d \leq 1$, every connected type in a forcing extension is isomorphic to a ground-model type. If $d \geq 2$, an extension has a connected type absent from the ground model if and only if it has a new real.*

Sources: source 04, Theorem 11.1.

Proof. The finitely generated abelian group G , its elements and its finite tuples are unchanged. Every subgroup is finitely generated, so there are no new subgroups of G . For $d \leq 1$, the proof of [proposition 30.5](#) describes each carrier by its subgroup and finitely many integer thresholds in a finite or virtually cyclic quotient. These finite data are already in the ground model. The invariance conditions on them are absolute.

For any d , a carrier in a countable quotient is coded by a real. With no new reals, the normal form introduces no new types. If $d \geq 2$ and R is a new real, use the fixed ground-model height and vector in [theorem 30.6](#) to construct S_R . Were it isomorphic to a ground-model type, that type could be put into a ground-model carrier $S' \subseteq G/U$; normal form would give $S_R = S' + t$ for a

ground-model group element t . Then S_R would be a ground-model set, and its height-zero trace would recover R , a contradiction. $\square$

The theorem does not assert that arbitrary individual expanded kernel models retain their schemes under every forcing. The cutoff's cardinality, its cofinality, and the sizes of the realized blocks can change. It identifies exactly when forcing can introduce new *types* for a fixed finite commutative presentation. The algebraic integer d itself is absolute.

[write] Relation to Part II's forcing theorem, 4 October 2026, batch 96. [Theorem 21.1](#) of Part II concerns a *fixed* ground action: if κ stays a cardinal, the expanded-Replacement verdict is unchanged, because component types and D_κ are absolute. [Theorem 35.1](#) concerns the *supply* of types for a fixed presentation: in rank $d \leq 1$ no new types appear, and in rank $d \geq 2$ new types appear exactly with new reals. The two complement each other: by the second, for $d \geq 2$ a forcing extension adding reals has new actions, built from new types, which is how the universal thresholds of [theorem 34.3](#) can change while every ground action keeps its verdict; this is the distinction [Question 23.11](#) and [Question 37.16](#) ask to make precise. (Source 04 does not cite [theorem 21.1](#).)

36 Verification, formalization, and the trust boundary

36.1 What has actually been checked

The delivered package includes the complete LaTeX source, the compiled PDF, the exact-arithmetic program, its JSON results, and a reproducible vector figure. Separate ChatGPT agents reviewed the written mathematical arguments along three routes: rooted-component coding and kernels; monoid localization and the type trichotomy; and the power-relation enumeration, including its torsion and period groups. The reviews found no remaining mathematical counterexample within the hypotheses stated here. This is an audit of written proofs, not a substitute for publication review or proof-assistant verification. (Source 03.)

The executed program performs the following distinct checks:

1. Fourteen relation matrices are evaluated by exact rational support-ray enumeration and Smith normal form. The predicted group ranks, cone dimensions, and finite torsion factors are asserted.
2. For all 64 pairs $1 \leq a, b \leq 8$, weak compositions are enumerated directly and grouped by canonical cyclic rotation. The program compares this count with the divisor formula ([33.1](#)) and with a separately computed average of fixed-word counts under every rotation.
3. Every enumerated cyclic class has its local stabilizer order computed. The code verifies orbit-stabilizer and divisibility of the stabilizer order by $\gcd(a, b)$.
4. Five coprime numerical semigroups are checked by enumerating subsets of their finite gap sets and imposing closure under both generators directly. The counts for $(2, 3)$, $(3, 4)$, $(3, 5)$, $(4, 5)$, $(4, 7)$ are 2, 5, 7, 14, 30, respectively.

The cyclic-composition grid contains 24,301 weak compositions and 3,521 rotation classes. For $(a, b) = (8, 8)$, it finds 810 classes: 800 with trivial stabilizer, 8 with stabilizer order 2, and one each with orders 4 and 8. These checks probe torsion and periodicity cases that a coprime-only experiment would miss.

The general proofs of infinite cardinalities and axiom schemes do not come from these finite enumerations. The JSON's cardinality labels are applications of [theorem 30.4](#), while its rays and finite counts are directly computed. This division of responsibilities is part of the artifact's documented verification boundary.

Mathematical review (source 04, Section 13.1). Separate internal proof reviews checked cancellativization, localization and translation equivalence, the torsion issue, positive heights, the rigid Cantor family, the fixed-stabilizer finite case, the common-power necklace and subgroup counts, automorphism groups, and the transfer to Replacement and reflection. The main conceptual corrections preserved in the final statements are:

- units belong to the cancellativization, not necessarily the original presented monoid;
- G/U can retain finite torsion;
- quotient stabilizers require recomputing the effective unit group;
- generator-preserving types differ from actions on labelled vertices;
- small entire type blocks differ from small individual components;
- reflection requires growing atom reservoirs and reflection of the totality assertion.

Exact computational companion (source 04, Section 13.2). The included `verify_results.py` uses only Python's standard library and exact integer or Fraction arithmetic. Its `verification.json` records inputs, witnesses, individual outcomes, and totals. All **148 cases** passed:

Independent check	Cases	Objects examined
Rank and unit certificates	8	8 circuits; 3 unit identities
Compositions versus necklace formula	36	1,709 compositions
Finite subgroups versus divisor sum	24	3,759 candidate subsets
Permutation pairs versus (33.8)	80	9,888 pair/exponent checks

The composition cases use $1 \leq m, n \leq 6$. The subgroup cases use $C_k \times C_D$, with $1 \leq D \leq 4$, $1 \leq k \leq 6$, and enumerate subgroups of order D . The permutation cases use $0 \leq N \leq 4$ and $1 \leq m, n \leq 4$. These checks exercise independent finite interpretations of the formulas; they do not replace proofs of the infinite or foundational theorems.

Run with Python 3.10 or later:

```
python verify_results.py
```

To build the document, run `pdflatex` twice on `commuting_injections.tex`, or use `latexmk -pdf`. The bibliography and figure are contained in the single source file.

[write] Shipped layout and replay, 4 October 2026, batch 96. In ProveIt the programs and records of this Part are `code/03-atom-actions-verify_presentations.py` (source 03's `verify_presentations.py`, SymPy), `code/03-atom-actions-make_figure.py` (its `figures/make_figure.py`, Matplotlib), `data/03-atom-actions-verification_results.json` (its `verification_results.json`) and `figures/03-atom-actions-antichain_upsets.pdf` with the preview `figures/03-atom-actions-antichain_upsets.png` (its `figures/antichain_upsets.*`, [figure 2](#)); and `code/04-commuting-injections-verify_results.py` with `data/04-commuting-injections-verification.json` (source 04's `verify_results.py` and `verification.json`). Source 04's figure is drawn by TikZ in this

article (figure 3). The text above names the delivered files; both programs, run without arguments, write their record *beside themselves*, under the delivered name, and the figure script writes `antichain_upsets.pdf` and `.png` beside itself, so all three must be run on copies or with `-output`; the commands `"latexmk -pdf atom_actions.tex"` and `"pdflatex ... commuting_injections.tex"` refer to the delivered articles, which are not shipped (this article is the merged text). At this write, on scratch copies: source 04's program passed all 148 cases in under a second (Python 3.14.4) and its record is byte-identical to the shipped one; source 03's program, run with SymPy 1.14.0 (Python 3.13.5) and `-output`, passed its assertions and printed the counts quoted above, and its record equals the shipped one except for line endings and the field `python` (3.13.5 against the delivered 3.12.14); the figure script, run with Matplotlib 3.10.8, reproduced the PDF except for its creation date (TrueType fonts, no Type 3) and the PNG pixel for pixel. The report's README gives the commands.

36.2 A concrete route into ProveIt

The repository already distinguishes exploratory computations from results connected to formal semantics by proved checkers [15]. This article fits that approach. A useful formalization order is (source 03):

1. **Finite signed paths.** Define the partial evaluation relation for a finite family of injections; prove endpoint uniqueness and the component equivalence relation. This portion is ordinary graph theory and does not need a universe of sets with atoms.
2. **Rooted codes.** Formalize the equality-diagram code, construct the unique pointed isomorphism from equality of codes, and prove the multiplicity formula. The key output is a theorem connecting a pure code to exactly one commuting-automorphism orbit.
3. **Algebraic localization.** Construct the quotient of pairs, prove the embedding and group action, and establish the coset-subset classification with label-preserving isomorphisms.
4. **Finite cone certificates.** Make the relation matrix and primitive rays executable data. Prove a checker for the ray-span claim and connect its output to d . The supplied exact script can generate candidate certificates; it need not be trusted as the checker.
5. **Power-relation boundaries.** Formalize the integer cutoff recurrence, cyclic weak compositions, and the finite group action on them. The period quotient is a particularly concrete algebraic target.
6. **Set-theoretic interpretation.** Add a universe-level or first-order-model presentation of atoms, actual kernels, and fixed-formula relativization. Prove the range-kernel lemmas before connecting the component classification to Replacement and reflection.

This ordering separates elementary finite certificates from the ambient set-theory assumptions. A cardinal-valued trichotomy should be stated in an explicit universe containing the type codes; a schema theorem should be stated over formula syntax and a defined satisfaction relation for the chosen model representation. Introducing an unrestricted truth predicate for the proper-class universe would change the foundations and is unnecessary.

[write] **Formal status, 4 October 2026, batch 96.** None of these six steps exists in ProveIt. As recorded in Section 1.4, the formula type shared by SetTheory/ZF and SetTheory/BoundedConsistency has no atom predicate and no function symbols, and no urelement theory is formalized; no monoid localization, cone certificate or necklace enumeration of this Part is formalized either.

36.3 Scope of the resolution

The paper gives a complete answer to the specified commutative-monoid cardinality question in the inspected repository. It also gives exact common-power enumeration and effective certificates that go beyond a cardinality answer. The older component and reservoir mechanisms are explicitly credited. No claim is made to settle Glazer and Yao's broader choiceless program, to resolve unrelated Glazer questions, or to have produced a kernel-checked formalization.

The finite certificates, the ordinary mathematical proofs, and a future proof-assistant development are distinct verification layers. The evidence provided here supports the stated theorems and their precise hypotheses; historical priority would require a broader specialist literature review. (Source 04, Section 13.3.)

37 Further questions and research

The following questions concern extensions beyond the proved statements. They are proposed research tasks, not assertions that every wording is a longstanding published open problem. In particular, neither the noncoprime power-relation enumeration nor the finite-presentation trichotomy is left as an open problem here. (Source 03.)

The questions below are directions produced by the proved classification. They are not all asserted to be longstanding open problems in the published literature. ProveIt's monoid cardinality question is answered under the hypotheses of this article and is not repeated as unresolved. (Source 04.)

[write] How the questions were merged, 4 October 2026, batch 96. Source 03 asks twelve questions (unnumbered by section, Questions 1–12 there, each followed by a paragraph of commentary), source 04 twelve (Questions 12.1–12.12). Eight of source 03's questions coincide with or contain one of source 04's; each such pair is printed once, with both wordings, and credited to both. The result is sixteen questions, in source 03's order with source 04's four unmatched questions after them, followed by one question added at the write under ProveIt's standing rule for unproved claims (docs/incoming/README.md, Section 4, item 11). Where a question continues one of Parts I–II or of Part VIII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (labels hset : ns :), the credit line says so; those questions are not reprinted here. None of the seventeen is answered in ProveIt.

Question 37.1 (Several positive relations and exact inventories; efficient finite boundary enumeration). For a finitely presented commutative monoid with rank-one group completion, give an efficient enumeration of all infinite connected types, including their local automorphism groups. The two-generator power relation admits the necklace formula above. With three or more generators, identify finite boundary data that replace the single cyclic composition.

Source 04's wording: for a rank-one effective monoid with three or more generators and a specified finite isotropy subgroup, compute the number of proper connected types efficiently from its conductor and gap poset. Determine which presentation families admit product formulas or rational generating functions comparable to the two-generator necklace formula.

Sources: source 03, Question 1; source 04, Question 12.2. For a fixed isotropy subgroup the number is finite and there is a finite enumeration method ([theorem 31.1](#)); efficiency and closed forms are what is asked.

The next concrete deliverable would be a certified enumerator for normalized ideals of a rank-one monoid with finite torsion, together with a proof of completeness. A type-count decision alone is

already supplied by [theorem 32.1](#).

Question 37.2 (Prescribed surjectivity). Add requirements that selected named injections be surjective or non-surjective. Classify when the remaining connected-type set is empty, finite, countably infinite, or continuum many, and determine the corresponding universal axiom thresholds.

Sources: source 03, Question 2.

For connected actions, non-surjectivity is a genuine extra condition: in the one-injection example it excludes every finite cycle and the bilateral chain, leaving just a ray. Global non-surjectivity of a disconnected action does not exclude finite-cycle components. Consequently the exact cardinality statement cannot simply be reused without modification. For the power relation, a first step is to read these restrictions from the drop sequence. (For $f^m = g^n$, [theorem 33.1](#) (iii) already shows that in a connected proper type both maps are non-surjective, and in a bilateral type both are bijective.)

Question 37.3 (Noncommutative monoids with common multiples; beyond commutativity). Extend the relation-sensitive trichotomy to specified finitely presented noncommutative monoids whose actions admit a useful group-of-fractions localization. Determine which group and order invariants control the number of invariant subsets of coset actions.

Source 04's wording: for finitely generated cancellative monoids with an Ore localization, identify hypotheses under which connected actions still admit a useful subset normal form. Determine a first family where nonabelian isotropy and boundary geometry can both be classified explicitly, before seeking an analogue of the general rank theorem.

Sources: source 03, Question 3; source 04, Question 12.9. Related: [Question 23.5](#) of Part II for groups.

The rooted-code and kernel theorems already apply to finite noncommuting injective signatures. The missing part is a usable classification of components. Group rank and an abelian cone no longer suffice.

Question 37.4 (Ordinary definability of rooted codes; definable atoms at the finite cutoff). For a fixed union of labelled components, characterize the pure reals definable from specified internal parameters. Combine that characterization with [proposition 27.4](#) to obtain a concrete description of ordinary finite-kernel definable closure.

Source 04's wording: develop an exact description of atom-valued definable closure at $\kappa = \omega$ for common-power actions and for higher-rank rigid components. Infinite type enumerations are not internal at this cutoff, so the finite boundary features detectable by a first-order formula must be identified explicitly.

Sources: source 03, Question 4; source 04, Question 12.5. With pure real parameters, source 04's question is answered by source 03: [theorem 27.3](#) (individually definable atoms), with [theorem 33.4](#) for common-power types (a proper type contributes exactly when its residual multiplicity and δ are both one) and [theorem 30.4](#) for higher-rank rigid components ($\delta_t = 1$, so exactly the types occurring once). What remains is ordinary definability, which is this question. It is also what remains of [Question 23.2](#) of Part II and of [Question 10.1](#) of Part I; Section 24.2.

This is the unresolved step between the exact reduction in [proposition 27.4](#) and an effective classification without arbitrary pure parameters. It should be studied first for finite unions and for arithmetically presented families of power-relation components.

Question 37.5 (Full fixed-object definability at larger cutoffs). Extend [theorem 27.5](#) to arbitrary infinite κ , with a complete internal description of the orbit of a small support enumeration. Determine the minimal additional pure data needed beyond its word-equality diagram.

Sources: source 03, Question 5. It contains what remains of [Question 23.1](#) of Part II for arbitrary set parameters at $\kappa > \omega$, inside the saturation of their kernels.

For infinite supports, equality of the generated-component diagrams can miss whether all copies of a type have been exhausted. A promising refinement records residual component multiplicities exactly below κ and records larger multiplicities by the failure of an internal transversal to exist. This is a route to a proof, not a theorem claimed in this manuscript.

Question 37.6 (Atom ideals beyond cardinal bounds; general atom ideals). Replace $|\ker(x)| < \kappa$ by $\ker(x) \in \mathcal{I}$ for an atom ideal. Find hypotheses under which the rooted-code formula gives an exact definable-kernel theorem and an exact Replacement criterion.

Source 04's wording: replace the cardinal bound on $\ker(x)$ by membership in a specified atom ideal. Separate the roles of closure under component saturation, unions of recognizable small blocks, and compression into reservoirs. Find a necessary and sufficient criterion for a substantial family of ideals beyond cardinal cutoffs.

Sources: source 03, Question 6; source 04, Question 12.6. The same question for named lifts is Question 10.7 of Part I, for commuting and group actions [Question 23.9](#) of Part II, and for group actions at the finite cutoff Question 65.1 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:q:ideals`). What is new here is the rooted-code and reservoir form.

Cardinal cutoffs automatically handle deletion of small families of copies. A general ideal need not. Closure under signed-path images, behavior under component swaps, and closure under definable unions of pointed fibers should be separated before attempting an equivalence.

Question 37.7 (Weak ambient choice; choice strength). Determine the weakest ambient choice principles sufficient for the type trichotomy, the kernel formula, and the universal axiom tables, separately.

Source 04's wording: audit the exact choice requirements of the algebraic normal forms, the type-cardinality statements, the atom-permutation extensions, and the reservoir construction. The explicit Cantor family and the finite certificates should be separated from the selection of families of component isomorphisms.

Sources: source 03, Question 7; source 04, Question 12.10. For named lifts this is Question 10.8 of Part I, for commuting actions [Question 23.8](#) of Part II.

The explicit continuum family needs no choice of translation representatives. The global extension of component isomorphisms and the common witness reservoirs do use ambient Choice. Isolating these uses would bring the construction closer to the choiceless setting of Glazer and Yao.

Question 37.8 (Syntactic complexity of failed instances). Bound the formula complexity of the type selector and the failed Replacement instances in terms of the presentation. For a power relation, replace the general rooted-code predicate by direct boundary recognition and determine whether the resulting complexity bound is optimal.

Sources: source 03, Question 8. The same question for the earlier negative examples is Question 10.9 of Part I, [Question 23.10](#) of Part II and Question 65.6 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:q:complexity`).

The problem should distinguish fixed numerical exponents, a pure code parameter, and a variable family of relation matrices. These different inputs permit different amounts of uniformity in a single formula.

Question 37.9 (Infinitely generated actions; countably generated monoids). Find additional hypotheses under which an infinitely generated commutative monoid has countably many connected action types. Clarify the effect of separate operation names versus one uniform evaluation symbol.

Source 04's wording: find a replacement for the rank criterion for selected countably generated commutative monoids. The example $\mathbb{Q}_{\geq 0}$ suggests controlling the index of positive cyclic subgroups or the cut structure of the induced order. Keep the algebraic classification separate from the choice of naming language in the model theory.

Sources: source 03, Question 9; source 04, Question 12.8. The language distinction is the one of Section 19 of Part II (separate names versus the evaluator E).

The rational-cut counterexample shows that finite rational rank is inadequate. A plausible starting class consists of unions of rank-one discrete monoids with controlled index growth; the language must be specified before an axiom-scheme assertion is made.

Question 37.10 (Noninjective maps and branching). Determine which parts of the rooted-code theorem extend when inverse images can have more than one element. Identify a replacement for the freeness of local automorphisms on a component.

Sources: source 03, Question 10. Compare Question 10.5 of Part I (other bounded-degree signatures) and Question 65.3 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:q:relational`).

With branching, one endpoint no longer determines an isomorphism. Codes based on finite partial substructures, rather than deterministic signed words, may require substantially more descriptive complexity.

Question 37.11 (Boundary statistics and finite certificates; refined common-power statistics). Refine the necklace enumeration by missing-predecessor counts, least period, and automorphism order. Seek generating functions and compact certificates for the resulting distributions, including several-relation cases.

Source 04's wording: refine the necklace count by the size of the intersection $(S \setminus fS) \cap (S \setminus gS)$, by distributions of consecutive boundary gaps, and by automorphism order simultaneously. Seek bijections and generating functions that distinguish these action statistics from classical area statistics on numerical-semigroup ideals.

Sources: source 03, Question 11; source 04, Question 12.3. The distribution by automorphism order alone is [theorem 33.5](#); the missing-predecessor counts $|S \setminus f(S)|$, $|S \setminus g(S)|$ alone are [proposition 33.6](#). The joint refinements are open.

The period computation already identifies the symmetry order of every individual cyclic composition. The further task is a systematic refined enumeration, with a proved connection to finite atom-fiber sizes in the corresponding model.

Question 37.12 (Machine-checked synthesis; formalization in ProveIt). Formalize the chain from a presentation certificate through localization and the type count to the exact Replacement verdict, without importing the final cardinality or model-theoretic conclusion as an axiom.

Source 04's wording: implement the localization and finite certificates first, then prove the common-power normal form and finite enumeration. A later layer should formalize the first-order model and its expanded schemes, with a separate theorem connecting semantic type codes to the internal type-recognition formula. No semantic shortcut should identify ambient Replacement with the scheme under study.

Sources: source 03, Question 12; source 04, Question 12.12. For the earlier results this is Question 10.12 of Part I, [Question 23.12](#) of Part II and Question 65.12 of Part VIII of birthday-cutoffs-and-hereditary-sets (hset:ns:q:formalization).

A useful first milestone is the complete $f^2 = g^3$ classification and its three infinite components, followed by the general boundary-necklace theorem. The finite computation is small; the important work is connecting it to the semantics of arbitrary components.

Question 37.13 (Complexity of the presentation test). Determine the bit complexity of deciding d , producing the grading, and certifying every unit generator. Rational linear programming avoids exhaustive circuit enumeration, but the costs of compact exact certificates and Smith normal forms should be tracked separately. A first concrete goal is a verified implementation with explicit size bounds.

Sources: source 04, Question 12.1. Compare [Question 23.5](#) of Part II, which asks the corresponding complexity for finitely presented groups.

Question 37.14 (Prescribed automorphism groups). For a general finite commutative presentation, a subgroup H , and a subgroup $K \leq G/H$, characterize when an invariant carrier S has exactly K as its translation automorphism group. The unit image must lie in K . Determine when that necessary condition and a quotient cone condition are sufficient.

Sources: source 04, Question 12.4. For $f^m = g^n$ the answer is [theorems 33.4](#) and [33.5](#).

Question 37.15 (Descriptive complexity of the type relation). On the natural coded space of invariant carriers, determine which monoids give smooth translation classification on particular definable subfamilies. The existence of a Cantor family of inequivalent rigid carriers does not by itself determine this descriptive complexity. Boundary restrictions, not cardinality alone, should be part of the question.

Sources: source 04, Question 12.7. Compare [Question 23.4](#) of Part II, on subgroup spaces.

Question 37.16 (Forcing and individual expansions). Combine [theorem 35.1](#) with precise preservation hypotheses on the cutoff and block cardinalities. Distinguish a new action using new real-coded types from a changed scheme in one fixed ground-model action. Investigate exact examples when the cutoff is collapsed or its cofinality changes.

Sources: source 04, Question 12.11. It continues [Question 10.10](#) of Part I and [Question 23.11](#) of Part II; the fixed-action half with the cutoff preserved is [theorem 21.1](#) (the note after [theorem 35.1](#)).

[write] A question added at the write, 4 October 2026, batch 96. Provel's standing rule for unproved claims moves the following claim of source 03 here; it is kept in place, with a dated note, in Section [27.2](#).

Question 37.17 (The parameter-free definable kernel of a rigid continuum family). *Source:* source 03, Section 4.3 (Section [27.2](#) here). *Claim:* for two commuting injections with $\mathfrak{c}$ pairwise nonisomorphic rigid countable components, one copy of each, at $\kappa = \omega$, every atom is definable from its pure real rooted code, while “the parameter-free definable kernel is at most countable”; so the pure-parameter qualification in [theorems 27.1](#) and [27.3](#) cannot be dropped. *Argument given:* there are countably many parameter-free formulas and each defines at most one object, with a finite kernel; source 03 calls this “external semantic reasoning, for example after placing the construction in a set-sized ambient model whose atom set is uncountable in a larger metatheory with access to its satisfaction relation”. *Missing:* a framework in which the claim is a theorem, and its proof. In

the scheme reading of [remark 25.4](#) the union over all formulas is not an object of the ambient theory (the note in [Section 27.2](#)). In a set-sized ambient model W inside a universe that has its satisfaction relation, the counting argument is immediate, but the contrast with $D_\omega^{\text{pt}} = A$ needs A to be uncountable in the outer universe (so W cannot be countable) and needs [theorems 26.2](#) and [30.4](#) to hold in W ; what hypotheses on W suffice, and can they be met without assuming a set model of the whole ambient theory? Is there a statement *inside* the ambient theory, one formula at a time, that expresses the same necessity, for instance that no single formula with a fixed finite tuple of parameters defines, as its parameter varies over a set of size below $\mathfrak{c}$, every atom of the family?

Sources: the write, from source 03, Section 4.3.

[write] Status of the questions, 4 October 2026, batch 96. None of these seventeen questions is answered in ProveIt. The ones that continue questions of Parts I–II or of Part VIII of birthday-cutoffs-and-hereditary-sets say so in their credit lines: [Question 37.4](#) is what remains of [Questions 10.1](#) and [23.2](#), [Question 37.5](#) of [Question 23.1](#); [Questions 37.6](#) to [37.8](#), [37.12](#) and [37.16](#) are Part I’s [Questions 10.7](#), [10.8](#), [10.9](#), [10.12](#) and [10.10](#) and Part II’s [Questions 23.8](#) to [23.12](#) asked again for monoid actions (with Part VIII’s [Questions 65.1](#), [65.6](#) and [65.12](#) for the first, third and fourth); [Question 37.10](#) meets Part I’s [Question 10.5](#). Conversely this Part answers Part II’s [Question 23.3](#) and, with pure real parameters only, its [Questions 23.1](#) and [23.2](#) and Part I’s [Question 10.1](#); dated notes after those four questions record it, and [Section 24.2](#) gives the details.

F Notation and reproducibility

Notation	Meaning
A, κ	External atom set and infinite kernel cutoff, with $ A \geq \kappa$.
$\ker(x)$	Atoms in $\text{TC}(\{x\})$; the actual transitive-closure kernel.
$M_\kappa(A)$	All objects with fewer than κ atoms in their kernel.
$\mathcal{Z}$	Atom permutations commuting with every named injection.
$\text{Sat}(B)$	Union of signed-path components meeting B .
D_κ^{pt}	Atoms in $\mathcal{Z}$ -orbits of size below κ .
D_κ	Union of entire unpointed type blocks of size below κ .
μ_t, δ_t	Number of copies of type t ; cardinality of its local automorphism group.
M, P, G	Presented commutative monoid; its cancellative image; its group completion.
U, Q, d	Units $P \cap (-P)$; quotient G/U ; its free rank.
$\tau(M)$	Number of connected labelled action types.
g_0	Greatest common divisor of the exponents in $f^m = g^n$.
e	Order of an infinite component’s torsion localization stabilizer.
δ	Repetition count of its boundary word; its local symmetry order.
$\text{Neck}(a, b)$	Rotation-class count of weak compositions of b into a parts.

Sources: source 03, Appendix A (its ak and $\mathcal{N}(a, b)$ printed $\ker$ and Neck). The full notation of this Part is in [Section 24.6](#).

To rebuild the article, run `latexmk -pdf atom_actions.tex` from the package directory. The vector figure is included, so compiling the PDF does not require Python. To reproduce the exact checks, run `python3 verify_presentations.py`; the dependency is SymPy. The figure can be regenerated with `python3 figures/make_figure.py` using Matplotlib. The accompanying `README.txt`

lists the commands and the precise verification scope. No external dataset or network request is needed to run the finite checks. (Source 03; in ProveIt the delivered article and `README.txt` are not shipped, the programs have prefixed names and write beside themselves by default, and the report's `README` gives commands that do not overwrite the shipped records: the note in Section 36.)

G Source audit and limits of the conclusion

The research-interest connection was checked against Glazer's thesis and Glazer–Yao's primary paper. The repository questions were read directly in the pinned current source, and the related bounded-kernel report was checked to avoid re-presenting its finite-orbit Replacement theorem as new. Literature searches also covered localization of monoid acts, torsion-free indecomposable acts, affine-monoid units and cones, and the rational Catalan special case. The standard sources listed below support those ingredients.

No inspected source was found stating the exact general type-count trichotomy in [theorem 30.4](#). This is a limited search result, not evidence that every relevant publication has been excluded. The strongest supported claim is that the article proves its stated results and answers the identified repository questions under explicit conventions. Questions about unrestricted choiceless set theory, noninitial elementary embeddings, arbitrary noninjective operations, and field-wide historical priority are outside that claim. (Source 03, Appendix B.)

[write] The cited papers, checked at the write, 4 October 2026, batch 96. Source 03's "answers the identified repository questions" is re-scoped in Section 24.2: Question 23.3 is answered, Questions 23.1 and 23.2 with pure real parameters. The arXiv records of [1] and [3] were checked at the write of Part I (Section 1.4); the new citations were not rechecked at this write, and the page and item references in them (Bailey–Renshaw's Lemmas 6.4–6.8, Boardman's Theorems 7–8, Koch's Lemmas 3.4.4–3.4.5, Glazer–Yao's Definition 18 and Theorem 19) are the sources' own.

References

- [1] Elliot Glazer and Bokai Yao, *Reflection Principles in ZFU*, arXiv:2602.21970v1, 25 February 2026. <https://arxiv.org/abs/2602.21970>.
- [2] Elliot Glazer, *Global choice is not conservative over local choice for Zermelo set theory*, arXiv:2312.11902v3, January 2024. <https://arxiv.org/abs/2312.11902>.
- [3] Bokai Yao, *Set Theory with Urelements*, Ph.D. dissertation, University of Notre Dame, June 2023; arXiv:2303.14274v3. In particular, Definition 24, Lemma 25, Theorem 26, and the kernel-bound examples in Theorem 27. <https://arxiv.org/abs/2303.14274>.
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- [5] Vladimir Reshetnikov and contributors, *ProveIt: Bounded-complexity consistency for ZFC*, SetTheory/BoundedConsistency/README.md, repository documentation consulted for the external-index and language-complexity distinctions. <https://github.com/VladimirReshetnikov/ProveIt/blob/2b7b388ba81a3355b19a7c2d2fe797e92f572355/SetTheory/BoundedConsistency/README.md>.
- [6] Vladimir Reshetnikov and contributors, *ProveIt: Self-Embeddings of the Surreal Numbers*, research-report README, marked unrefereed and not formalized. <https://github.com/VladimirReshetnikov/ProveIt/blob/2b7b388ba81a3355b19a7c2d2fe797e92f572355/Algebra/SurrealNumbers/docs/surreal/surreal-self-embeddings/README.md>.
- [7] Vladimir Reshetnikov and contributors, *Naming Elementary Embeddings*, AI-assisted, unrefereed research report in the ProveIt repository, filed 3 October 2026. SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings/article.tex. Pinned source at commit a9ab9a69872c2fb7464097af32fd7b76c75cb030. Theorems 5.1, 6.1, 7.5, and 7.7; Questions 10.1, 10.4, and 10.10. [write] This is Part I of the present report, as it stood at the pin; its numbers are unchanged (Section 12.4).
- [8] Vladimir Reshetnikov and contributors, *ProveIt: first-order ZF development*, SetTheory/ZF/Lean/ZF/Zf.lean, same pinned commit as [7]. Pinned Lean source. [write] Part II's entry RepoZF, renamed; the file is the same blob as at the pin of [4].
- [9] Stephen A. Mitchell, *Group actions and other topics in group theory*, University of Washington lecture notes, 11 October 2014, especially Section 5 on transitive G -sets. <https://sites.math.washington.edu/~mitchell/Alg/gset.pdf>.
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- [11] Howard Smith, *Groups with finitely many conjugacy classes of subgroups with large subnormal defect*, Glasgow Mathematical Journal **37** (1995), 69–71.

- <https://doi.org/10.1017/S0017089500030408>. The existence statement used in [remark 18.4](#) is recorded on p. 69.
- [12] S. V. Ivanov and A. Yu. Ol’shanskii, *Some applications of graded diagrams in combinatorial group theory*, in C. M. Campbell and E. F. Robertson (eds.), *Groups St Andrews 1989*, Vol. 2, London Mathematical Society Lecture Note Series 160, Cambridge University Press, 1991, pp. 258–308. <https://doi.org/10.1017/CBO9780511661846.004>. Original construction reference; the existence fact needed here was checked in [11].
- [13] Vladimir Reshetnikov and contributors, *Naming Elementary Embeddings*, ProveIt research report, Parts I–II, including *Commuting Actions and Replacement*. Snapshot 715a716a3002b1e9f28daa2d81c009a392851b94, inspected 4 October 2026. Questions 23.1–23.3 have source labels `nee:ca:q:nonabelian`, `nee:ca:q:injections`, and `nee:ca:q:monoids`; source 04 also cites Question 23.5, the component criterion in Parts I–II and the reservoir arguments in Part I. [Pinned report source](#). This is a repository research manuscript, not a claim of peer-reviewed publication. [write] Sources 03 and 04’s entries `RepoNaming` and `Repo`; this is Parts I–II of the present report, as they stood at the pin, with the numbers they have here (Section 24.5).
- [14] Vladimir Reshetnikov and contributors, *Birthday Cutoffs and Hereditary Sets*, ProveIt research report, Part VIII, especially Theorem 59.1, source label `hset:ns:thm:graded`. Same snapshot as [13]. [Pinned report source](#).
- [15] Vladimir Reshetnikov and contributors, *ProveIt*, repository overview and formal mathematics. <https://github.com/VladimirReshetnikov/ProveIt>.
- [16] Elliot Glazer, *Interactions Among Principles of Choice, Forcing, and Measure Theory*, Ph.D. dissertation, Harvard University, 2023. <https://nrs.harvard.edu/URN-3:HUL.INSTREPOS:37378151>.
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